

# **Ada - Programming Learning Platform**

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**Class:- 12NSTK**

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**Version 1.0.0**

| REVISION HISTORY |         |                                                                                                                                                                                |                         |
|------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| DATE             | VERSION | DESCRIPTION                                                                                                                                                                    | AUTHOR                  |
| 04/10/2025       | 1.0.0   | First full version of the technical documentation, including UI designs, ER diagram, SQL queries, routes, colour & fonts, user interactions, and detailed component breakdown. | G. A. Pasindu Vidunitha |
|                  |         |                                                                                                                                                                                |                         |
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# INTRODUCTION

This document gives the necessary technical requirements to develop this learning platform for programming. This platform helps programmers improve their coding skills via text-based tutorials, real-time feedback, and AI-powered support. The project serves as an educational resource for users who need to learn programming languages and concepts, as well as solve coding problems, and they are assisted by both AI tools and community members.

## PURPOSE

The main purpose of this project is to develop an all-in-one learning platform. This project allows users to:

1. Access well-documented programming courses and tutorials.
2. Solve coding challenges and get instant feedback on them.
3. Ask questions from other programmers who are in this community.
4. Answer to other programmers' questions.
5. Get AI-powered code review and debugging assistance.

## INTENDED AUDIENCE AND PERTINENT SECTIONS

| AUDIENCE   | DESCRIPTION                                                            | PERTINENT SECTIONS                                         |
|------------|------------------------------------------------------------------------|------------------------------------------------------------|
| Developers | Responsible for building and maintaining the platform.                 | System Overview, Code structure, API Endpoints, Deployment |
| Testers    | Responsible for resolving bugs and issues associated with this project | Testing and debugging, Code structure                      |

## PROJECT SCOPE

This Project aims to create the best platform for both new and advanced programmers to develop their programming knowledge while keeping their motivation for programming. The main goal of this platform is to make coding education more accessible and engaging for people.

# DESCRIPTION

## PRODUCT PERSPECTIVE

This learning platform gives an interactive learning environment for programmers. This product originates from the need for an all-in-one, dynamic, and engaging platform for new programmers. It is difficult to get a solid knowledge of programming at the start of programming and to keep that motivation consistently. Expected functionalities are Side-by-side tutorials and a code editor, instant feedback, AI assistance for code review, and a gamification system that encourages students.

## FEATURES

1. **Well-documented and Structured text-based tutorials**
  - Learning path for new programmers with correct guidelines
2. **Real-Time coding challenges**
  - Users can solve interactive coding challenges and get feedback on them
3. **AI-powered code review**
  - AI assistance to analyze your code and help users with errors and bugs
4. **Gamification**
  - Points, Badges, and leaderboards to motivate the users and check their progress
5. **Community Interaction**
  - Users can discuss their problems with other users
6. **Integration with External Resources**
  - User can enhance their courses, tutorials, and resources if they like.

## USER OVERVIEW

- **Beginners:**  
People who are new to programming and who are looking for tutorials that are easy to understand.
- **Intermediate Users:**  
Users who have some knowledge of programming.
- **Advanced Users**  
Users who want to improve their coding knowledge further.

## OPERATING ENVIRONMENT

- **Hardware platform:**  
This Platform is web-based, therefore, anyone who has a device with an internet connection and a web browser can access this platform.
- **Operating Systems:**  
This platform works on most operating systems, including Windows, macOS, and Linux.
- **Software requirements:**  
This platform needs an internet connection for functionality, and it will integrate with external APIs for additional content.

## CONSTRAINTS: IMPLEMENTATION / DESIGN

Describe limitations impacting development.

- **Limited Backend Database:**  
As this platform uses SQLite, there might be limitations in the database. This might be a problem when the number of users increases.
- **Real-time feedback Performance:**  
AI will take some time to give as much accurate feedback as possible, and because of that, there might be a problem when it comes to performance.
- **Cross-Browser Compatibility:**  
The platform should work on all browsers as expected without having problems with the design, etc.

## ASSUMPTIONS / DEPENDENCIES

Detail all assumed factors (not known facts) that could potentially impact the technical specifications set forth. Include external factors.

- **User Internet Connection:**  
This platform assumes that users will have a stable internet connection.
- **AI assistance:**  
The successful AI-powered features depend on the machine-learning model's accuracy.
- **Third-party APIs for additional Content:**  
As the platform depends on external AI APIs to expand its learning materials, this might be a problem if third-party providers discontinue or change

# SYSTEM FEATURES

## SYSTEM FEATURE 1: User Authentication

|                               |                                                                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION AND PRIORITY      | <ul style="list-style-type: none"><li>• The system allows users to create accounts, log in, and manage their profiles</li><li>• <b>Priority:</b> High</li></ul>                                                                                                                           |
| STIMULUS / RESPONSE SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> User provides login details or registers a new account</li><li>• <b>Response:</b> The system verifies account details and gives access to the platform</li></ul>                                                                 |
| FUNCTIONAL REQUIREMENTS       | <ul style="list-style-type: none"><li>• The System must validate the login process by a secure authentication method.</li><li>• Users should have a customizable profile, such as a username and custom avatar.</li><li>• User accounts should be allowed to recover passwords.</li></ul> |

## SYSTEM FEATURE 2: Courses and Learning Routes

|                               |                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION AND PRIORITY      | <ul style="list-style-type: none"><li>• This system offers a list of available courses categorized by language.</li><li>• <b>Priority:</b> High</li></ul>                                                   |
| STIMULUS / RESPONSE SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> The user selects a course.</li><li>• <b>Response:</b> The system displays the selected course.</li></ul>                                           |
| FUNCTIONAL REQUIREMENTS       | <ul style="list-style-type: none"><li>• The platform must display the relevant course when the user selects that course.</li><li>• Users should be able to track their progress with each course.</li></ul> |

### SYSTEM FEATURE 3: Code editor and Compiler integration

|                                     |                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• An integrated code editor that allows users to write code and compile it on this platform.</li><li>• <b>Priority:</b> High</li></ul>                                                                                                                            |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> Users write code in the given code editor.</li><li>• <b>Response:</b> The system compiles those codes and shows the relevant output</li></ul>                                                                                                  |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• The code editor must support multiple programming languages.</li><li>• The platform must provide real-time feedback on code errors and output.</li><li>• Users should be able to submit their codes in the challenges section to check their answers.</li></ul> |

### SYSTEM FEATURE 4: Coding challenges

|                                     |                                                                                                                                                                                                                                                                                 |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• The platform provides various coding challenges for users at different difficulty levels.</li><li>• <b>Priority:</b> High</li></ul>                                                                                                     |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> Users select a coding challenge and submit the answer.</li><li>• <b>Response:</b> The system presents the selected coding challenge, then evaluates the user's solution and gives feedback on that using AI.</li></ul> |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• This platform must allow users to submit their solutions and get detailed feedback on them.</li></ul>                                                                                                                                   |

## SYSTEM FEATURE 5: AI assistance for code reviews and debugging

|                                     |                                                                                                                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• This platform uses AI to get feedback and debug Users' codes and assist them.</li><li>• <b>Priority:</b> High</li></ul>                                           |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> Users submit their code for review or debugging</li><li>• <b>Response:</b> The AI assistance analyzes that and gives feedback on that.</li></ul> |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• The system must provide real-time debugging assistance for users.</li><li>• AI must give suggestions to the user's code if they ask.</li></ul>                    |

## SYSTEM FEATURE 6: Chat with other programmers

|                                     |                                                                                                                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• Users can ask questions and give answers</li><li>• <b>Priority:</b> High</li></ul>                                                                                                                                      |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> Users can post answers and questions</li><li>• <b>Response:</b> The system allows users to post questions and answers, and gives a notification to the original poster when someone responds</li></ul> |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• The platform must allow for the posting of questions and answers</li><li>• Users should be able to search for them by keywords</li></ul>                                                                                |

## SYSTEM FEATURE 10: Integration with External APIs for Additional Resources

|                               |                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION AND PRIORITY      | <ul style="list-style-type: none"><li>• This platform integrates with external APIs (e.g, Gemini, ChatGPT, or any other AI model) to provide extra learning materials</li><li>• <b>Priority:</b> High</li></ul> |
| STIMULUS / RESPONSE SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> User accesses external resources</li><li>• <b>Response:</b> The system retrieves data from the external API and displays it in the platform</li></ul>  |
| FUNCTIONAL REQUIREMENTS       | <ul style="list-style-type: none"><li>• Users should be able to access external learning materials according to their wishes.</li><li>• The system should handle errors with those external APIs</li></ul>      |

## SYSTEM FEATURE 9: Notifications and Reminders

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION AND PRIORITY      | <ul style="list-style-type: none"><li>• User receives notifications on responses, updates, etc</li><li>• <b>Priority:</b> Low (If the developer has time)</li></ul>                    |
| STIMULUS / RESPONSE SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> A new update, challenge, or message is available</li><li>• <b>Response:</b> The system sends a notification on that</li></ul> |
| FUNCTIONAL REQUIREMENTS       | <ul style="list-style-type: none"><li>• Users should get those notifications</li></ul>                                                                                                 |

## SYSTEM FEATURE 7: Progress Analytics and Feedback

|                                     |                                                                                                                                                                                                               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• The platform tracks the user's progress using courses and challenges that are completed by users</li><li>• <b>Priority:</b> Low (If the developer has time)</li></ul> |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> User completes a course or a challenge</li><li>• <b>Response:</b> The system generates a detailed progress report</li></ul>                          |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• Progress reports should be represented visually</li><li>• The system must detect the user's progress</li></ul>                                                        |

## SYSTEM FEATURE 8: Customizable User Interface

|                                     |                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION<br>AND PRIORITY         | <ul style="list-style-type: none"><li>• This platform allows users to switch from light themes to dark themes or vice versa</li><li>• <b>Priority:</b> Low</li></ul> |
| STIMULUS /<br>RESPONSE<br>SEQUENCES | <ul style="list-style-type: none"><li>• <b>Stimulus:</b> User change their interface</li><li>• <b>Response:</b> The system should update that immediately</li></ul>  |
| FUNCTIONAL<br>REQUIREMENTS          | <ul style="list-style-type: none"><li>• The platform must support light themes and dark themes</li><li>• The system must remember the user's decision</li></ul>      |



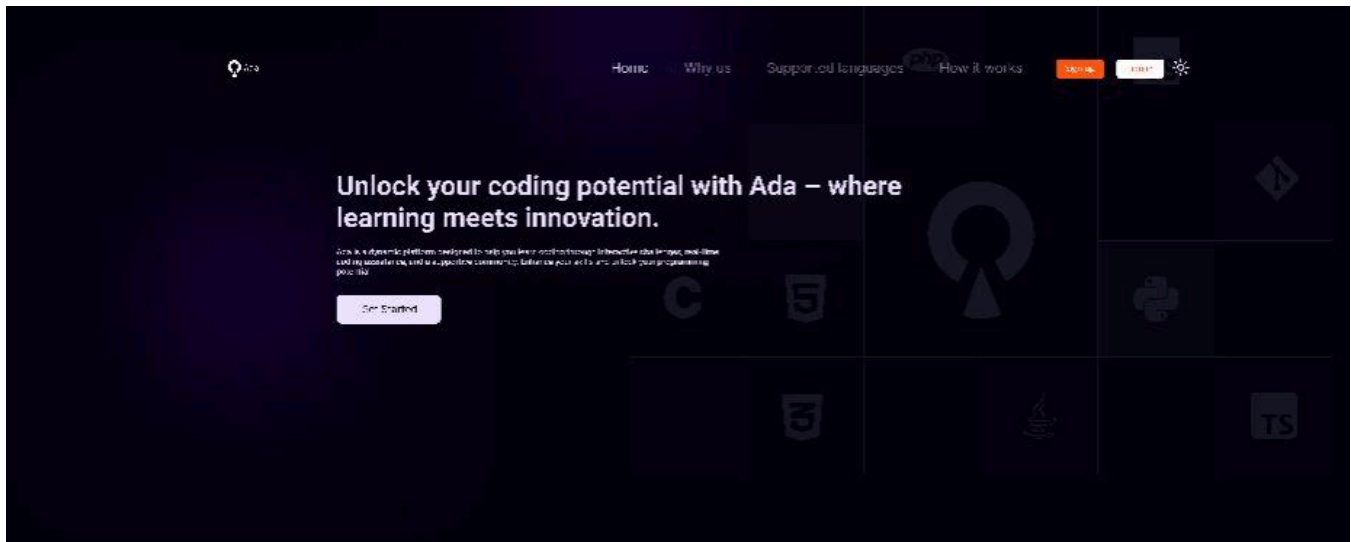
# REQUIREMENTS OF EXTERNAL INTERFACE

## USER INTERFACE BREAKDOWN WITH PURPOSE, SQL & ROUTES

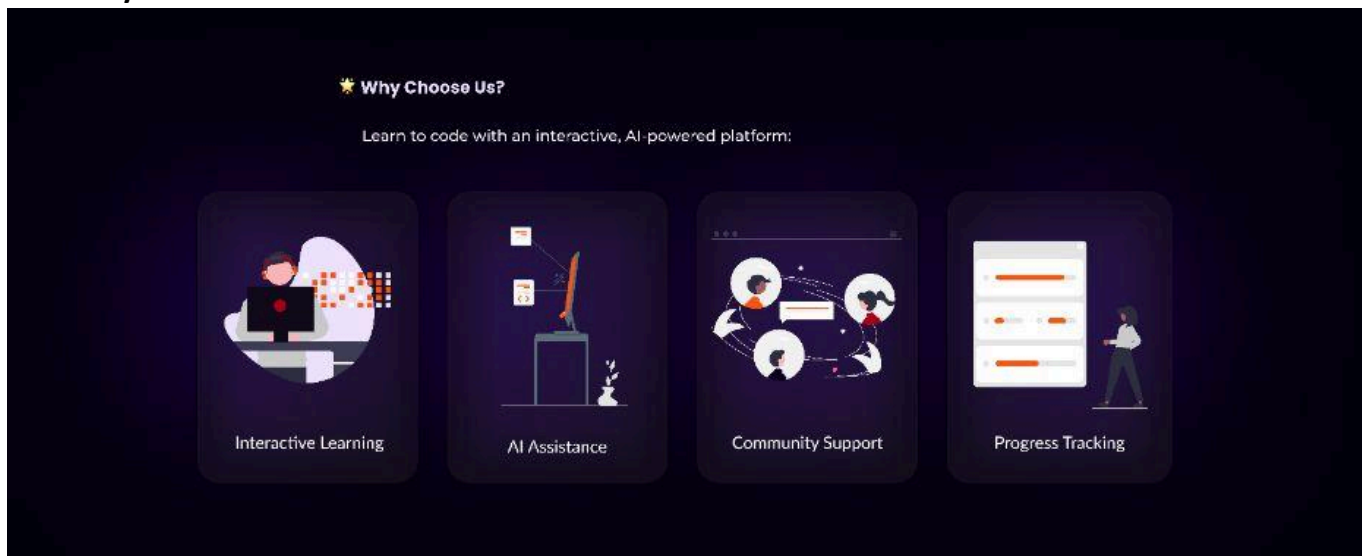
### Landing Page (Desktop - Dark)

---

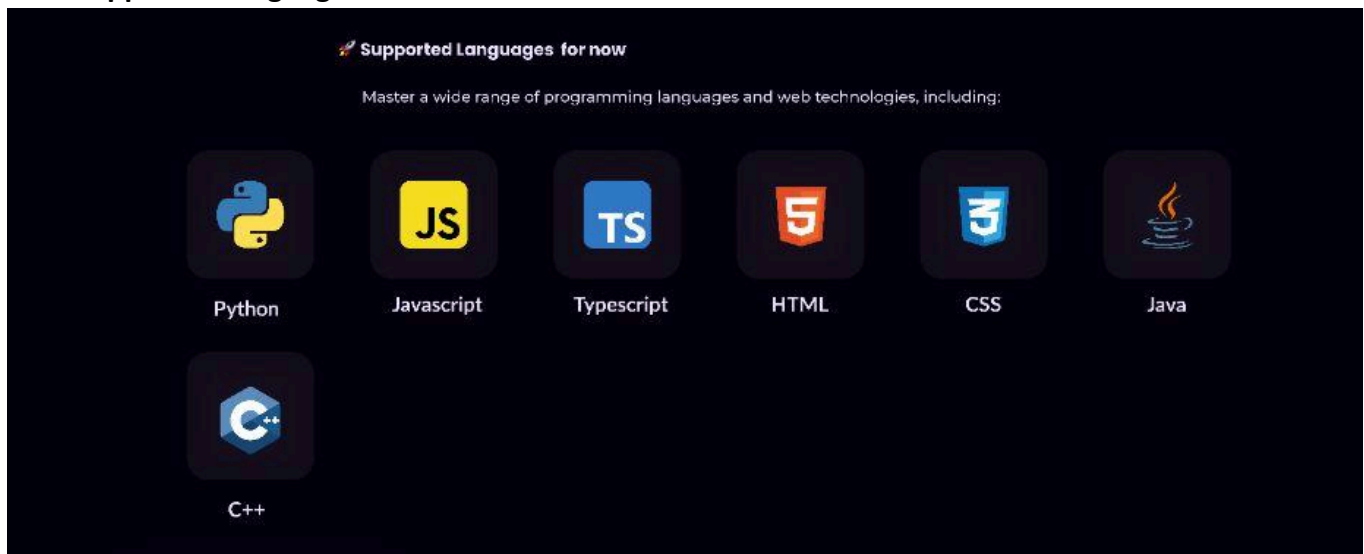
#### 1. Home Section



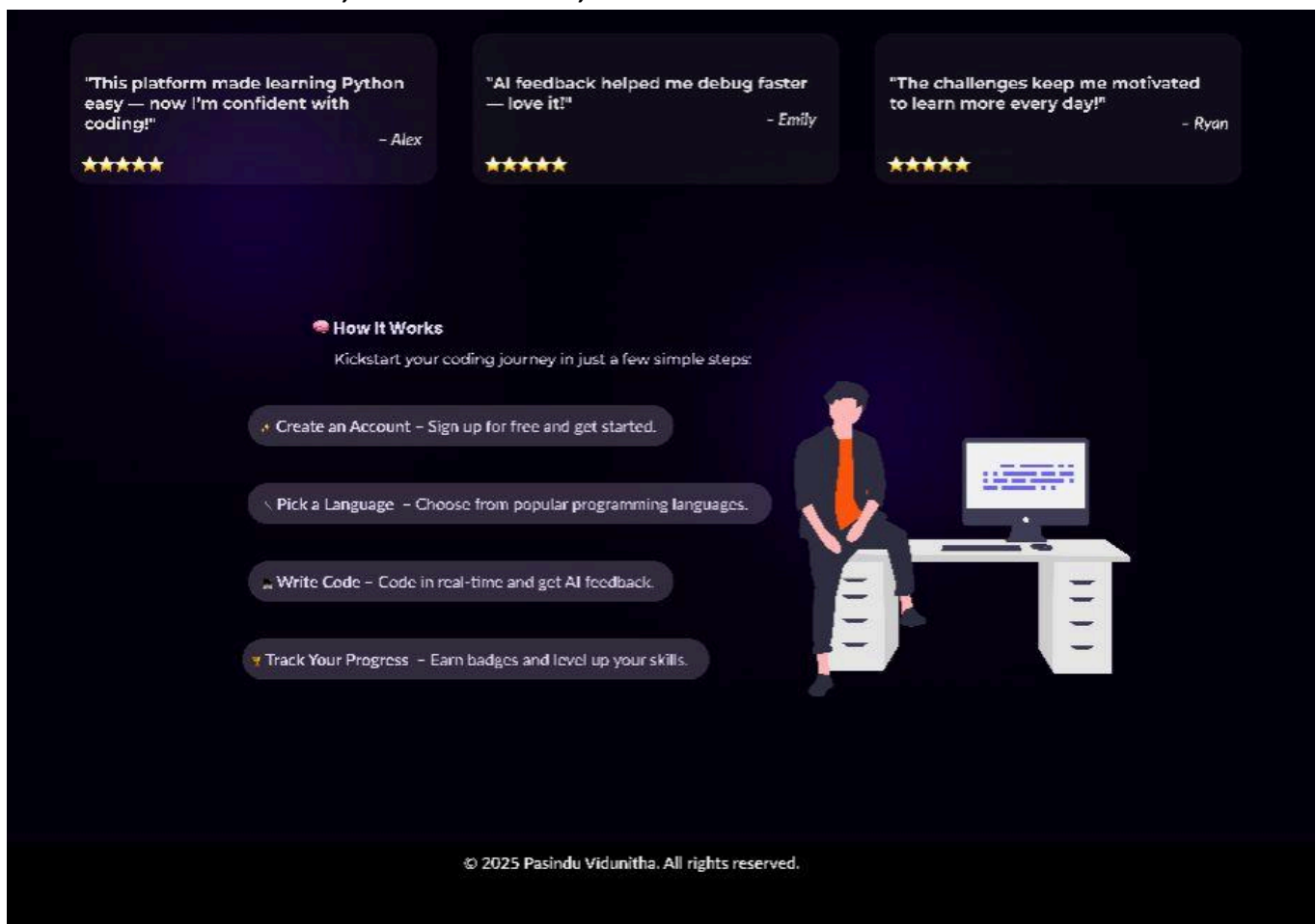
#### 2. Why Us Section



### 3. Supported Languages Section

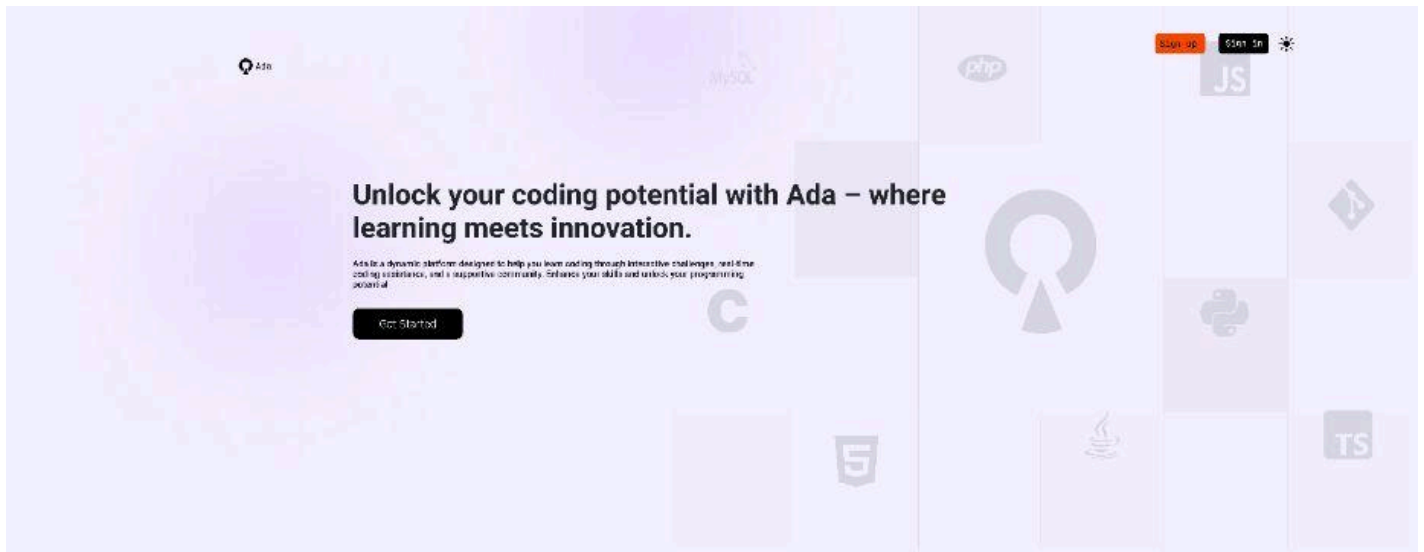


### 4. How it works Section, comment section, and the footer



## Landing Page (Desktop - Light)

### 1. Home Section



### 2. Why Us Section



### 3. Supported Languages Section

 **Supported Languages for now**

Master a wide range of programming languages and web technologies, including:

  
Python

  
Javascript

  
Typescript

  
HTML

  
CSS

  
Java

  
C++

### 4. How it works Section, comment section, and the footer

"This platform made learning Python easy — now I'm confident with coding!"  
— Alex, Beginner Programmer  
★★★★★

"AI feedback helped me debug faster — love it!"  
— Emily, Web Developer  
★★★★★

"The challenges keep me motivated to learn more every day!"  
— Ryan, Coding Enthusiast  
★★★★★

 **How It Works**

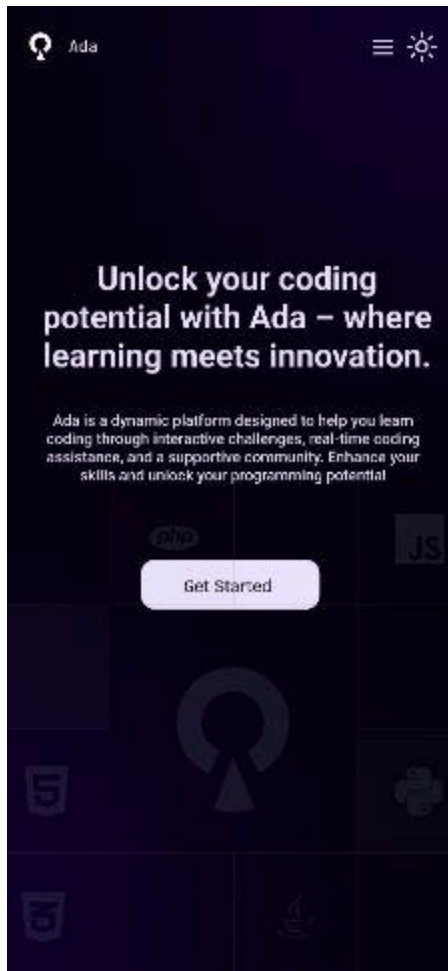
Kickstart your coding journey in just a few simple steps:

- Create an Account – Sign up for free and get started.
- Pick a Language – Choose from popular programming languages.
- Write Code – Code in real-time and get AI feedback.
- Track Your Progress – Earn badges and level up your skills.



## Landing Page (Mobile)

### 1. Home Section



### 2. Why Us Section



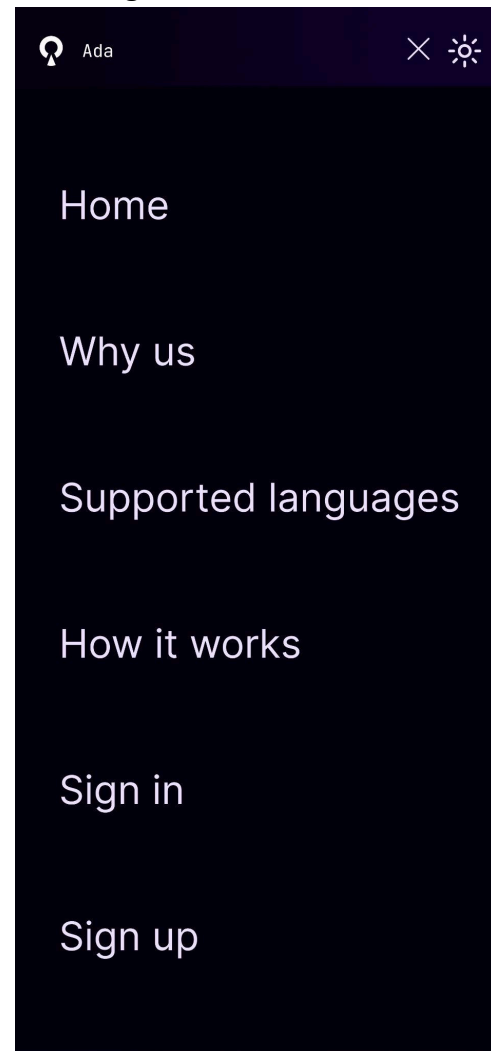
### 3. Supported Languages Section & comment section



#### 4. How it works Section



#### 5. Menu Page

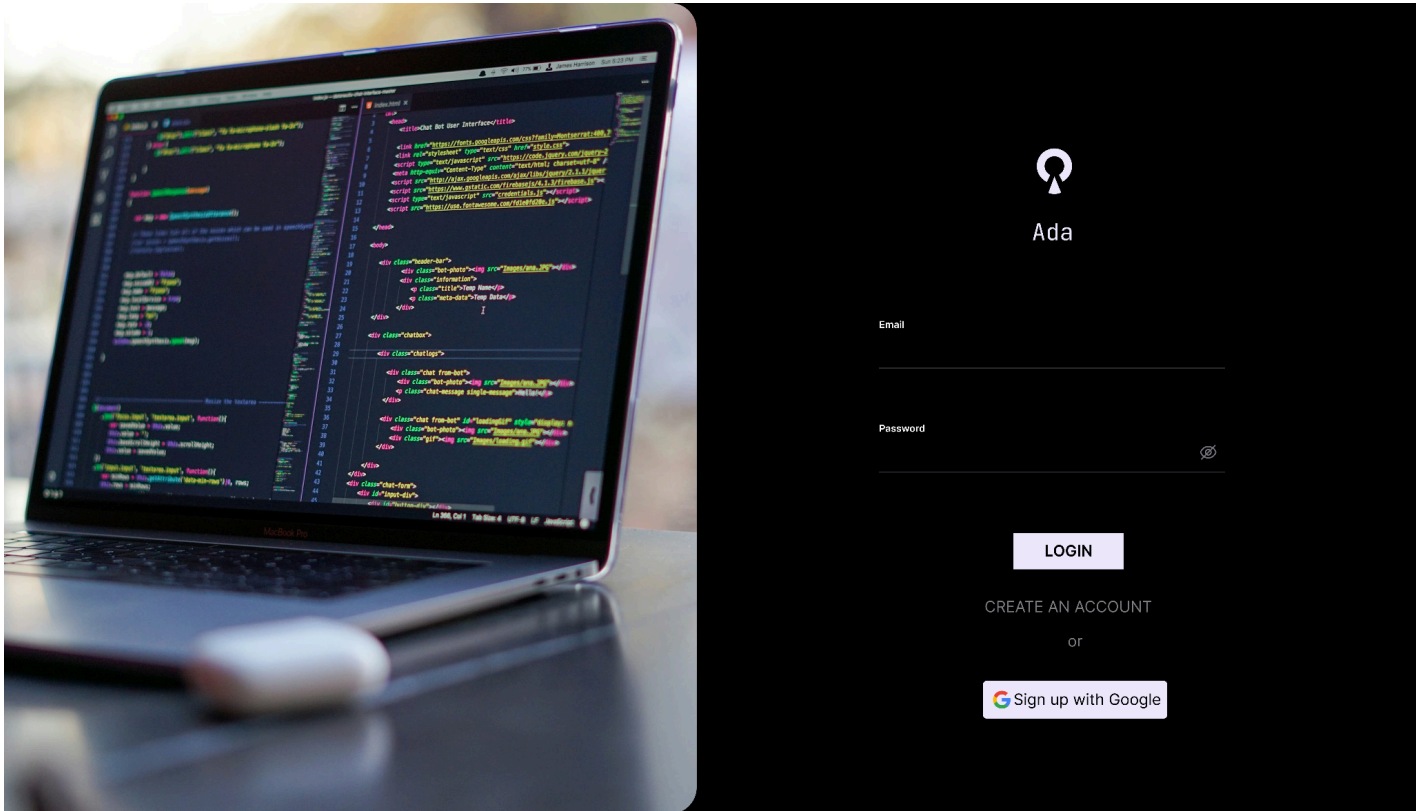


#### Routes & Function Signatures:

```
@app.route('/')
def landing():
    pass
```

## Login Page

### Dark theme Desktop UI





## Dark theme Mobile UI



Ada

Username

Password

LOGIN

CREATE AN ACCOUNT

or

 Sign up with Google



**Purpose:** This allows users to log in with Google or create accounts in the system.

### SQL Queries:

- Normal Login

```
SELECT user_id, profile_image, theme_preference FROM User WHERE email= ? AND password = ? AND auth_provider = "manual"
```

- Log in with a Google account.

```
SELECT user_id, profile_image, theme_preference FROM User WHERE google_id= ? AND auth_provider = "google"
```

- If the user doesn't exist (This is for those who are using Google accounts)

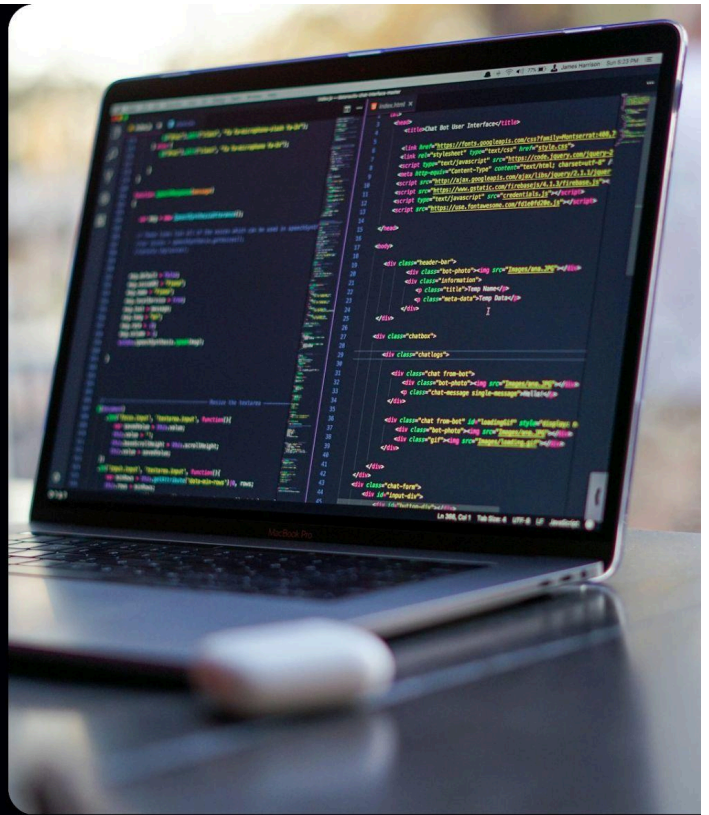
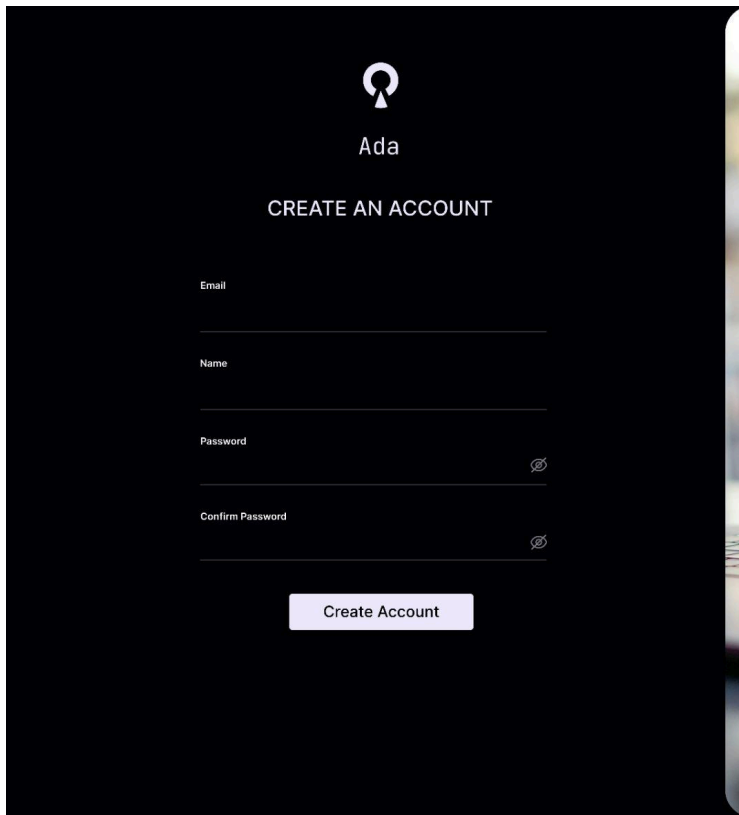
```
INSERT INTO User (user_id, email, full_name, profile_image, auth_provider, google_id, theme_preference) values (?, ?, ?, ?, "google", ?, "dark")
```

### Routes & Function Signatures:

```
@app.route('/login', methods=['GET', 'POST'])
def login():
    pass
```

## Sign Up page

### Dark theme Desktop UI



**Purpose:** To create new accounts for new users

**SQL Query:**

```
INSERT INTO User (user_id, email, full_name, profile_image, auth_provider, password, theme_preference) VALUES (?, ?, ?, "default_img.png", "manual", ?, "dark")
```

**Routes & Function Signatures:**

```
@app.route('/signup', methods=['GET', 'POST'])
def signup():
    pass
```



Ada

## CREATE AN ACCOUNT

Username

Email

Password

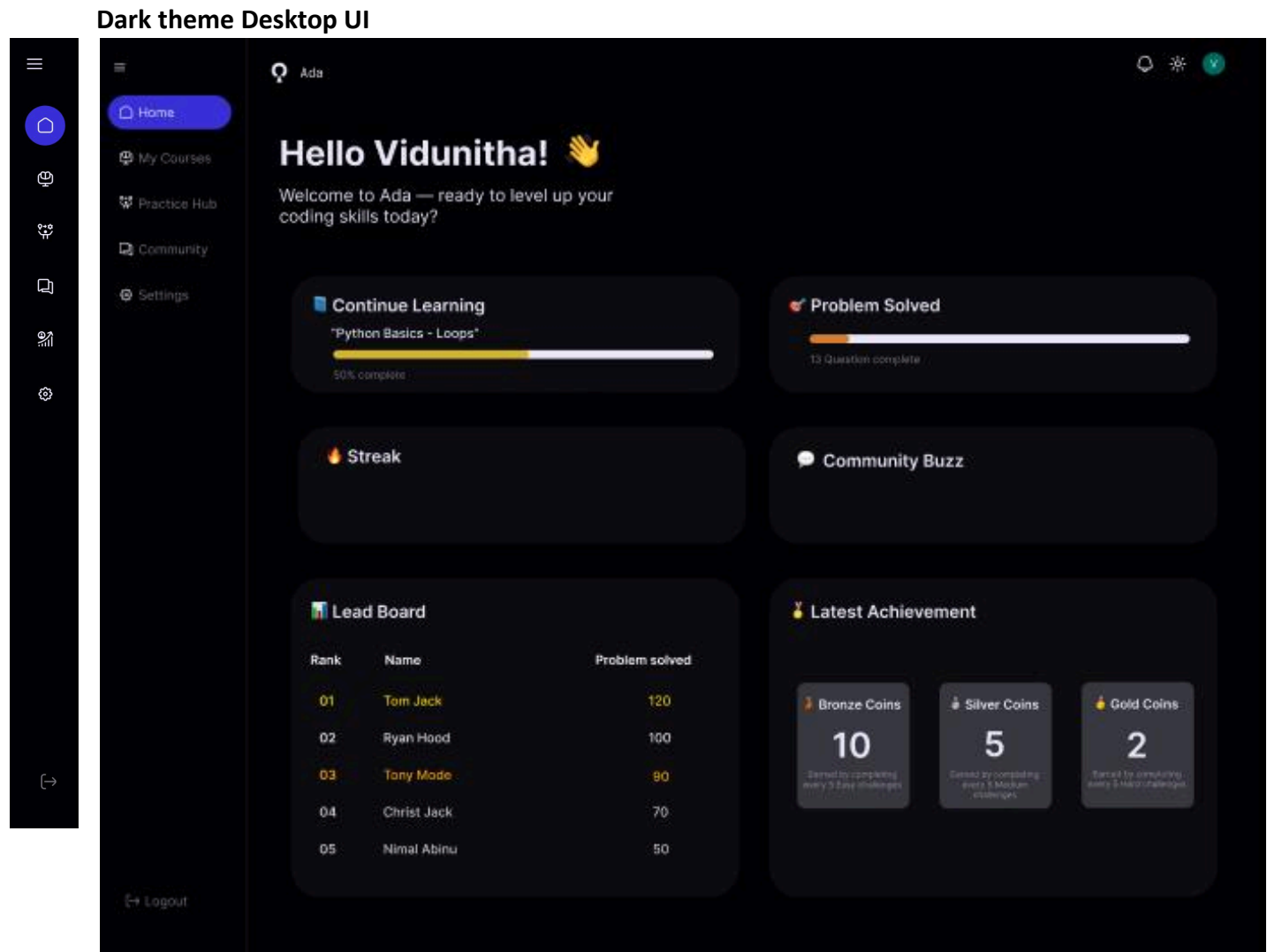


Confirm Password



Create Account

## Dashboard



**Purpose:** To display the User's name, the user's progress and etc.

### SQL Queries:

1. User Welcome Section - Get the Full name of the logged user  
**(SAVE THE USER\_ID, AS WE NEED THAT)**

```
SELECT full_name FROM User WHERE user_id = ?;
```

2. Continue Learning Progress - Get the user's current Course and the percentage of it completed

```
SELECT c.course_name,  
ROUND(COUNT(ul.lesson_id) * 100.0 /
```

```
(SELECT COUNT(*) FROM Lesson WHERE course_id = c.course_id), 0) AS progress_percent FROM
user_lesson ul
JOIN Lesson l ON ul.lesson_id = l.lesson_id
JOIN Course c ON l.course_id = c.course_id WHERE ul.user_id = ?
GROUP BY c.course_id ORDER BY ul.completed_at DESC LIMIT 1;
```

3. Count the number of challenges that have been completed by the user

```
SELECT COUNT(*) AS problems_solved FROM Challenge_attempt WHERE user_id = ?;
```

4. Streak Tracking - Count the challenges completed per day

```
SELECT DATE(completed_at) AS day, COUNT(*) AS challenges_done FROM Challenge_attempt
WHERE user_id = ? GROUP BY DATE(completed_at) ORDER BY DATE(completed_at) DESC;
```

5. Community Buizz - Retrieve the most recent question asked in the community.

```
SELECT question FROM Chat ORDER BY created_at DESC LIMIT 1;
```

6. Display the top 5 users with the highest number of solved challenges.

```
SELECT u.full_name, COUNT(ca.id) AS problems_solved FROM Challenge_attempt AS ca
JOIN User u ON ca.user_id = u.user_id GROUP BY ca.user_id ORDER BY problems_solved DESC
LIMIT 5;
```

7. Coin Rewards (Based on difficulty levels in Challenges)

Bronze Coins (Every 5 Easy Challenges):

```
SELECT COUNT(*) / 5 AS bronze_coins FROM Challenge_attempt AS ca JOIN Challenge c ON
ca.challenge_id = c.challenge_id WHERE ca.user_id = ? AND c.difficulty_level = 'easy';
```

Silver Coins (Every 5 Medium Challenges):

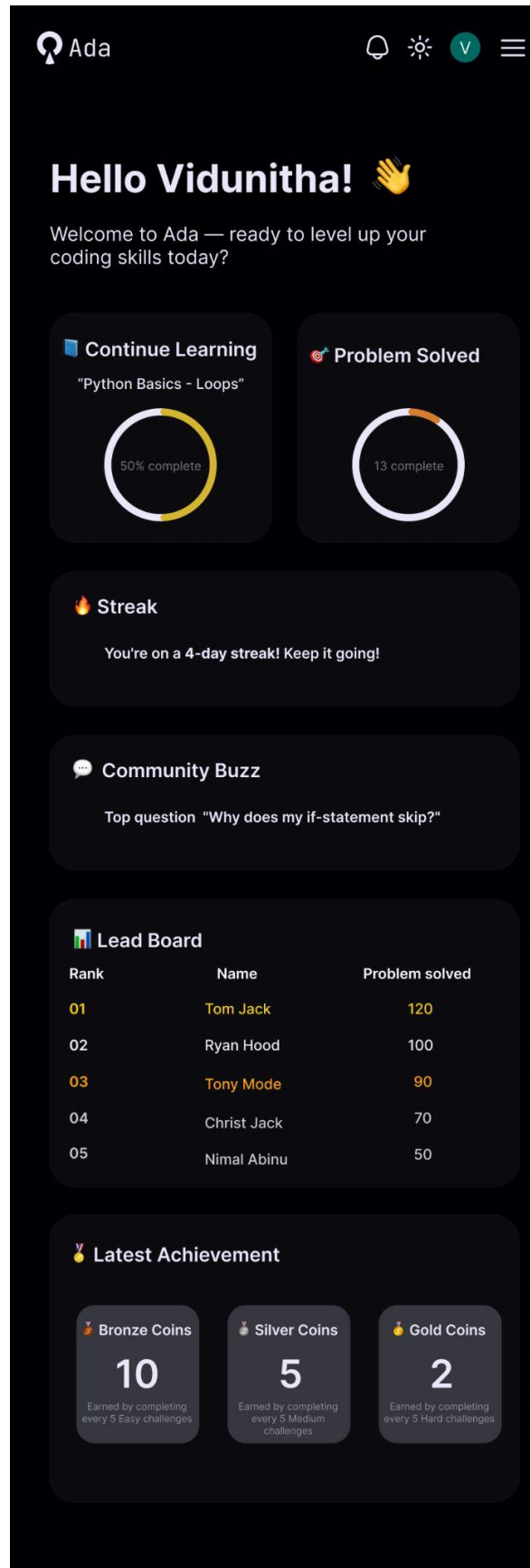
```
SELECT COUNT(*) / 5 AS silver_coins FROM Challenge_attempt AS ca JOIN Challenge c ON  
ca.challenge_id = c.challenge_id WHERE ca.user_id = ? AND c.difficulty_level = 'medium';
```

Gold Coins (Every 5 Hard Challenges):

```
SELECT COUNT(*) / 5 AS gold_coins FROM Challenge_attempt AS ca JOIN Challenge c ON  
ca.challenge_id = c.challenge_id WHERE ca.user_id = ? AND c.difficulty_level = 'hard';
```

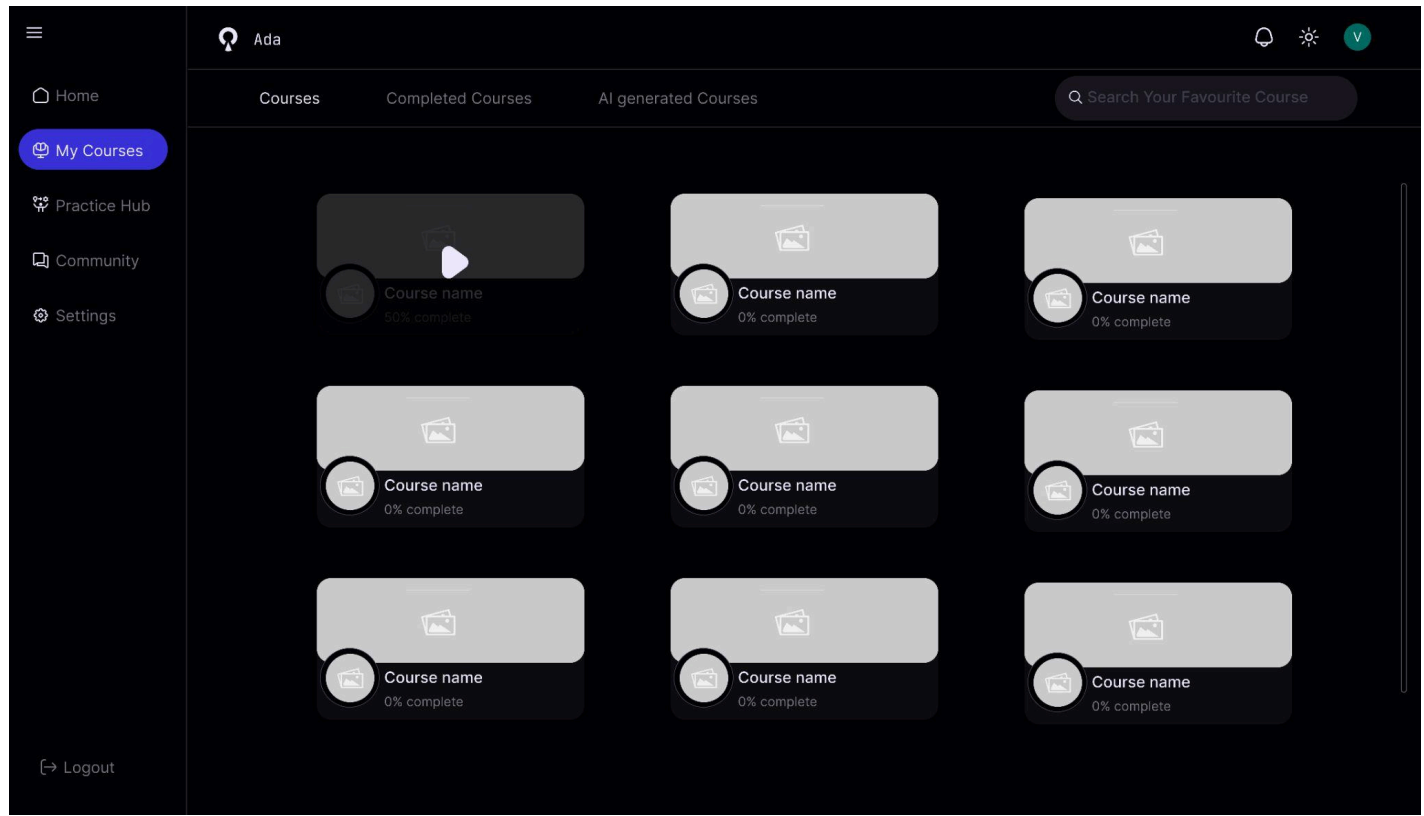
### Routes & Function Signatures:

```
@app.route('/dashboard')  
def dashboard():  
    pass
```



## Course Page

### Dark theme Desktop UI



**Purpose:** To show the courses that users haven't started, and have started but not completed.

**User Interaction:** In the desktop version, when the user hovers over a course, it will change to the first course card.

All the course details are passed in JSON format to the frontend, and when the user clicks on a specific course, Relevant JSON data is sent to the backend, saved for future use.

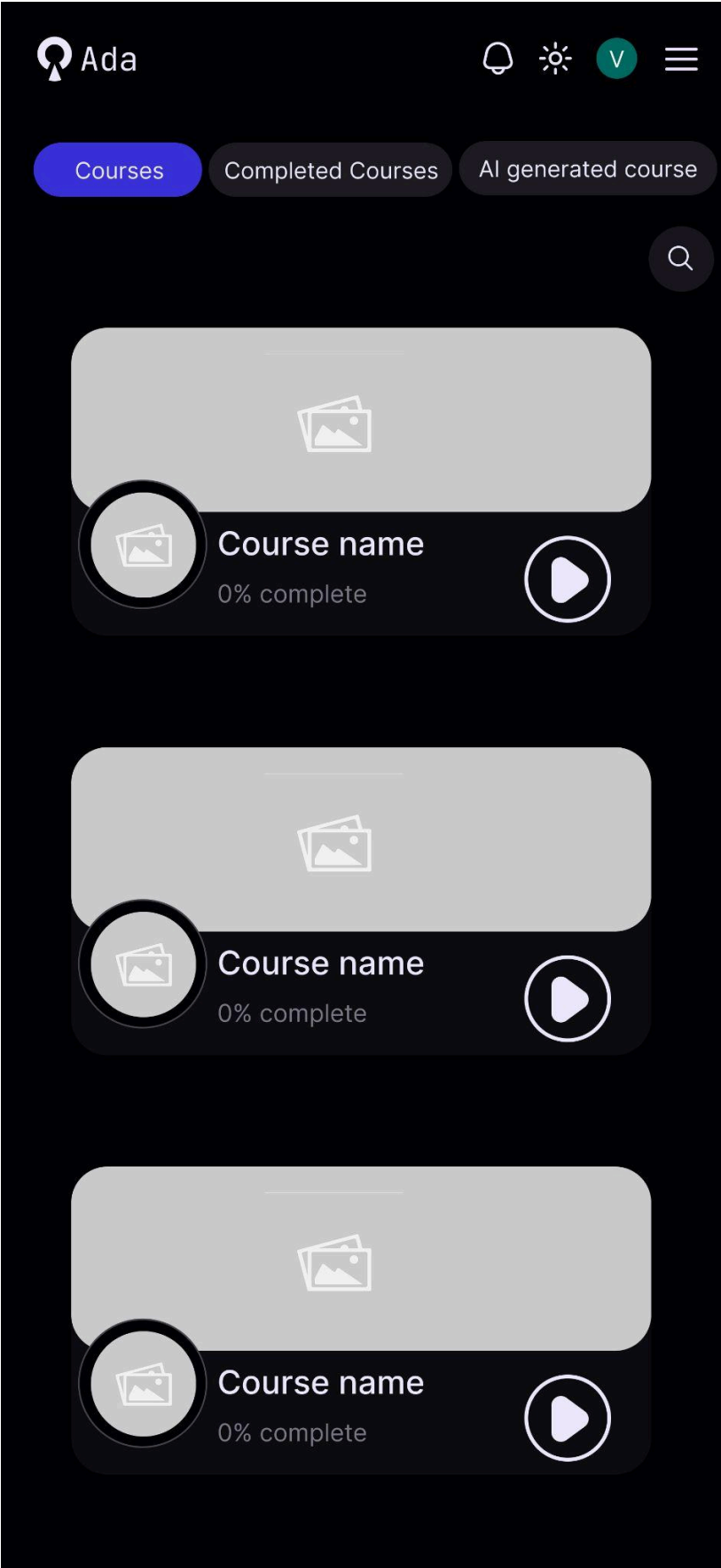
### SQL Query:

```
SELECT c.course_id, c.course_name, c.language_image, c.course_image,
       ROUND(COUNT(ul.lesson_id) * 100.0 /
             (SELECT COUNT(*) FROM Lesson WHERE course_id = c.course_id), 0) AS
progress_percent FROM Enrollment AS e
JOIN Course AS c ON e.course_id = c.course_id
LEFT JOIN Lesson l ON c.course_id = l.course_id
LEFT JOIN user_lesson ul ON ul.lesson_id = l.lesson_id AND ul.user_id = e.user_id
WHERE e.user_id = ?
GROUP BY c.course_id HAVING progress_percent < 100;
```

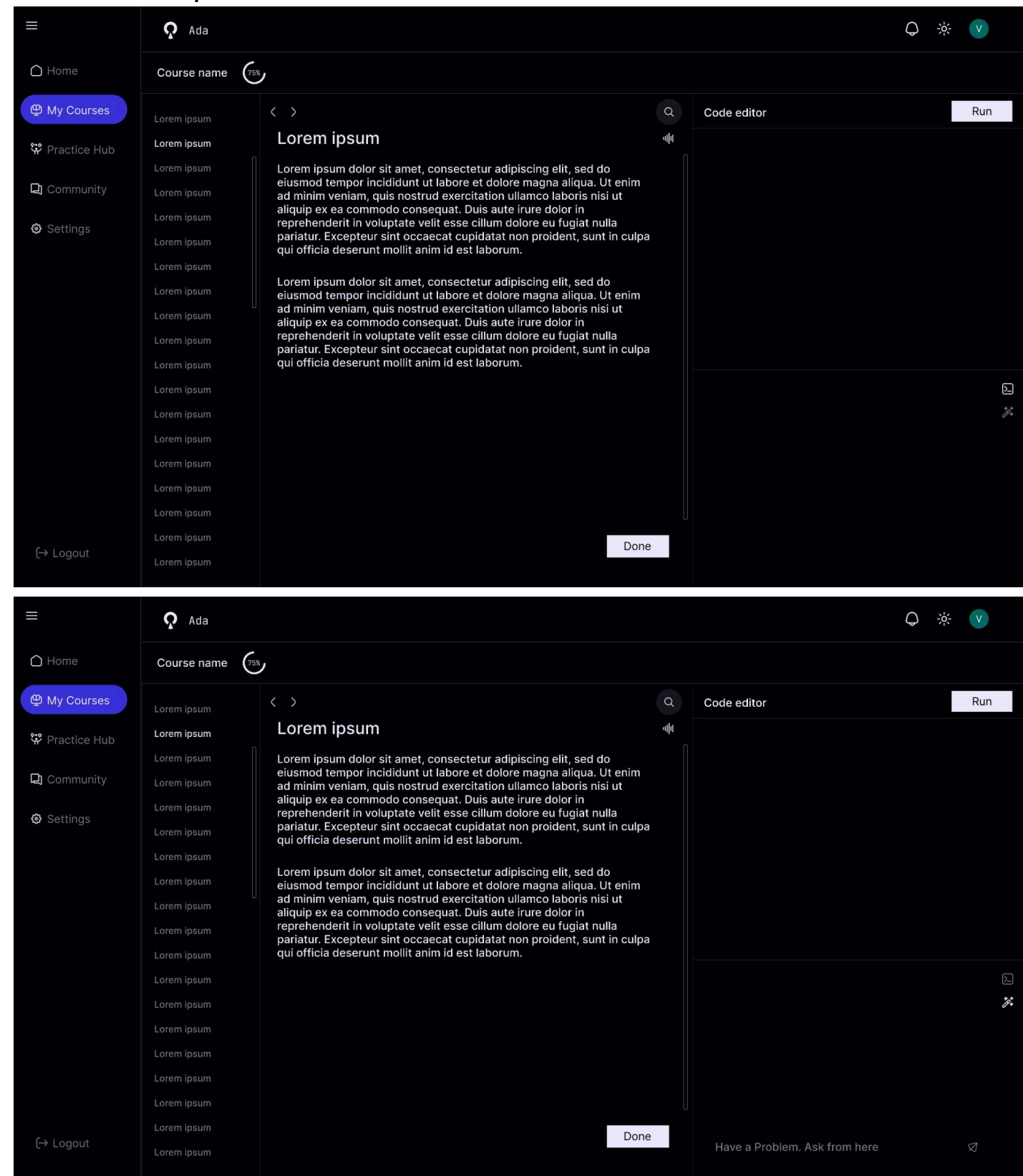


## Routes & Function Signatures:

```
@app.route('/my_courses')  
def courses():  
    pass
```



Dark theme Desktop UI



**Purpose:** This page shows the lessons which are related to the course that is selected by the user.

### User Interactions:

- On the bottom right side, there is a box where the user can switch from AI assistance to a code terminal.
- When the user clicks on the done button, it is marked as “completed” in the *lesson table*.
- “Run” button to run the code that is typed by the user.
- When the user clicks on the “sound wave” button, the user will be able to hear the text that is displayed on the page.
- The search button is used to type keywords to find things in the text.
- In the left corner menu, which is between the text and the main menu, the user will be able to see all the lessons that are related to the selected course.
- From the back and front arrows, the user can go to the next and previous lessons.

### SQL Queries:

1. Get all the lessons related to the selected course

```
SELECT l.lesson_id, l.lesson_title, l.content FROM Lesson AS l WHERE l.course_id = ?  
ORDER BY l.lesson_order;
```

2. Display the selected lesson title and the content.

```
SELECT l.lesson_title, l.lesson_content FROM lesson as l WHERE l.lesson_id = ?;
```

Frontend receives all lesson details in JSON format → User clicks on a specific lesson → The selected lesson ID is sent to the backend → Backend retrieves the lesson and stores it in a variable for further use.

3. Mark as completed when the user clicks on the “Done” Button

```
UPDATE user_lesson SET status = 'completed', completed_at = CURRENT_TIMESTAMP WHERE  
user_id = ? AND lesson_id = ?;
```

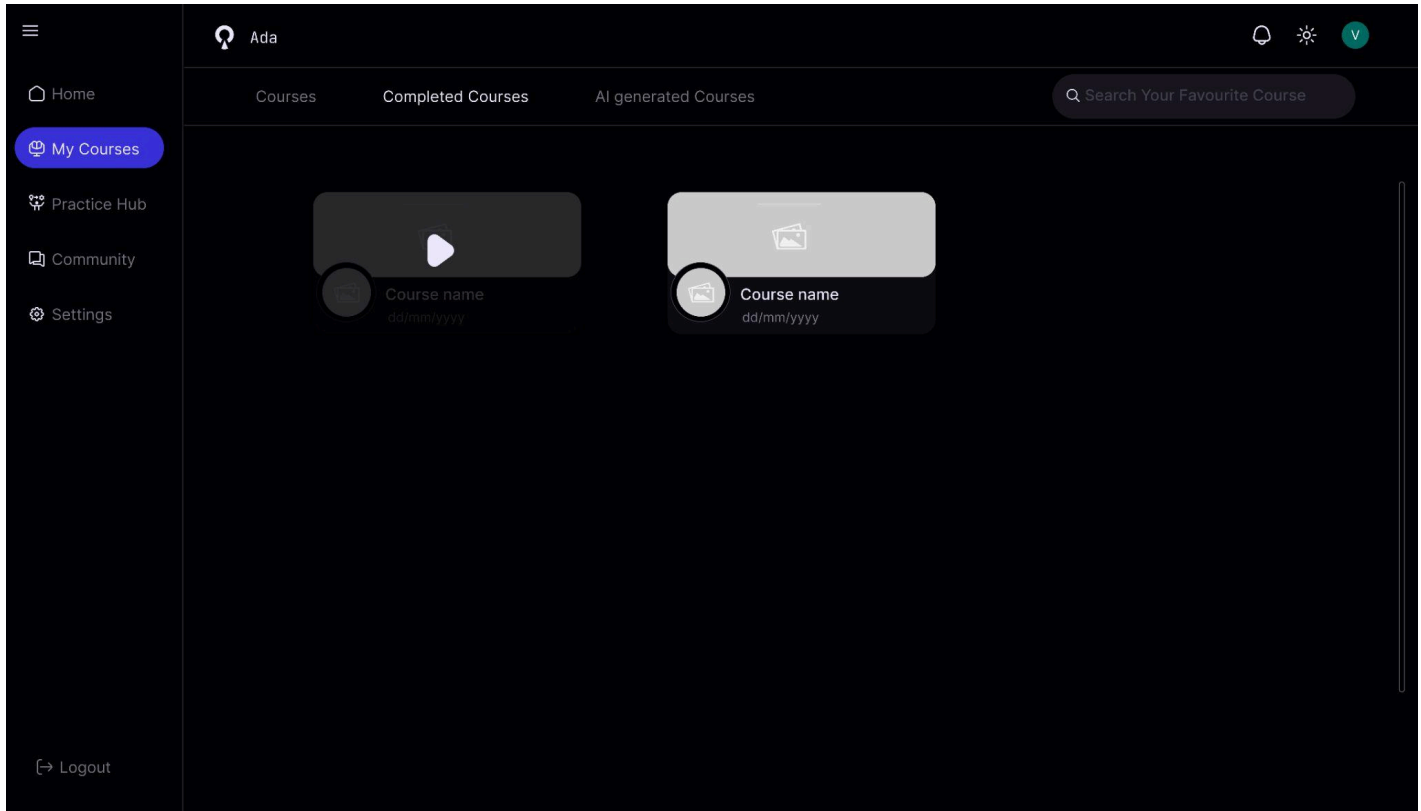
**HINT: Don't use a separate table to save AI chat; save it in local storage in the browser**

## Routes & Function Signatures:

```
@app.route('/my_courses/<course_id>/lesson/<lesson_id>')  
def view_lesson(course_id, lesson_id):  
    pass
```

## Completed Course Page

### Dark theme Desktop UI



**Purpose:** To show courses that have been completed by the user; if they like, they can do those courses again. It shows the course title and the completion date.

### User Interactions:

- In the desktop version, when the user hovers over a course, it will change to the first course card.

### SQL Query:

```
SELECT c.course_id , c.course_name, e.completed_at FROM Enrollment AS e
JOIN Course c ON e.course_id = c.course_id WHERE e.user_id = ? AND e.completed_at IS
NOT NULL ORDER BY e.completed_at DESC;
```

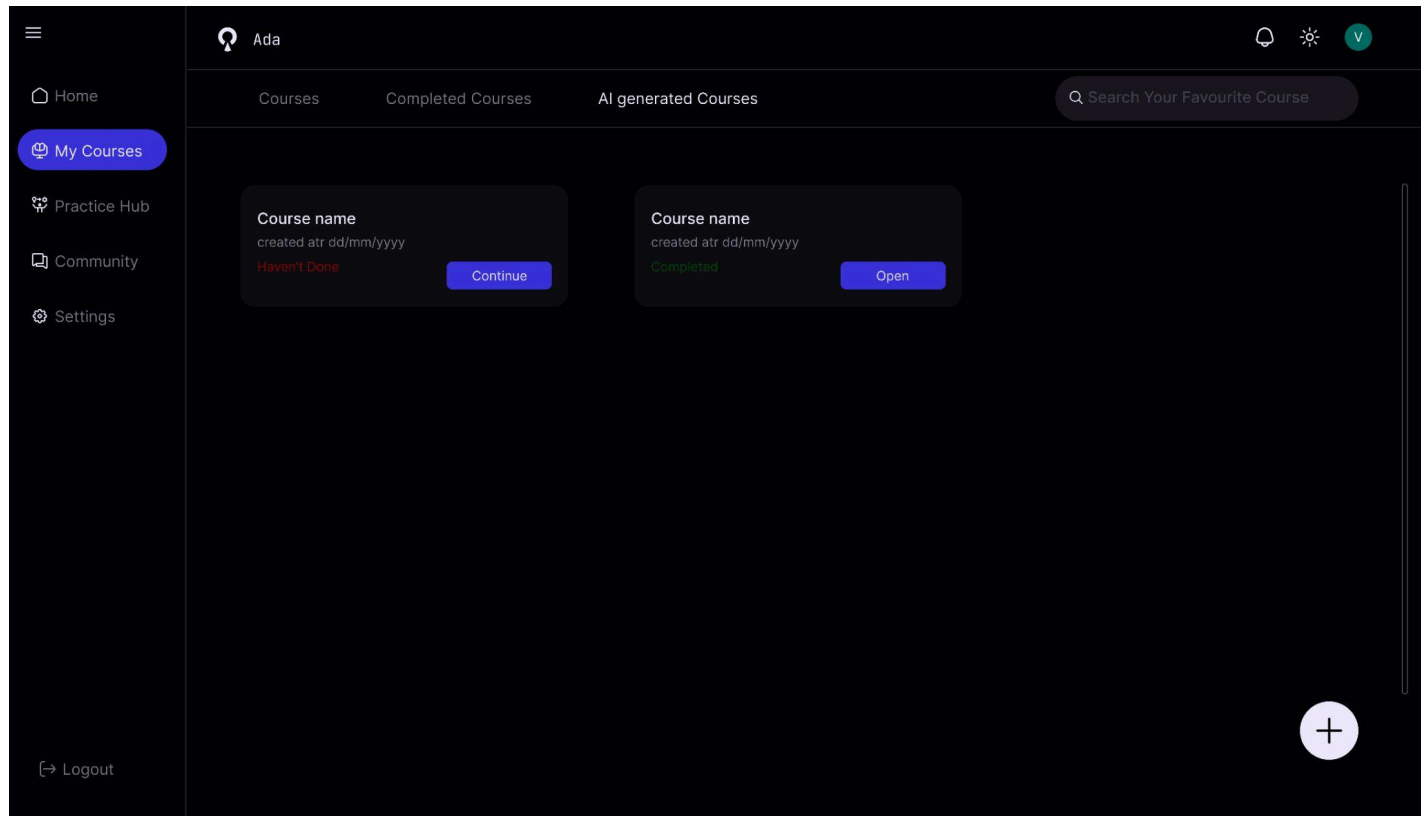
**NOTE:** The `completed_at` field is automatically updated in the backend once all the lessons are completed by the users in a course

### Routes & Function Signatures:

```
@app.route('/my_courses/completed')
def completed_courses():
    pass
```

## AI Genrated Course Page

### Dark theme Desktop UI



**Purpose:** To show AI-generated courses that are created by users

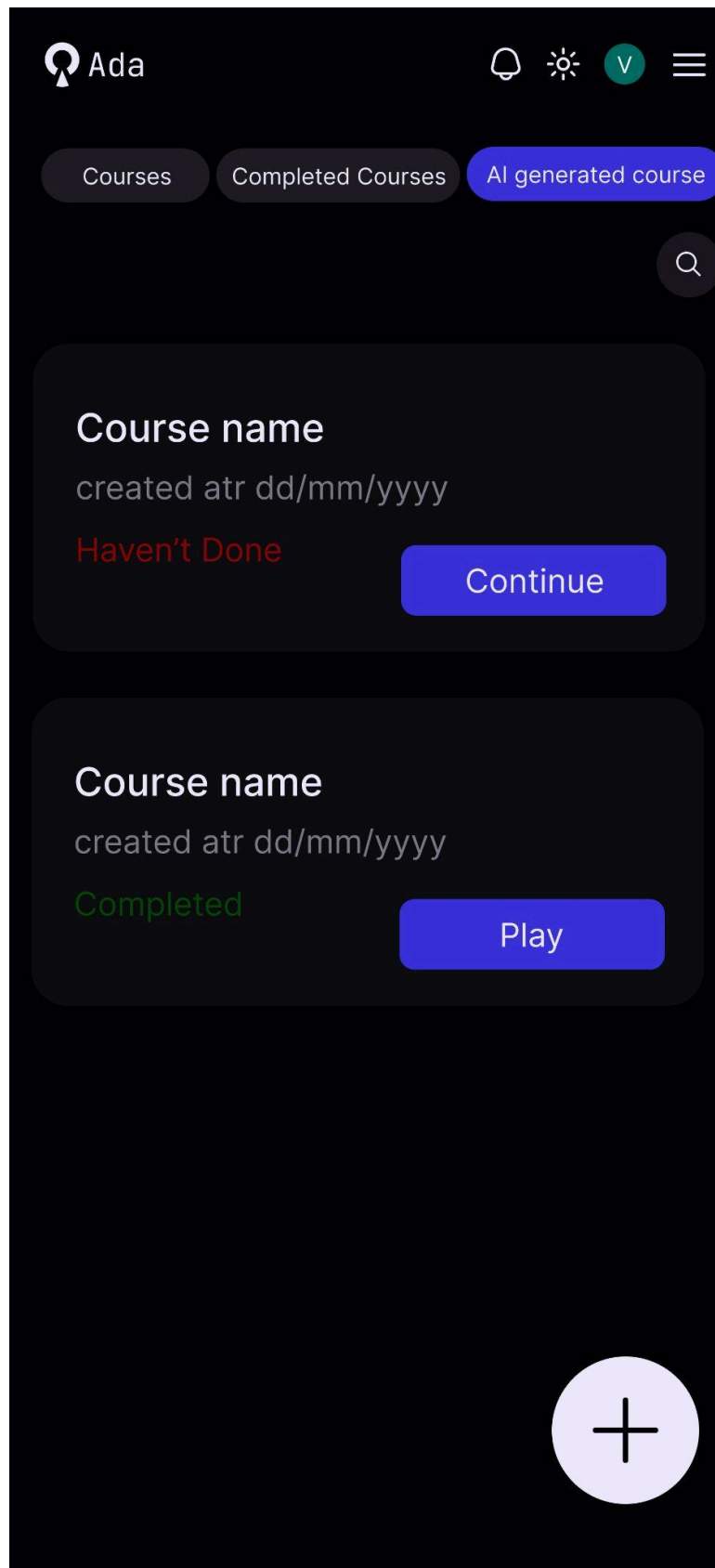
**User Interactions:** All the AI courses' details are passed in JSON format to the frontend, and when the user clicks on a specific course, Relevant JSON data is sent to the backend, saved for future use.

#### SQL Query:

```
SELECT ai.title, ai.user_id , status, generated_at FROM Ai_resource AS ai WHERE  
ai.user_id = ?;
```

#### Routes & Function Signatures:

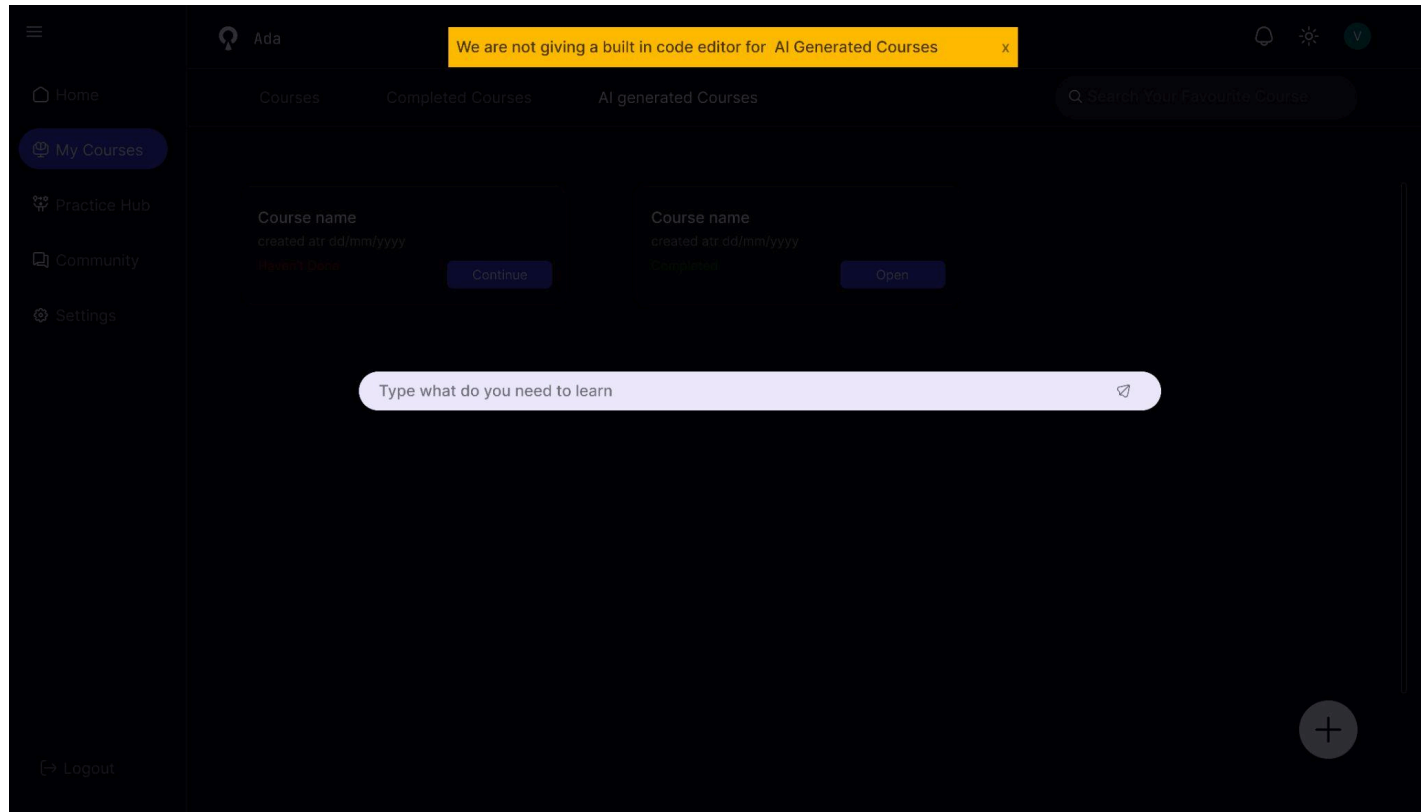
```
@app.route('/my_courses/ai_generated')  
def ai_courses():  
    pass
```





## AI Course Request Panel– Pop-up window

### Dark theme Desktop UI



**Purpose:** Users can request new courses from the AI

#### User Interactions:

- When the user clicks on the plus button text input box appears, and the rest of the part will become darker to highlight the text input box, and the user can go back to the normal screen by clicking the background.
- The user input passes to the third-party AI, and it generates the title and the course and saves it into the database.

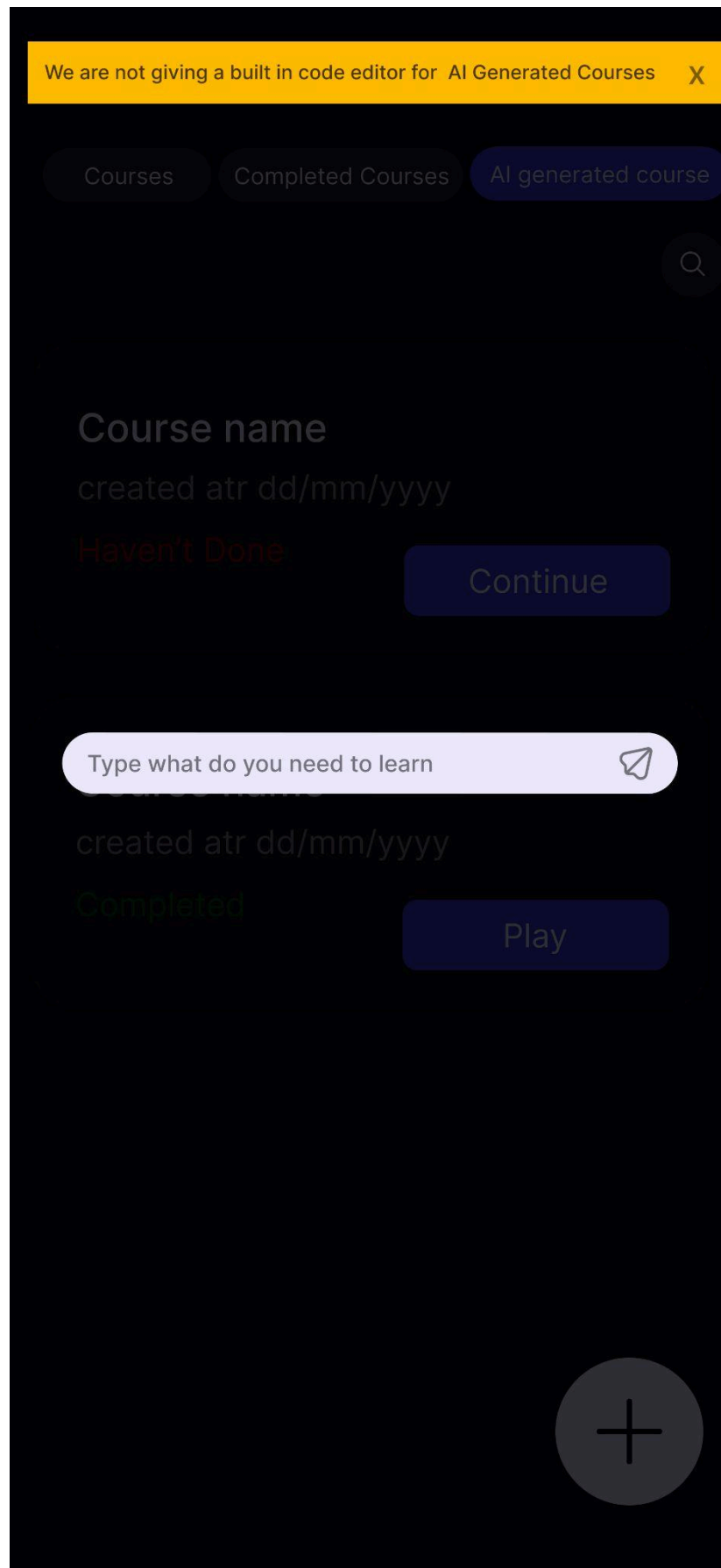
#### SQL Query:

1. Insert a new AI-generated course

```
INSERT INTO Ai_resource ( resource_id, user_id, title, content generated_at, status )  
VALUES (?, ?, ?, ?, CURRENT_TIMESTAMP, "not_completed");
```

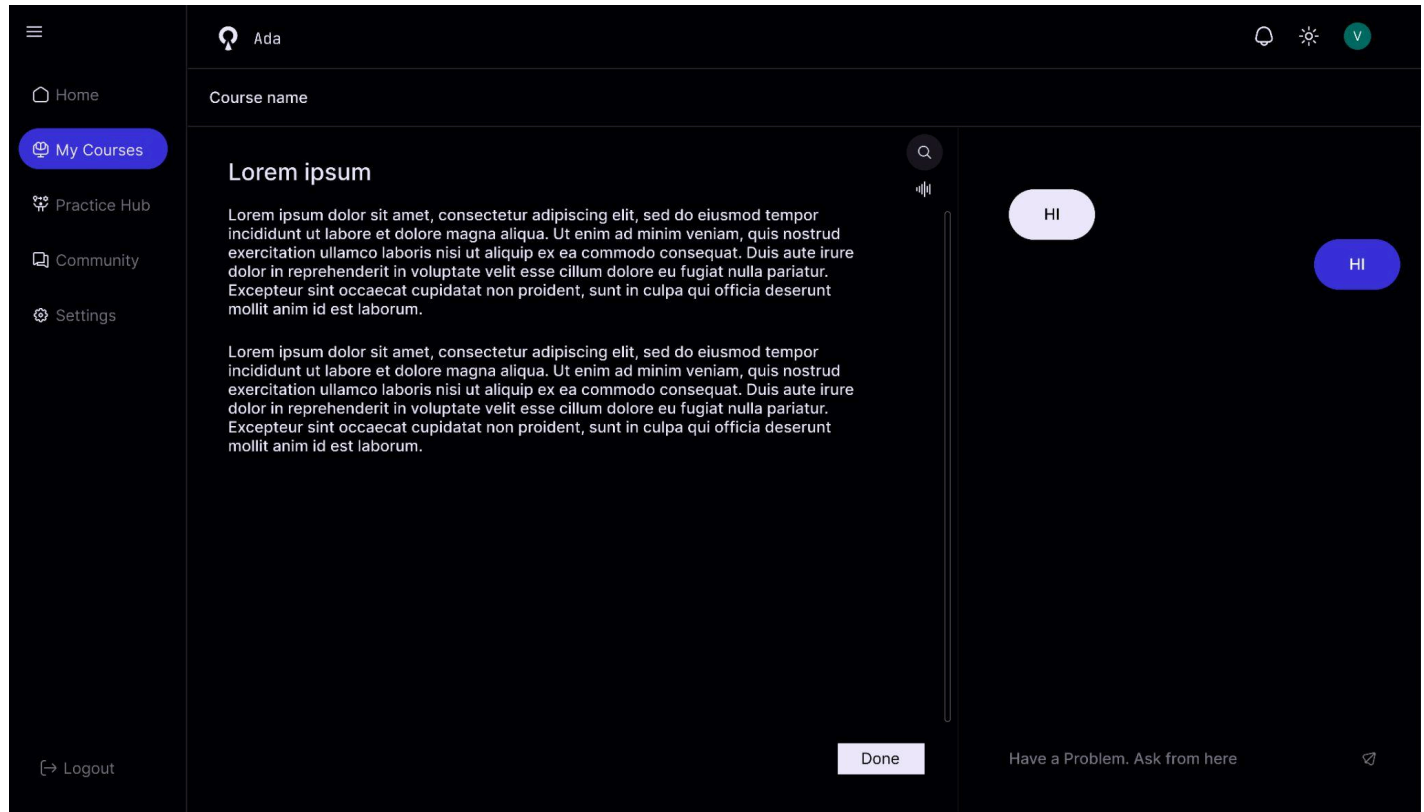
#### Routes & Function Signatures:

```
@app.route('/my_courses/ai_generated')  
def ai_courses():  
    pass
```



## AI Courses Content Page

### Dark theme Desktop UI



**Purpose:** This page shows AI-generated course content, but no built-in code editor for this, On the left side, the user can chat with the AI assistant

### User Interactions:

- When the user clicks on the done button, it is marked as “completed” in the *lesson table*.
- When the user clicks on the “sound wave” button, the user will be able to hear the text that is displayed on the page.
- The search button is used to type keywords to find things in the text.

**NOTE: NO NEED TO SAVE USER AND AI CHAT IN A TABLE, SAVE IT IN LOCAL STORAGE**

### SQL Queries:

1. Get all the challenge details

```
SELECT title, content FROM Ai_resource WHERE user_id = ? AND resource_id = ?;
```

2. When the user clicks on the “Done” button

```
UPDATE Ai_resource SET status = 'completed' WHERE user_id = ? AND resource_id = ?;
```

## Routes & Function Signatures:

```
@app.route('/my_courses/ai_generated/<resource_id>')
def ai_course_detail(resource_id):
    pass
```

## Uncompleted Challenges Page

### Dark theme Desktop UI

Ada

V

Uncompleted Questions

Completed Questions

Q

Search

Home

My Courses

Practice Hub

Community

Settings

Logout

| Status  | No. | Title                                                    | Difficulty | Solution |
|---------|-----|----------------------------------------------------------|------------|----------|
| Started | 1   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          |
| -       | 2   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          |
| -       | 3   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          |
| -       | 4   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Medium     |          |
| -       | 5   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Medium     |          |
| -       | 6   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Medium     |          |
| -       | 7   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Hard       |          |
| -       | 8   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Hard       |          |
| -       | 9   | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Hard       |          |

**Purpose:** This display displays all the challenges that the user hasn't started, and has started but hasn't completed

### User Interactions:

- Users can click on those questions, a new window will open
- Users can click on the solution icon to see the solution for the relevant challenge, which pops up a screen with the solution

### SQL Query:

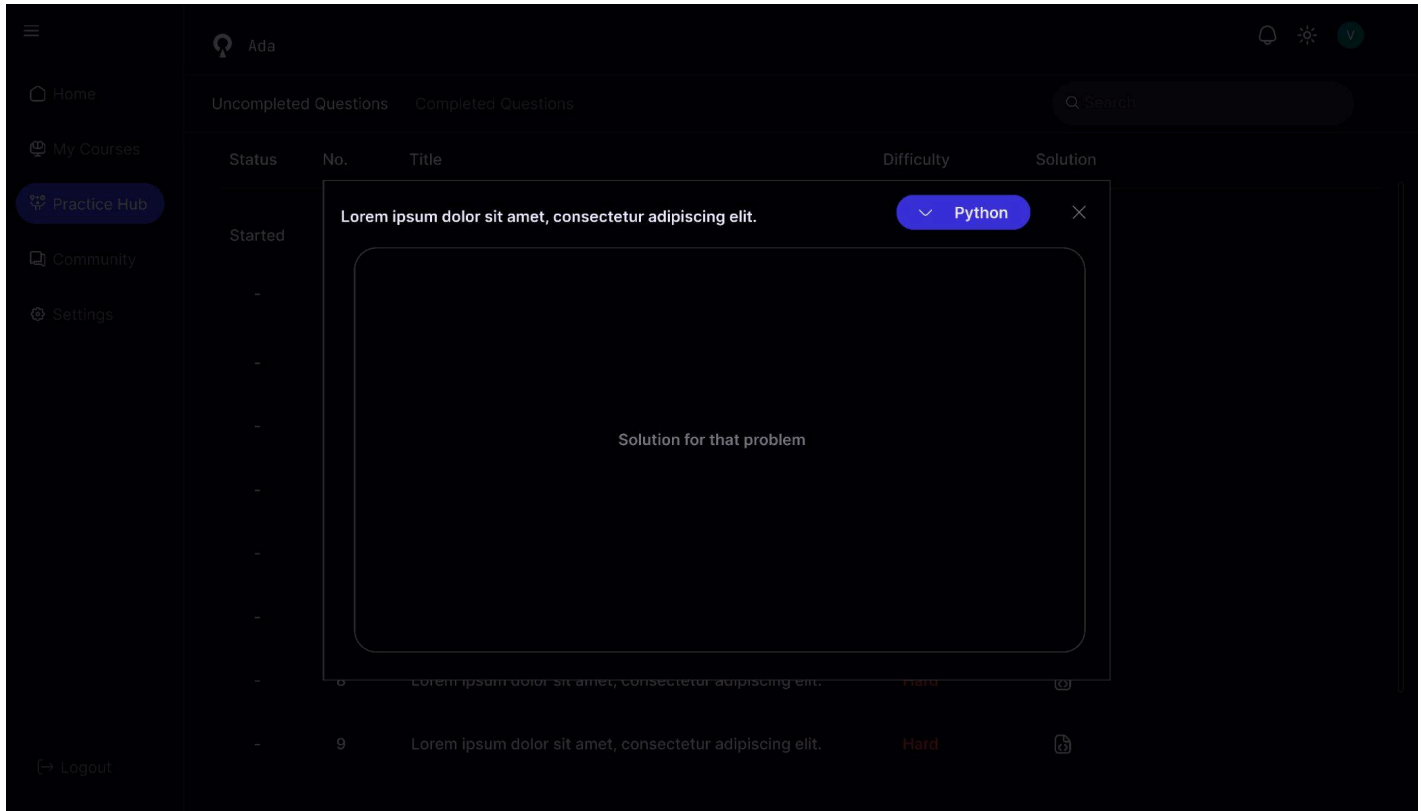
```
SELECT ch.question, ch.status, ch.difficulty_level, ch.challenge_id, ch.number FROM
Challenge AS ch WHERE ch.user_id = ?;
```

### Routes & Function Signatures:

```
@app.route('/practice_hub/uncompleted')
def uncompleted_challenges():
    pass
```

## Pop-up window in Uncompleted Question Page

### Dark theme Desktop UI



**Purpose:** This is a pop-up window that appears when the user clicks the solution button on a challenge.

### User Interactions:

- Users can change the programming language to get the solution according to their needs.
- Users can close the window by clicking the “x” button.

### SQL Query:

1. Displays relevant solutions for the selected challenge and the selected language

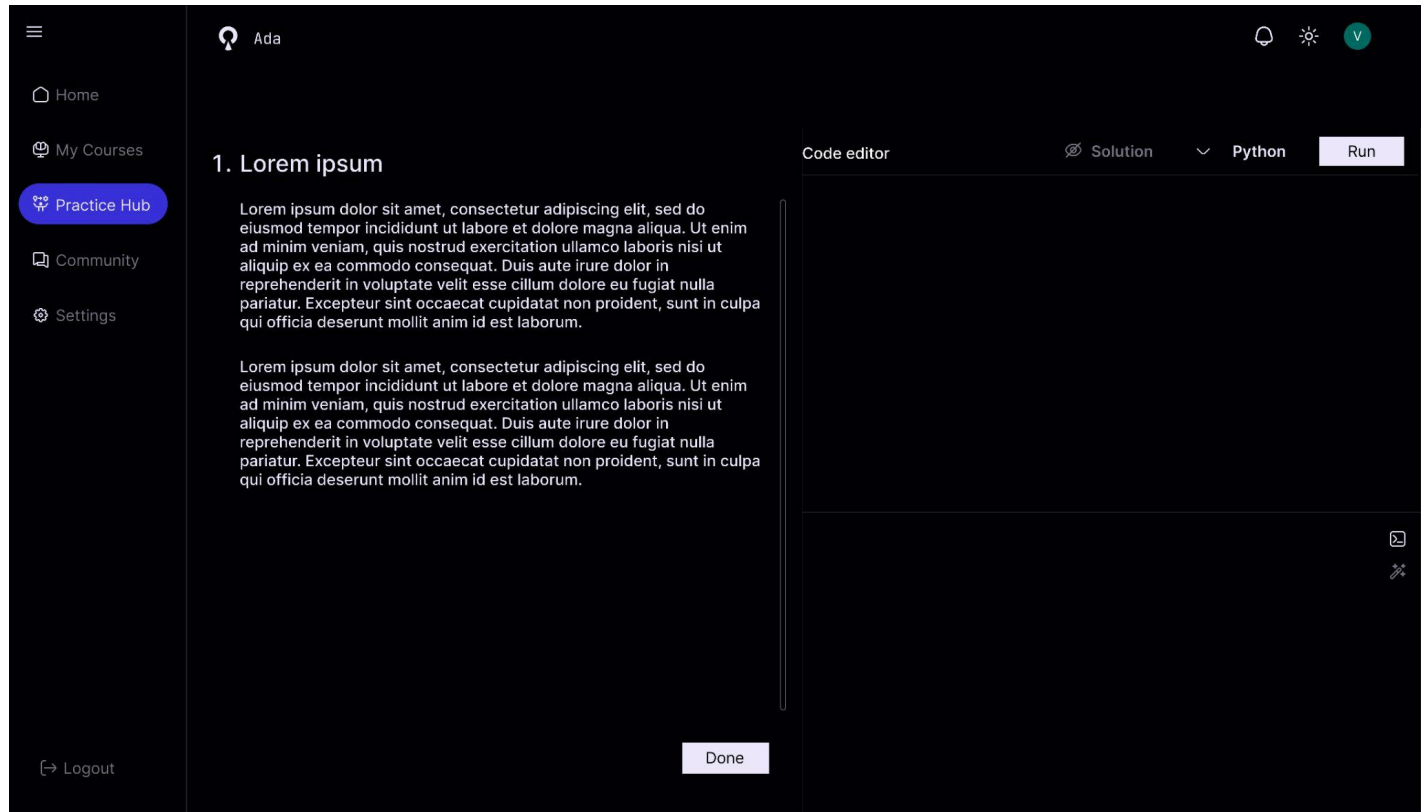
```
SELECT s.answer FROM Solution as s WHERE s.challenge_id = ? AND s.language = ?;
```

### Routes & Function Signatures:

```
@app.route('/practice_hub/uncompleted')
def uncompleted_challenges():
    pass
```

## Code Editor page for the Challenges

### Dark theme Desktop UI



**Purpose:** This page displays the code editor, the challenge title and its content and the terminal, and the AI assistance similar to *the lesson page*.

### User interactions:

- Users can run their code, and they can change their programming languages.
- When the user clicks on the “Done” button, it marks as compelled the challenge if the answer is correct. (Use an AI model to check if the user's answer is correct or not.)
- When the user clicks on the solution button, they can see the solution in a pop-up window similar to the previous page (**Pop-up window in Uncompleted Question Page**)

### SQL Queries:

1. Display the Challenge title and the content.

```
SELECT ch.question, ch.challenge_title, ch.number FROM Challenge AS ch WHERE  
ch.user_id = ? AND ch.challenge_id = ?;
```

2. Mark as completed the challenge if the answer is correct

```
UPDATE User_challenge SET completed_at = CURRENT_TIMESTAMP WHERE user_id = ? AND  
challenge_id = ?;  
UPDATE Challenge SET status = 'completed' WHERE challenge_id = ? AND user_id = ?;
```

### 3. Displays the Solution

```
SELECT answer FROM Solution WHERE challenge_id = ? AND language = ?;
```

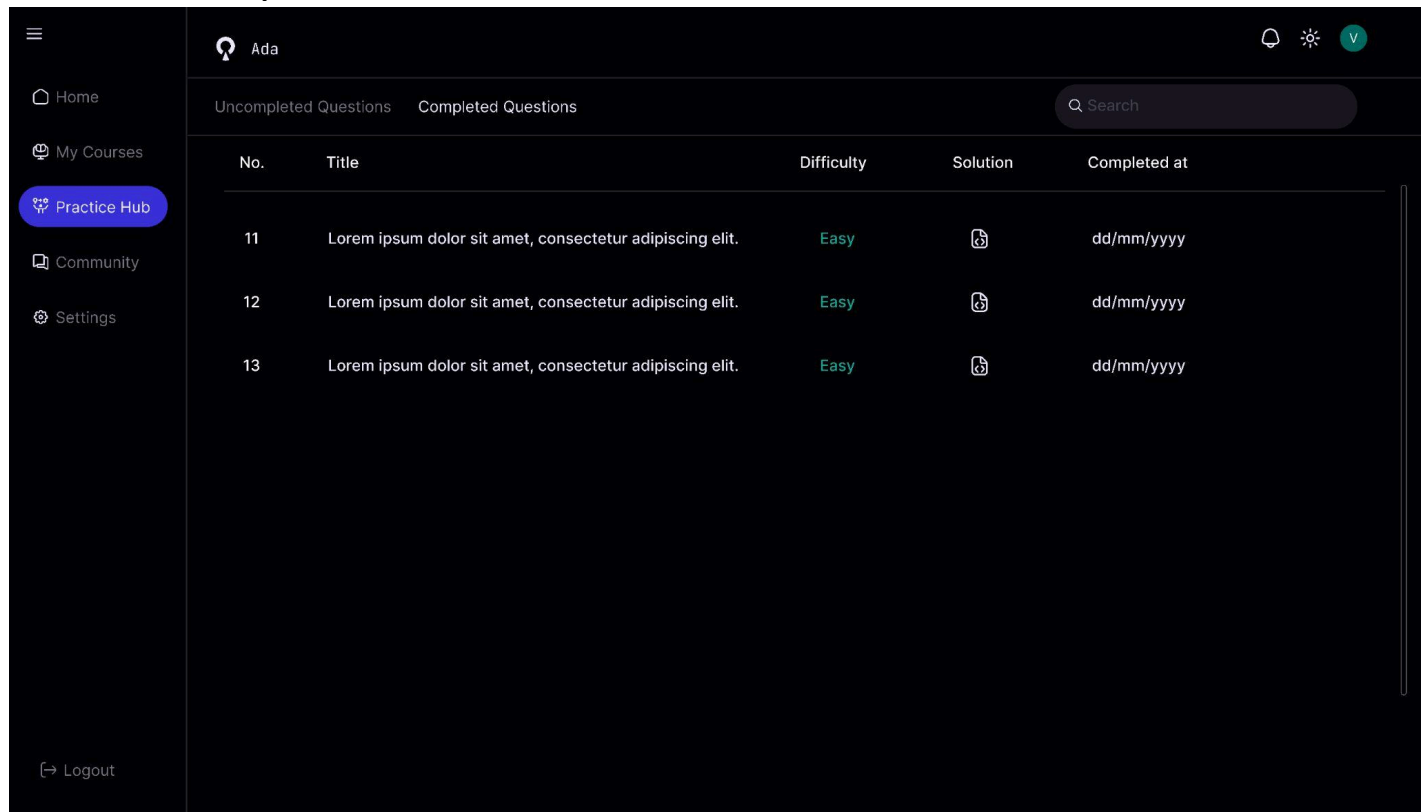
### Routes & Function Signatures:

```
@app.route('/practice_hub/<challenge_id>')  
def challenge_editor(challenge_id):  
    pass
```



## Completed Challenges Page

### Dark theme Desktop UI



The image shows a dark-themed desktop user interface for a 'Completed Challenges' page. On the left is a sidebar with navigation links: Home, My Courses, Practice Hub (highlighted in blue), Community, and Settings. At the bottom of the sidebar is a 'Logout' button. The main content area has a header with a user profile 'Ada', notification and settings icons, and a search bar. Below the header are two tabs: 'Uncompleted Questions' and 'Completed Questions'. The 'Completed Questions' tab is active, displaying a table with three rows of completed challenges. Each row includes a challenge number, title, difficulty level (Easy), a solution button, and the completion date.

| No. | Title                                                    | Difficulty | Solution | Completed at |
|-----|----------------------------------------------------------|------------|----------|--------------|
| 11  | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          | dd/mm/yyyy   |
| 12  | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          | dd/mm/yyyy   |
| 13  | Lorem ipsum dolor sit amet, consectetur adipiscing elit. | Easy       |          | dd/mm/yyyy   |

**Purpose:** This page displays challenges that are completed by the user. If they like, they can do those things again

#### -User Interactions:

- When the user clicks on the solution button, they can see the solution in a pop-up window similar to the previous page (**Pop-up window in Uncompleted Question Page**)
- They can click on the challenge and do it again.

#### SQL Query:

1. Display the completed challenges

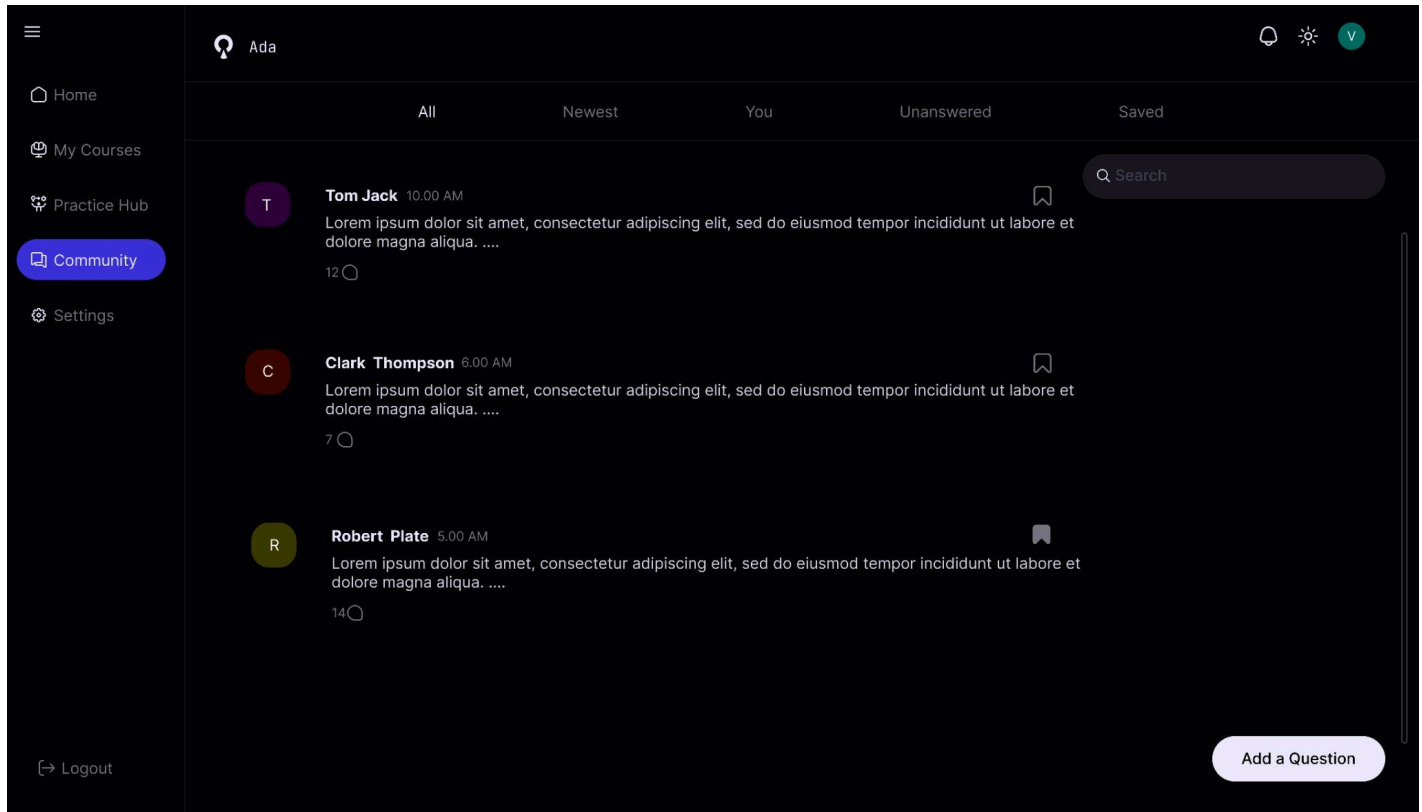
```
SELECT ch.challenge_id, ch.challenge_title, ch.question, ch.difficulty_level,
ca.completed_at, ch.number FROM Challenge_attempt AS ca JOIN Challenge AS ch ON
ca.challenge_id = ch.challenge_id WHERE ca.user_id = ? AND ca.status = 'completed';
```

## Routes & Function Signatures:

```
@app.route('/practice_hub/completed')  
def completed_challenges():  
    pass
```

## Community Page - All Chat

### Dark theme Desktop UI



**Purpose:** This page displays all community questions and discussions that have happened throughout time.

**User interactions:** Users can save community discussions if they like.

#### SQL Query:

1. Represent community discussion ordered by time, the latest discussions at the top oldest discussions at the bottom

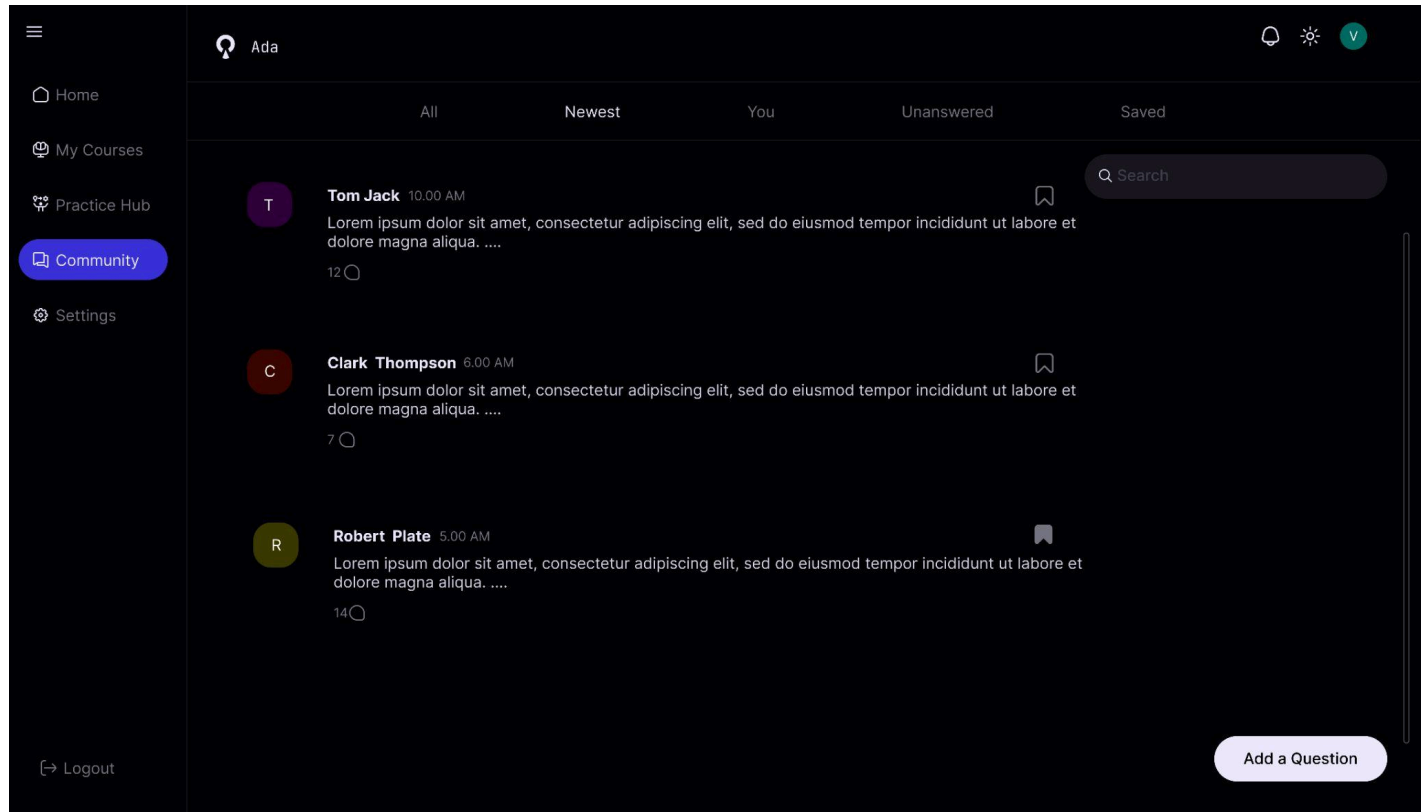
```
SELECT ch.chat_id, u.full_name, u.profile_image, ch.question, ch.created_at,
ch.is_saved, (SELECT COUNT(*) FROM Reply AS r WHERE r.chat_id = ch.chat_id) AS
nu_reply FROM Chat AS ch JOIN User AS u ON ch.user_id = u.user_id ORDER BY
ch.created_at DESC;
```

#### Routes & Function Signatures:

```
@app.route('/community')
def community_all():
    pass
```

## Community Page - Newest

### Dark theme Desktop UI



**Purpose:** This page displays all community questions and discussions that have happened in a day.

### SQL Query:

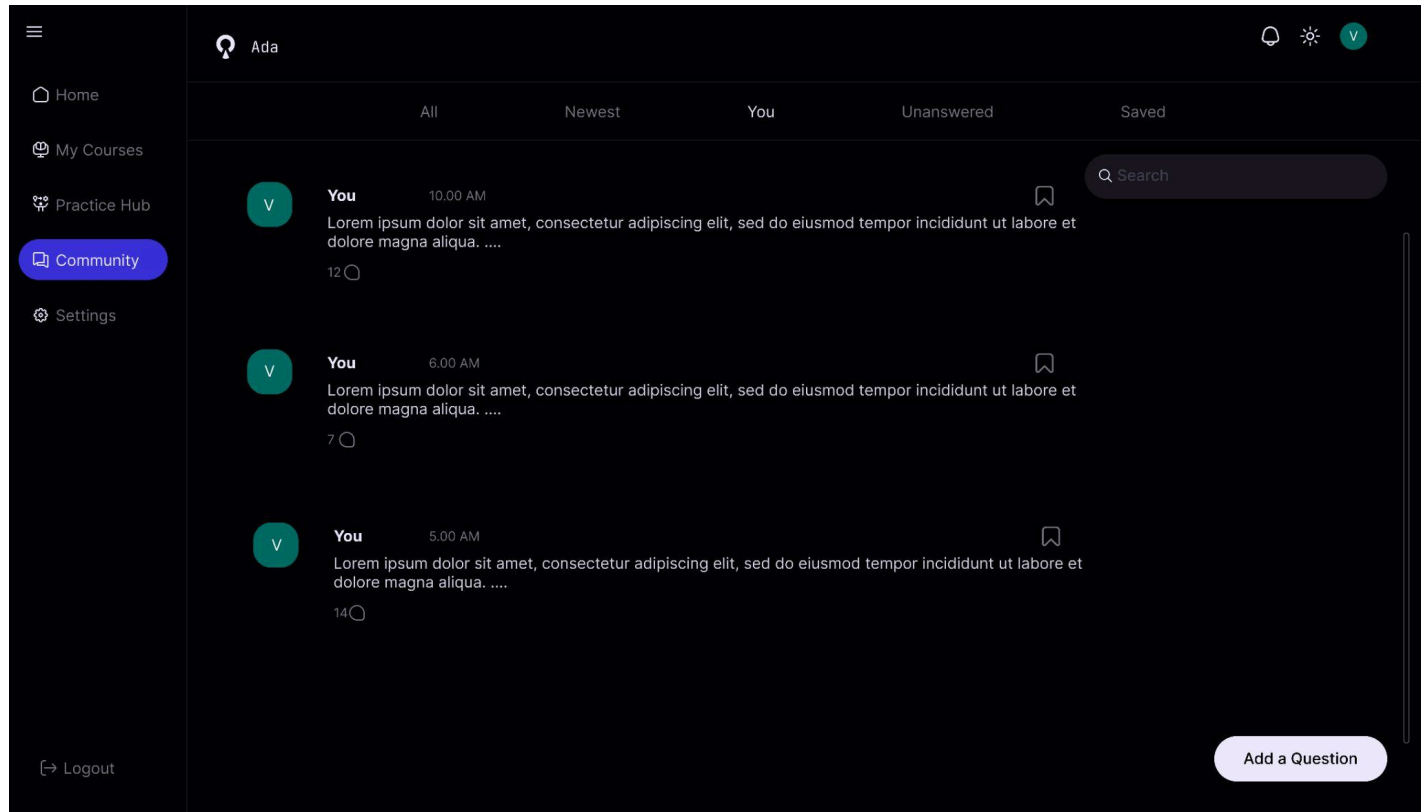
```
SELECT ch.chat_id, u.full_name, u.profile_image, ch.question, ch.created_at,
ch.is_saved,
(SELECT COUNT(*) FROM Reply AS r WHERE r.chat_id = ch.chat_id) AS nu_reply FROM Chat
AS ch JOIN User AS u ON ch.user_id = u.user_id
WHERE ch.created_at ≥ DATETIME('now', '-1 day') ORDER BY ch.created_at DESC;
```

### Routes & Function Signatures:

```
@app.route('/community/newest')
def community_newest():
    pass
```

## User's Question Page

### Dark theme Desktop UI



**Purpose:** This page displays all community questions and discussions that have been done by the user.

### SQL Query:

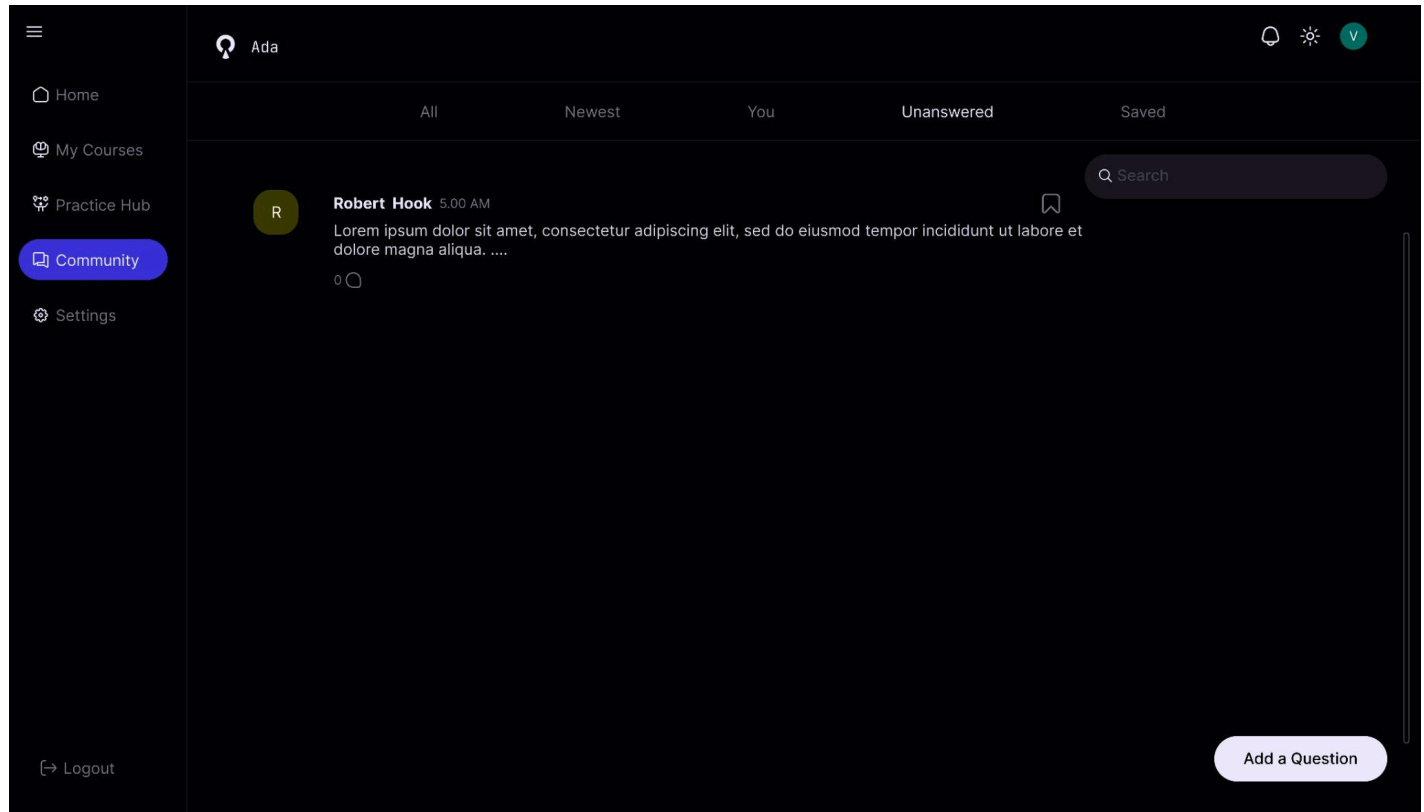
```
SELECT ch.chat_id, u.full_name, u.profile_image, ch.question, ch.created_at,
ch.is_saved,
(SELECT COUNT(*) FROM Reply AS r WHERE r.chat_id = ch.chat_id) AS nu_reply FROM Chat
AS ch JOIN User AS u ON ch.user_id = u.user_id WHERE ch.user_id = ? ORDER BY
ch.created_at DESC;
```

### Routes & Function Signatures:

```
@app.route('/community/you')
def your_question():
    pass
```

## Not Answered Chat Page

### Dark theme Desktop UI



**Purpose:** This page displays all the unanswered questions that are asked by users.

### SQL Query:

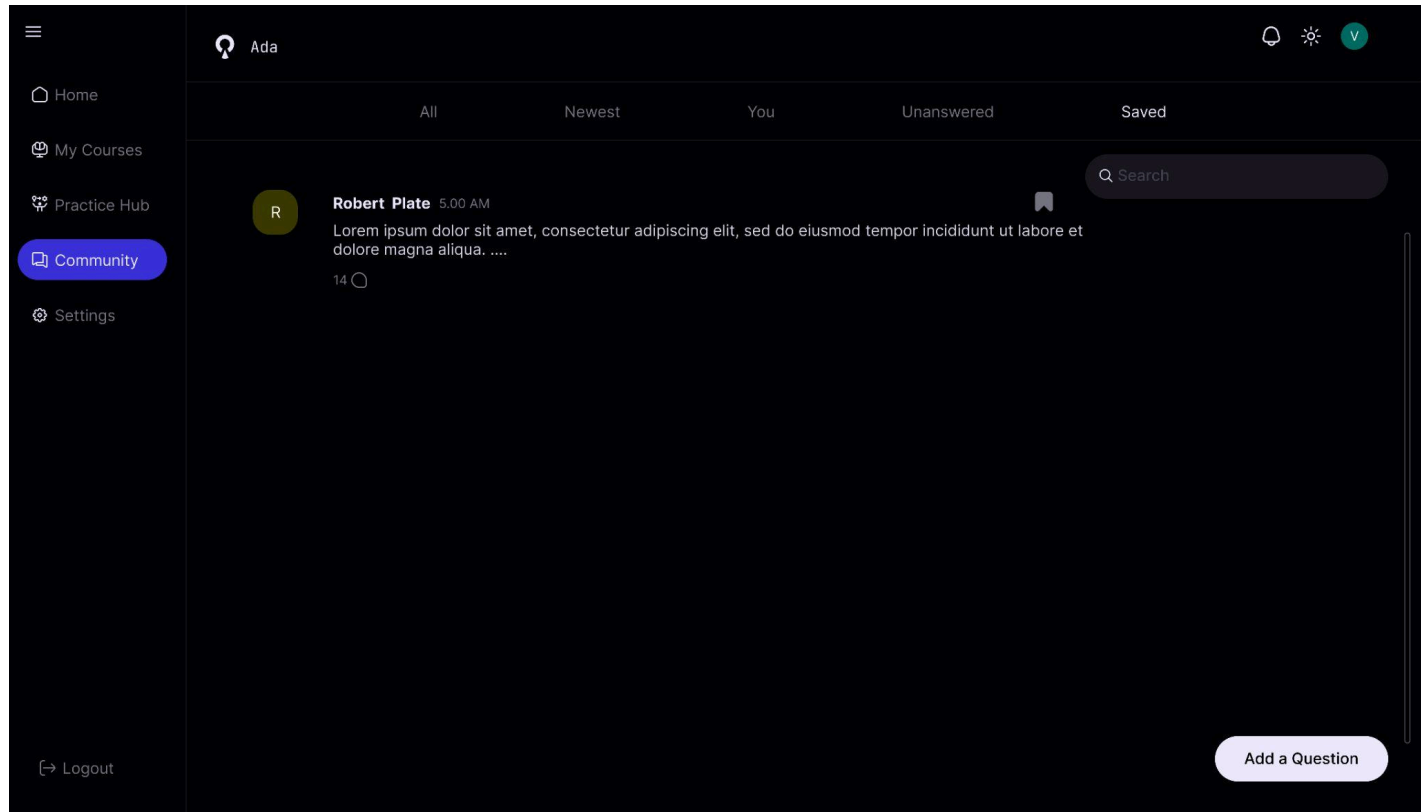
```
SELECT ch.chat_id, u.full_name, u.profile_image, ch.question, ch.created_at,
ch.is_saved, COUNT(r.reply_id) AS no_reply FROM Chat AS ch JOIN User AS u ON
ch.user_id = u.user_id LEFT JOIN Reply AS r ON ch.chat_id = r.chat_id GROUP BY
ch.chat_id, ch.question, ch.created_at, u.full_name HAVING no_reply = 0 ORDER BY
ch.created_at DESC;
```

### Routes & Function Signatures:

```
@app.route('/community/unanswered')
def community_unanswered():
    pass
```

## Saved Chat Page

### Dark theme Desktop UI



**Purpose:** This page displays all the saved questions and discussions.

**User Interactions:** Users can save or unsave a chat by clicking the bookmark icon

### SQL Queries:

1. Displays saved chat

```
SELECT ch.chat_id, u.full_name, u.profile_image, ch.question, ch.created_at,
(SELECT COUNT(*) FROM Reply AS r WHERE r.chat_id = ch.chat_id)
AS nu_reply FROM Chat AS ch JOIN User AS u ON ch.user_id = u.user_id
WHERE ch.saved = 1 ORDER BY ch.created_at DESC;
```

2. Save a Chat

```
UPDATE Chat SET is_saved = 1 WHERE chat_id = ? AND user_id = ?;
```

3. Unsave a Chat

```
UPDATE Chat SET is_saved = 0 WHERE chat_id = ? AND user_id = ?;
```

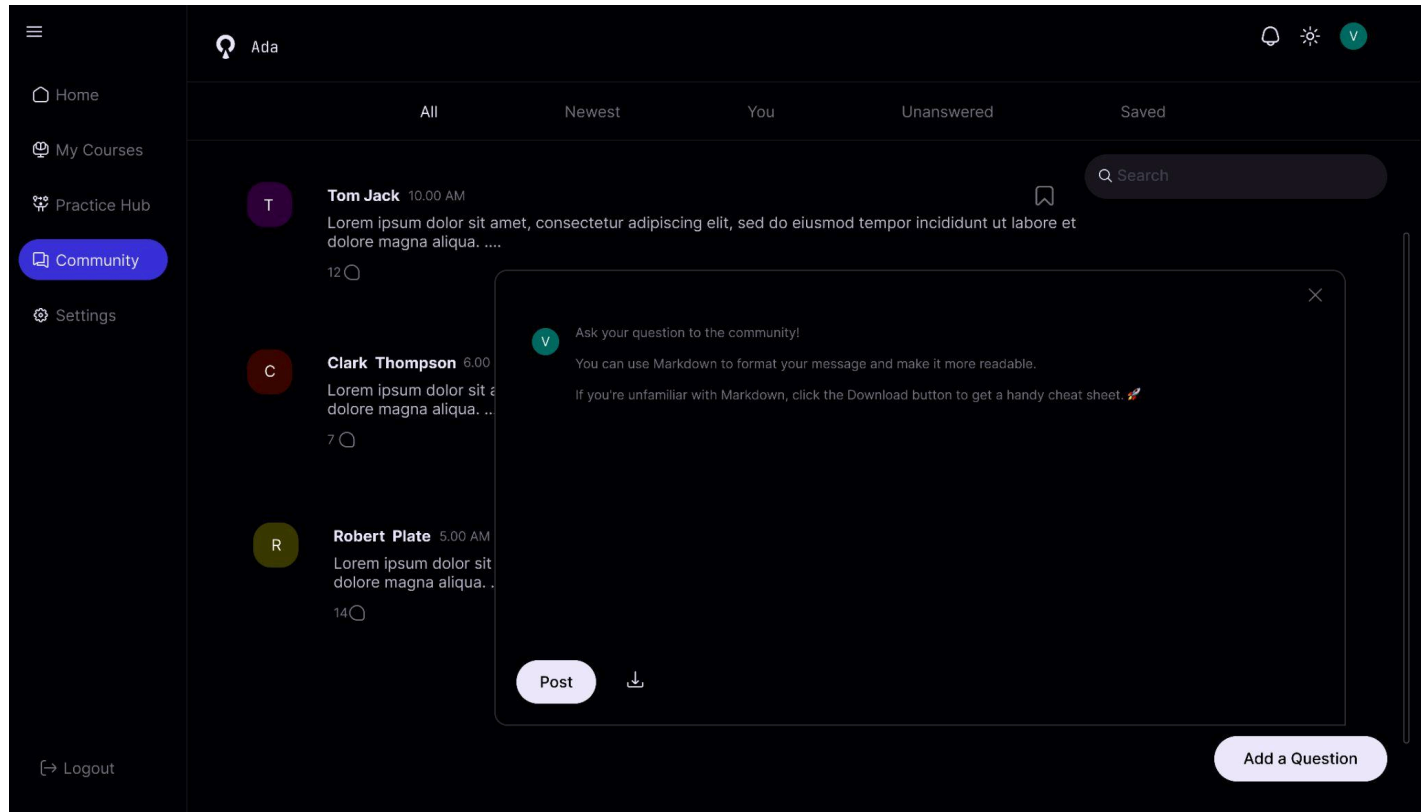
### Routes & Function Signatures:

```
@app.route('/community/saved')  
def community_saved():  
    pass
```



## Pop-up window to Post Questions

### Dark theme Desktop UI



**Purpose:** This pop-up window allows users to post their questions.

#### User Interactions:

- When the user clicks on the “Add a Question” Button, this window will appear.
- Users can use Markdown to write more stylishly, and they can learn to write Markdown.
- Users can download a cheat sheet on markdown through the download button.
- Users can click “x” to exit from this window.
- Users can click the “post” button to post it.
- Clicking the “post” button submits the question to the backend route ``/community/post``.

**NOTE:** This input box should support markdown

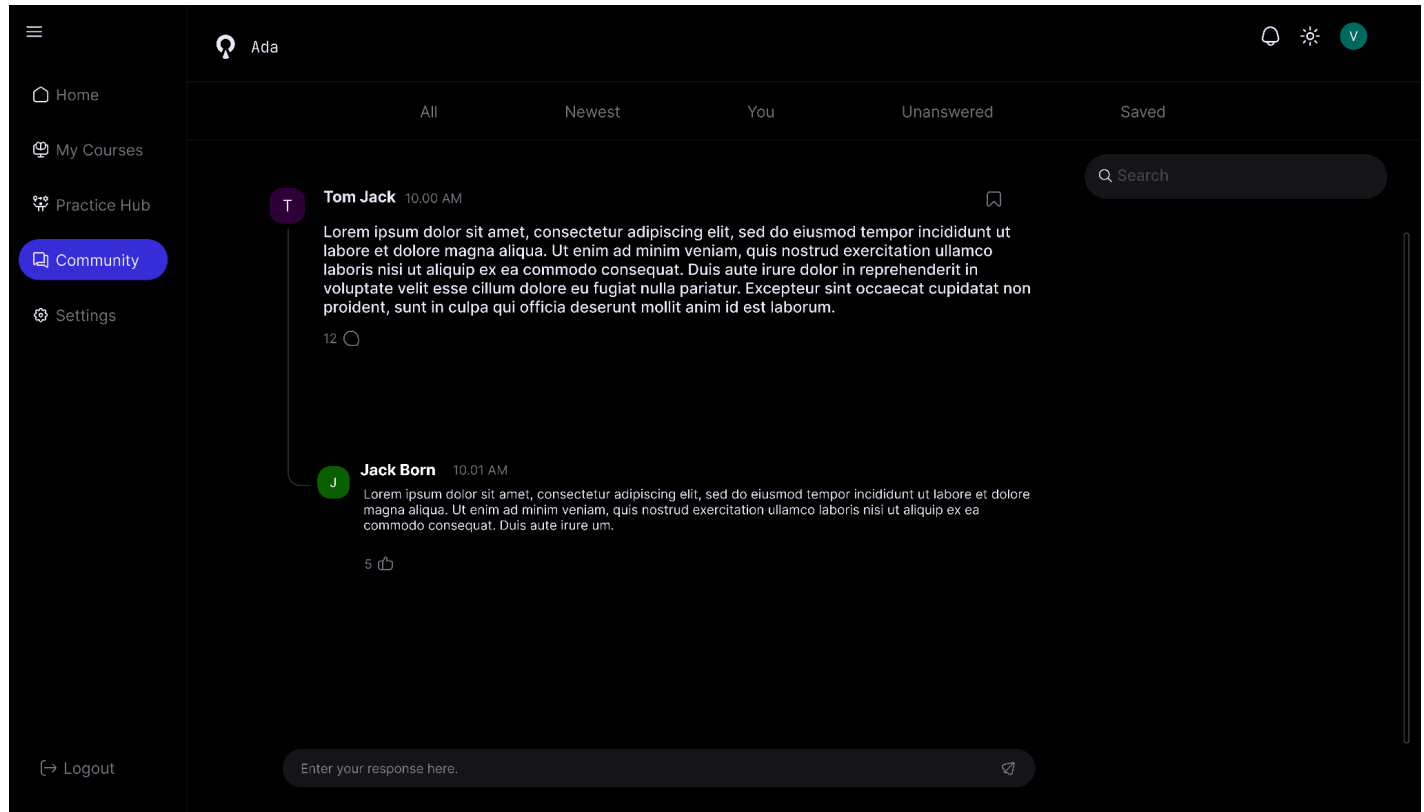
#### SQL Query:

```
INSERT INTO Chat (chat_id, user_id, question, created_at, is_saved) VALUES (?, ?, ?,  
CURRENT_TIMESTAMP, 0);
```

## Routes & Function Signatures:

```
@app.route('/community/post', methods=['POST'])
def post_question():
    pass
```

### Dark theme Desktop UI



**Purpose:** This page displays the selected question and relevant replies for that question.

#### User Interaction:

- The user can put a like on a reply.
- The user can write replies or a reply to that question

#### SQL Queries:

1. Displays the question and replies to it, Here the oldest message appears first and the newest message appears last

```
SELECT ch.chat_id, ch.question AS chat_question, ch.created_at AS chat_time,
u.full_name AS asked_by, u.profile_image AS asked_by_image, r.reply_id, r.reply,
r.created_at AS reply_time, ru.full_name AS replied_by, ru.profile_image AS
replied_by_image, r.likes,
(SELECT COUNT(*) FROM Reply AS r2 WHERE r2.chat_id = ch.chat_id) AS total_reply_count
FROM Chat AS ch JOIN User AS u ON ch.user_id = u.user_id LEFT JOIN Reply AS r ON
ch.chat_id = r.chat_id LEFT JOIN User AS ru ON r.user_id = ru.user_id WHERE
ch.chat_id = ? ORDER BY r.created_at ASC;
```

## 2. Answer to that question

```
INSERT INTO Reply (reply_id, chat_id, user_id, reply, created_at, like) VALUES (?, ?, ?, ?, CURRENT_TIMESTAMP, 0)
```

## 3. Get the number of likes per reply

```
SELECT r.reply_id, COUNT(rl.like_id) AS like_count FROM Reply AS r  
LEFT JOIN Reply_like AS rl ON r.reply_id = rl.reply_id GROUP BY r.reply_id;
```

## 4. Add a like (One Like Per User Per Reply)

```
INSERT INTO Reply_like (like_id, user_id, reply_id, liked_at) VALUES (?, ?, ?,  
CURRENT_TIMESTAMP);
```

## 5. Remove a like

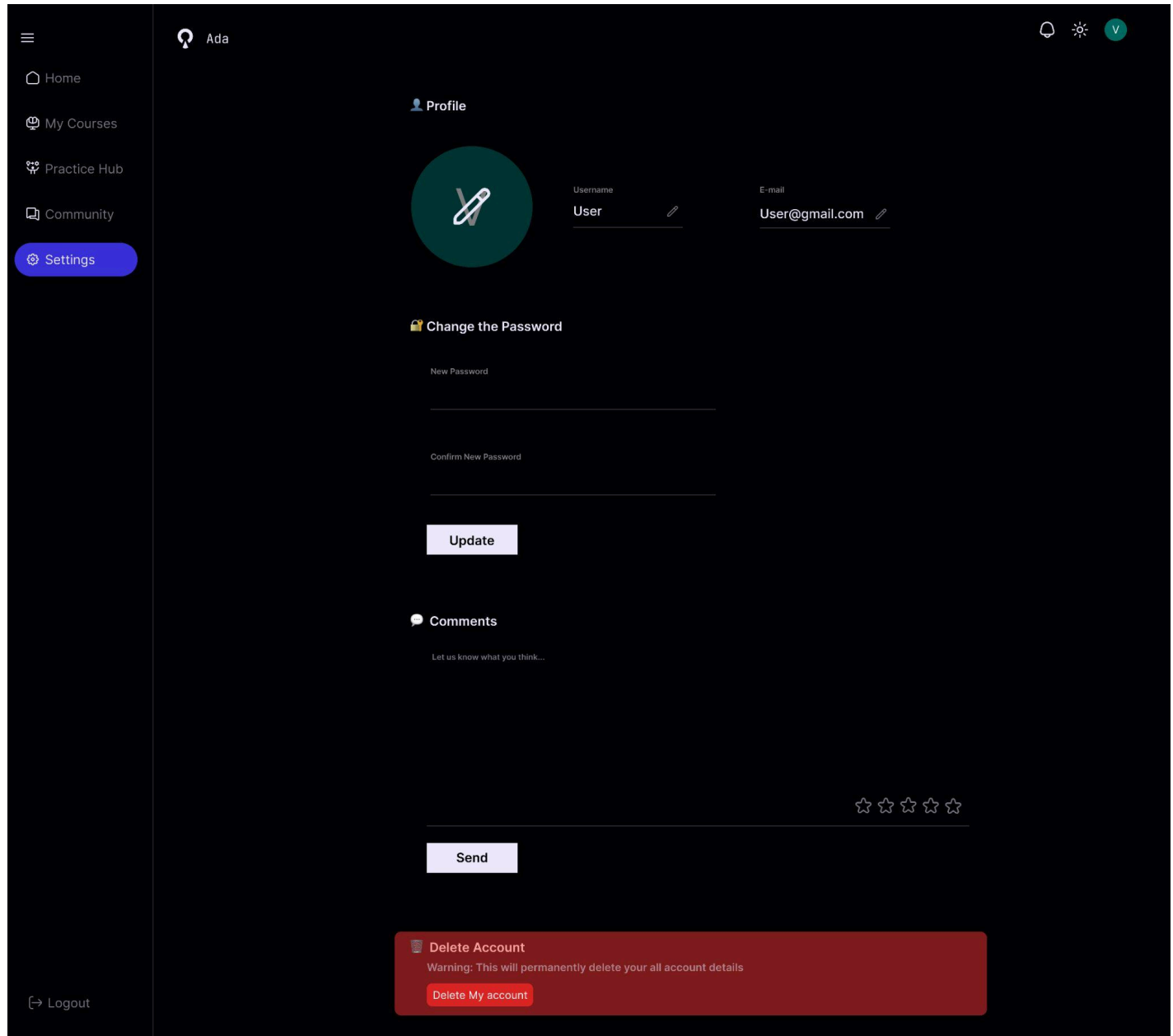
```
DELETE FROM Reply_like  
WHERE user_id = ? AND reply_id = ?;
```

## Routes & Function Signatures:

```
@app.route('/community/chat/<chat_id>')  
def view_chat(chat_id):  
    pass
```

## Setting Page

### Dark theme Desktop UI



The image shows a dark-themed desktop user interface for a settings page. On the left is a sidebar with a hamburger menu icon and links to Home, My Courses, Practice Hub, Community, and Settings (which is highlighted in blue). The main content area has a header with a user profile icon and the name 'Ada', along with notification, theme, and verification icons. The 'Profile' section includes a profile picture placeholder, a username 'User' with an edit icon, and an email 'User@gmail.com' with an edit icon. Below this is the 'Change the Password' section with input fields for 'New Password' and 'Confirm New Password', and an 'Update' button. The 'Comments' section has a text input field with the placeholder 'Let us know what you think...' and a 'Send' button. At the bottom right, there is a 'Delete Account' warning box with a 'Delete My account' button. A 'Logout' link is in the bottom left corner.

**Purpose:** The User can change their profile pictures, username, email, passwords, and they can put a comment on this product. And users can delete their accounts

### SQL Queries:

1. Change the Profile Picture

```
UPDATE User SET profile_image = ? WHERE user_id = ?;
```

## 2. Change the username (Full Name)

```
UPDATE User
SET full_name = ?
WHERE user_id = ?;
```

## 3. Change Email

```
UPDATE User
SET email = ?
WHERE user_id = ?;
```

## 4. Change the Password

```
UPDATE User
SET password = ?
WHERE user_id = ?;
```

**NOTE: Hash the password**

## 5. Add a comment on the product

```
INSERT INTO User_feedback (feedback_id, user_id, feedback, star)
VALUES (?, ?, ?, ?);
```

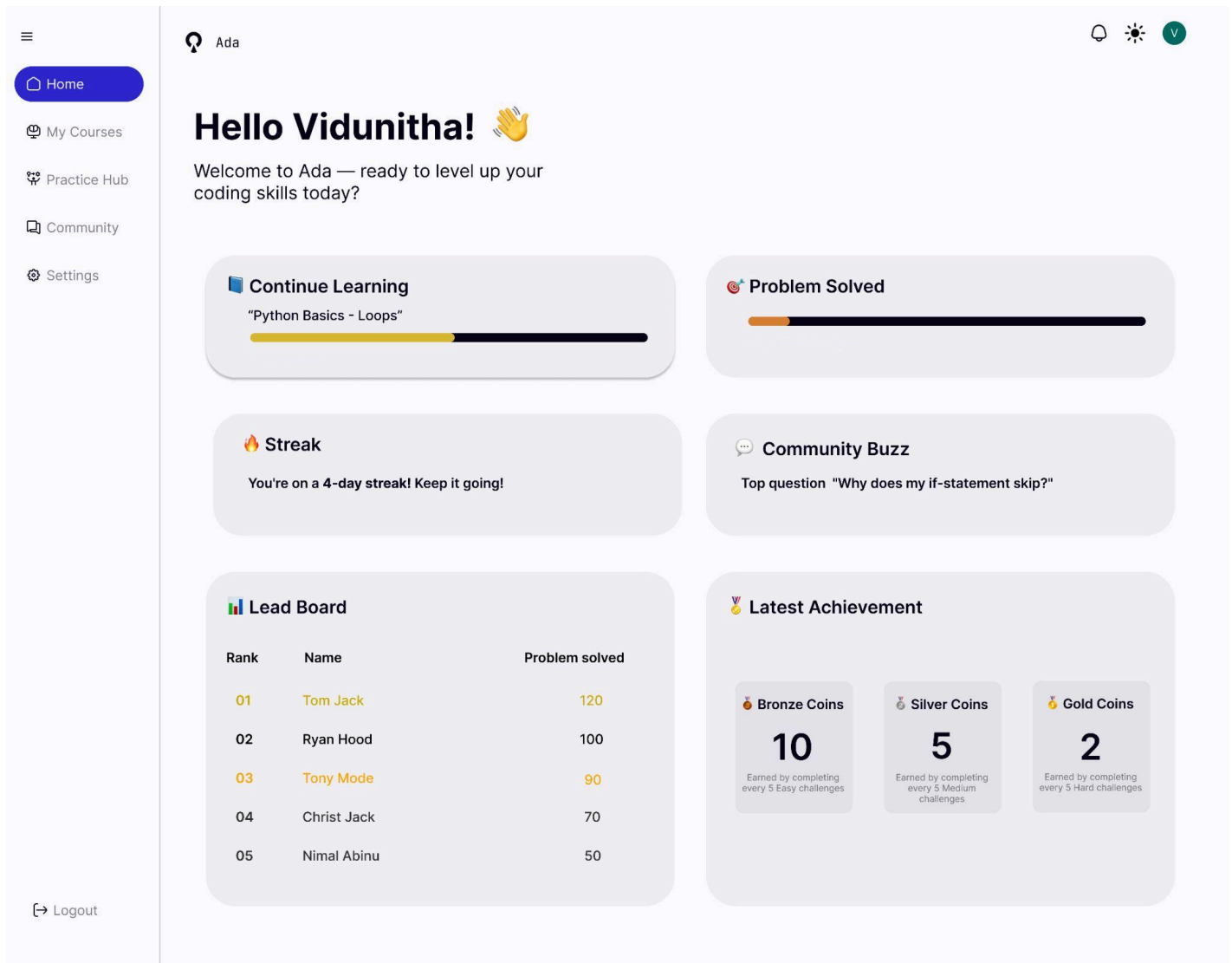
## 6. Delete the Account

```
DELETE FROM User
WHERE user_id = ?;
```

## Routes & Function Signatures:

```
@app.route('/settings', methods=['GET', 'POST'])
def settings():
    pass
```

## Sample White theme UI



**NOTE:** There are sample UIs for the Mobile version on the above pages

## SOFTWARE INTERFACES

### Core Technologies and Frameworks

| LAYER      | TECHNOLOGY                  |
|------------|-----------------------------|
| Frontend   | HTML + CSS + Jinja          |
|            | JavaScript (when necessary) |
|            | React (Optional)            |
| Backend    | Python + flask              |
| Templating | Jinja                       |
| Database   | SQLite + SQLite Studio      |

### Libraries and Tools

| Tool / Library | Purpose                                    | Status   |
|----------------|--------------------------------------------|----------|
| Flask          | Web backend framework                      | In use   |
| Jinja          | Templating engine for HTML                 | In use   |
| Monaco Editor  | In-browser code editor                     | In use   |
| SQLite         | Lightweight local database                 | In use   |
| Tailwind CSS   | Utility-first CSS framework for UI styling | Optional |
| React          | To increase the speed of the app           | Optional |
| bcrypt         | Password hashing for secure login          | Planned  |
| OpenAI API     | AI-powered code feedback (future feature)  | Planned  |
| Judge0 API     | Remote code execution engine               | Planned  |

**NOTE:** These Libraries and tools can change within the development process, and there will be more libraries.

### Operating Environment

- Operating System Supported:
  - **Development:** Windows/macOS/Linux
  - **Deployment:** Any system capable of running Python/Flask apps



## Development Environment

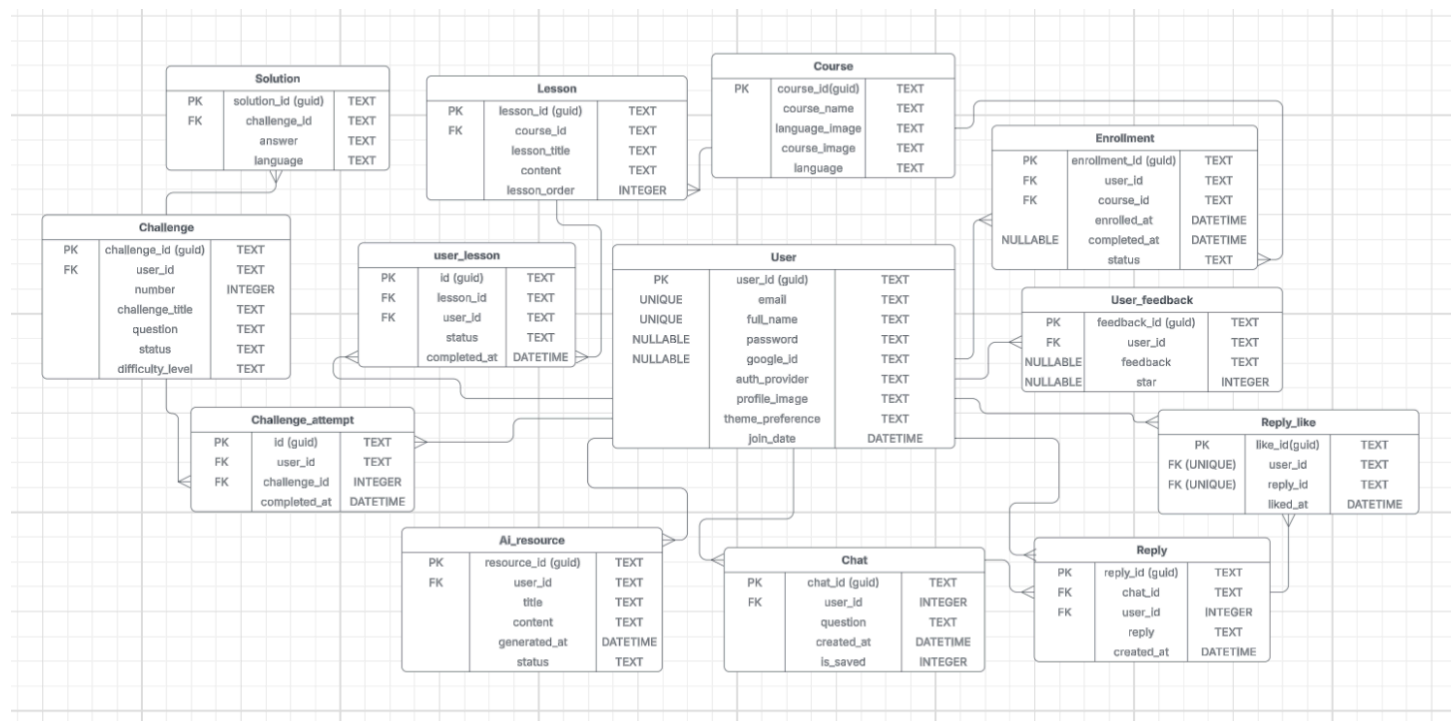
- **Code Editor:** Visual Studio Code
- **Python Version:** Python 3.x (latest stable version)
- **Package Manager:** **pip** (To install libraries)
- **Virtual Environment:** **venv** (optional)
- **Browser:** Chrome, Firefox, or Microsoft Edge (Test for frontend)
- **Command Line:** Terminal / Command Prompt / VS Code Terminal
- **SQLite Studio:** For database design

## ADDITIONAL NONFUNCTIONAL REQUIREMENTS

### SECURITY

- User Passwords should be stored securely.
- API keys will be stored in environment variables and never exposed to the frontend.
- Form data is validated to avoid injection attacks.
- HTTPS must be used to communicate between the client and server when deployed.

## ER DIAGRAM FOR THE APPLICATION



# COLOUR AND FONTS

## Dark theme Colour Palette - Web Application

|                 |                       |                    |                      |                  |
|-----------------|-----------------------|--------------------|----------------------|------------------|
| Text<br>#EBE9FC | Background<br>#010104 | Primary<br>#3A31D8 | Secondary<br>#020024 | Cards<br>#0D0D11 |
|-----------------|-----------------------|--------------------|----------------------|------------------|

## White theme Colour Palette - Web Application

|                 |                       |                    |                      |                  |
|-----------------|-----------------------|--------------------|----------------------|------------------|
| Text<br>#050316 | Background<br>#fbfbfe | Primary<br>#2F27CE | Secondary<br>#dddbff | Cards<br>#EEEEF2 |
|-----------------|-----------------------|--------------------|----------------------|------------------|

## Dark theme Colour Palette - Landing Page

|                 |                       |                    |                      |
|-----------------|-----------------------|--------------------|----------------------|
| Text<br>#ede3ff | Background<br>#05000E | Primary<br>#0D0024 | Secondary<br>#fd550e |
|-----------------|-----------------------|--------------------|----------------------|

## White theme Colour Palette - Landing Page

|                 |                       |                    |                      |
|-----------------|-----------------------|--------------------|----------------------|
| Text<br>#09001a | Background<br>#f5f0ff | Primary<br>#e8dbff | Secondary<br>#f24a02 |
|-----------------|-----------------------|--------------------|----------------------|

I use a Blue colour theme for the application, because blue and violet tones are psychologically associated with **trust, relaxation, less stress, intelligence, and stability**, and it gives a professional look to the product, which is best for a Developer-Focused Interface. Most applications use blue colour themes, some examples are VS Code, GitHub

## Fonts I use for the Web Application

1. Konkhermer Sleokchher:- For the greeting part in the dashboard
2. Inter:- For all other texts except above

## Fonts I use for the Landing Page

1. Roboto:- Landing Page "Unlock your coding potential with Ada ..." and "Ada is a dynamic platform designed to help you learn coding ..."
2. Inter:- Navigation Bar and sign-in button, sign-up button
3. Krub:- "Get started" button
4. Poppins:- Heading of the Why Us section. supported language section, how it works section
5. Montserrat:- description in those sections
6. Lato:- Other texts except above

# IMPLEMENTATION AND ITERATIVE DEVELOPMENT

**Project Link :**

<https://github.com/AlgorithmicPV/Ada-Learning-Platform.git>

## ITERATIVE DEVELOPMENT SUMMARY

This project has been developed using an iterative development methodology. I divided the project into nine sprints. Each sprint focuses on building and refining features, followed by testing, feedback and improvements. This method helps me to improve the quality of my application through feedback, manage my time efficiently and increase the flexibility of my application.

| Sprint   | Work                       |
|----------|----------------------------|
| Sprint 1 | Planning                   |
| Sprint 2 | Landing Page               |
| Sprint 3 | Database Setup             |
| Sprint 4 | Authentication             |
| Sprint 5 | Core Learning Features     |
| Sprint 6 | Community and Practice Hub |
| Sprint 7 | Dashboard & Settings       |
| Sprint 8 | UI/UX Refinements          |
| Sprint 9 | Testing & Documentation    |

## Sprint 1 - Planning

### What I did

- Brainstormed some ideas for this digital outcome, and got some feedback on ideas that I thought. I got a few ideas, to make a web application for the business AI chat, a travelling web application and a Learning web application for new programmers.
- From my feedback, I decided to create a learning web application, which includes lessons, challenges, a custom AI-generated course, an in-built code editor and community features.
- Then, I designed the initial wireframes for my UI and ER diagram.
- I chose the colour plates and fonts for the application.
- Also, I wrote rough SQL queries for each UI, which can be helpful for my development process.

### Why

- Planning was helpful for me, because it gave me a clear direction before the development. By identifying key features early, I was able to stay focused. Wireframes helped me to design the UI quickly without taking time for the design. The ER diagram helped me to design the database soon, supporting

data integrity before implementation. Overall, planning helped me to move into the development process with confidence and saved time later in the project.

## **Sprint 2 - Landing Page and the frontend of the Dashboard**

### **What I did**

- The landing page was created with light and dark themes, enabling users to choose according to their preference or accessibility needs, and it created a professional, attractive first impression by adding branding elements, including a logo, custom fonts, and animated visuals.
- I also created the landing page to be mobile responsive.
- Created the frontend of the dashboard, but could not create the backend as the data was not enough at that moment

### **Why**

- Users will be first attracted by the landing page, and therefore, it should be visually appealing, modern, and accessible. I believe Dark/light themes can improve usability for different lighting conditions.
- Dashboard gives a rough idea to the user where they are.

## **Sprint 3 - Database Setup**

### **What I did**

- Designed all the tables with correct connections using sqliteStudio.
- Populated the tables with data using a Python Script

### **Why**

- A structured database confirms that the learning platform stores lessons, progress, and user activity reliably.

## **Sprint 4 - Authentication**

### **What I did**

- I built signup and login functionality using Argon2 password hashing for strong security.
- Added Google login integration for convenience.
- Implemented validation (reject empty email/password, etc).

### **Why**

- Authentication is the foundation for user trust and privacy. Argon2 is stronger than SHA256/MD5 and prevents brute-force attacks.

## **Sprint 5 - Core Learning Features**

### **What I did**

- I made the code editor with compiler support
- Added AI chatbot
- Implemented progress tracking
- Created a user interface for all the pages related to the My Course section with all the functionality

- Made all the UIs mobile responsive
- Search functionality

#### Why

- As learning features are the core purpose of the platform. Without these, the application would not meet its outcome.

## Sprint 6 - Community and Practice Hub

#### What I did

- I made the code editor with compiler support for the practice hub
- Added an AI chatbot for the practice hub section
- Implemented progress tracking
- Created a user interface for all the pages related to the Community section and Practice Hub section, with all the functionality
- Made all the UIs mobile responsive
- Search functionality for both sections
- Added functionalities for the community hub section, such as adding new questions, answering posted questions, saving questions, and liking answers.

#### Why

- Social features increase collaboration and engagement. The Practice Hub ensures learners don't just skip content but track their progress.

## Sprint 7 - Backend of the Dashboard & Settings

#### What I did

- I made the backend of the Dashboard
- Settings allow updating username, email, password, profile picture, and deleting the account.

#### Why

- Personalisation and control are essential for user engagement and ethical data management (deletion).

## Sprint 8 - UI/UX Refinements

#### What I did

- Improved responsiveness for mobile devices.
- Built 404 and 403 error pages with animations.
- Improved profile image handling and navigation bar usability

#### Why

- Polishing UI ensures the app is not only functional but enjoyable. Error pages prevent confusion.

## Sprint 9 - Testing & Documentation

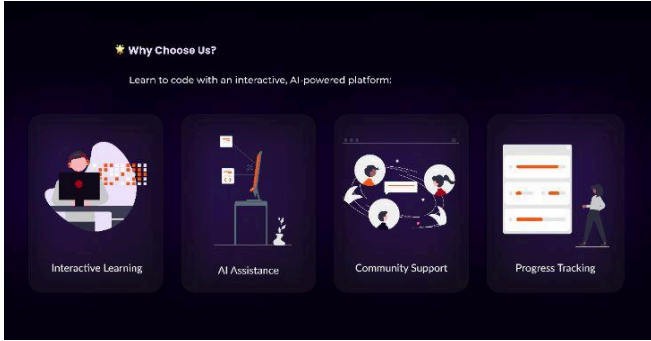
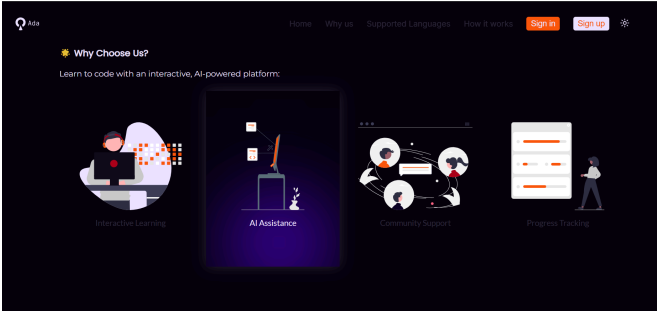
### What I did

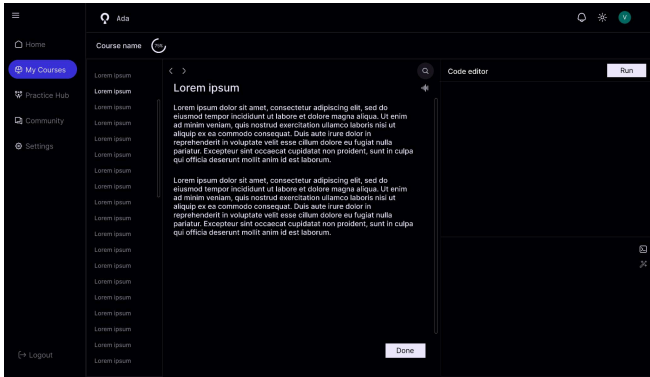
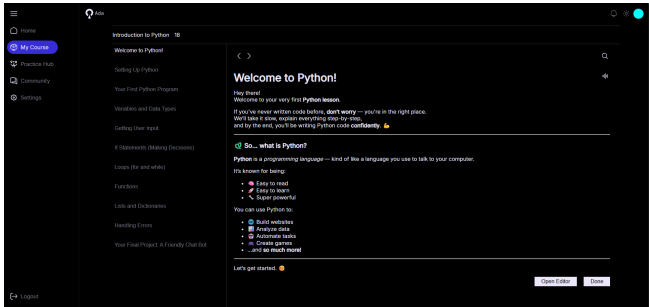
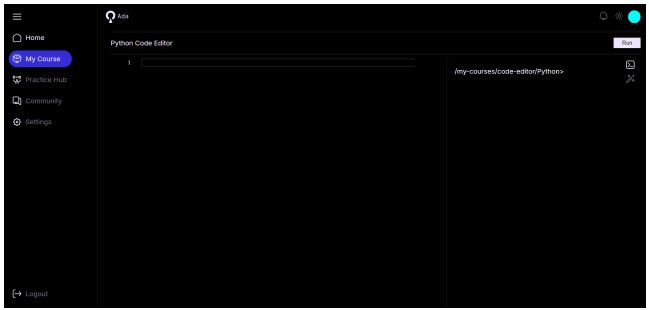
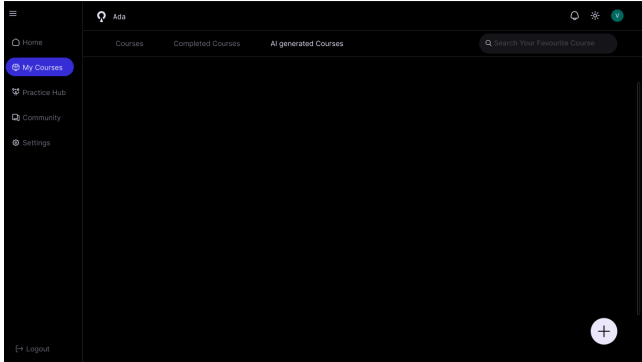
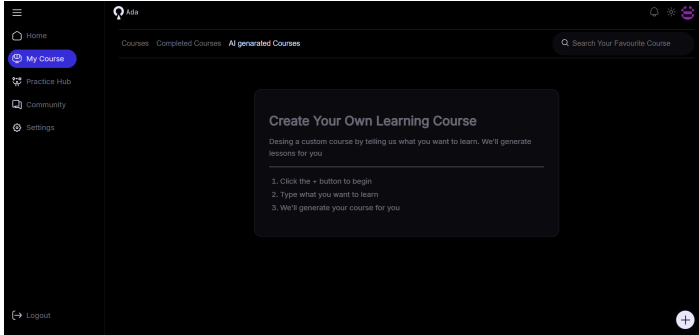
- Added Python docstrings and inline comments to functions and Flask routes.
- Refactored into Flask blueprints, reducing code repetition.
- Browser tested on Chrome, Edge..
- Done other Application testing

### Why

- Confirmed app worked across different browsers. Linted code with Flake8 and Pylint.

### Iterative Improvements Table (Web)

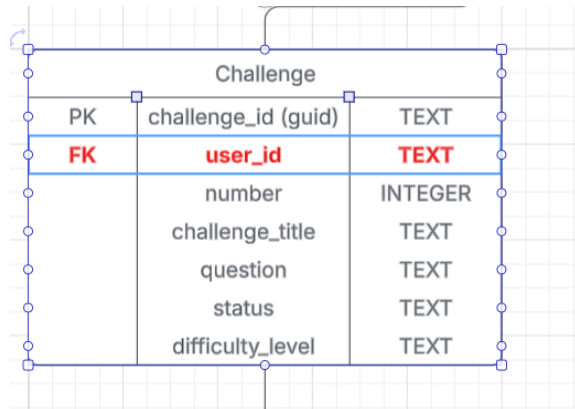
| No. | Date                                                                                                   | Area                                       | Description of change                                                                                                                             | Reason for change                                                                                                                                                                                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 06/08/2025                                                                                             | Frontend - Landing Page - Why Us Section   | Make the borders a little bit rounded, not a lot, and add a gradient overlay that moves with the mouse pointer, and add a box shadow to each card | It gives a modern look to the landing page                                                                                                                                                                                                                                                                    |
|     | <b>Before:-</b><br> |                                            |                                                                                                                                                   | <b>After:-</b><br>                                                                                                                                                                                                        |
| 2   | 06/23/2025                                                                                             | Frontend - Web app - My course-lesson page | Remove the code editor part from the lesson page and keep it on a separate page.                                                                  | With the code editor, it is difficult to manage the layout for the mobile version; mobile users cannot utilise the code editor effectively. Even on larger screens, it is challenging to maintain user-friendly elements, and users may feel overwhelmed by the excessive amount of content on a single page. |
|     | <b>Before:-</b>                                                                                        |                                            |                                                                                                                                                   | <b>After:-</b>                                                                                                                                                                                                                                                                                                |

|   |                                                                                                     |                                         |                                                    |                                                                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                    |                                         |                                                    |   |
| 3 | 08/02/2025                                                                                          | AI-generated courses page               | Add a user guide for the AI-generated courses page | Due to the feedback from Sprint 1, I think it is better now because it increases user friendliness. (This only shows when the user hasn't created any courses)        |
|   | <p>Before:-</p>  |                                         |                                                    | <p>After:-</p>                                                                    |
| 3 | 09/15/2025                                                                                          | Pages that include all the code editors | change the Run button text -> Test Code            | Due to the feedback from my mentor, He said that it is not clear to the user                                                                                          |

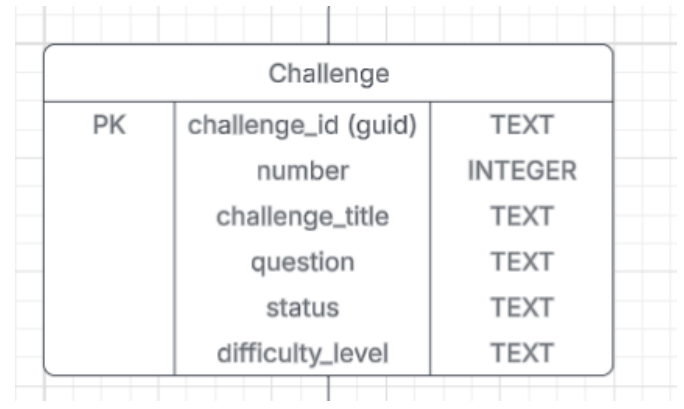
## Iterative Improvements Table (Database)

| No. | Date       | Area                       | Description of change                                               | Reason for change                                                                                                                                                                 |
|-----|------------|----------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 05/05/2025 | Database - Challenge table | Delete the <b><i>user_id</i></b> from the <b><i>Challenge</i></b> . | Because it was a mistake, we don't need that, as we have the <b><i>challenge_attempt</i></b> table to connect the <b><i>Challenge table</i></b> and the <b><i>User</i></b> table. |

Before



After



2

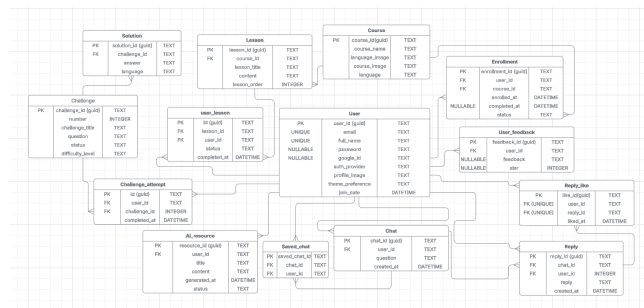
07/02/2025

Database -  
Saved\_chat  
table

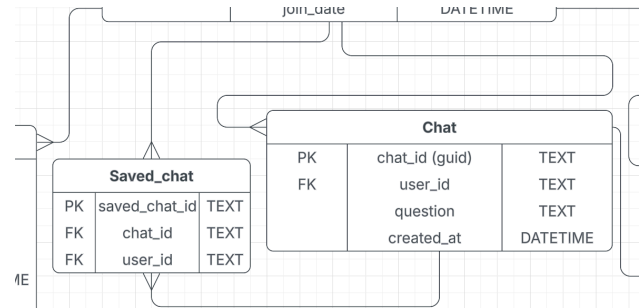
Add a new table  
called  
"Saved\_chat".

Previously, saved\_chat was in the "Chat" table as a column. This is not working as many users can select one discussion and one discussion can be selected by many users; therefore, put that into a separate table.

Before:-



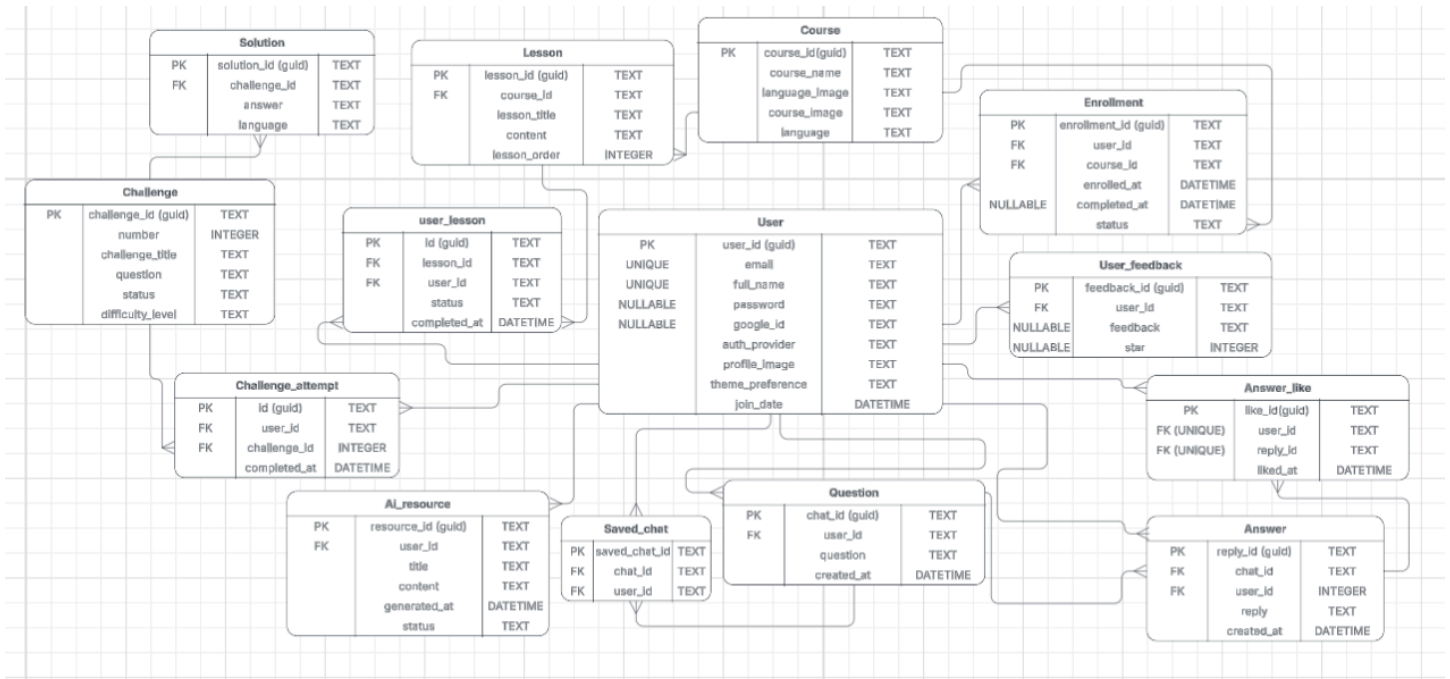
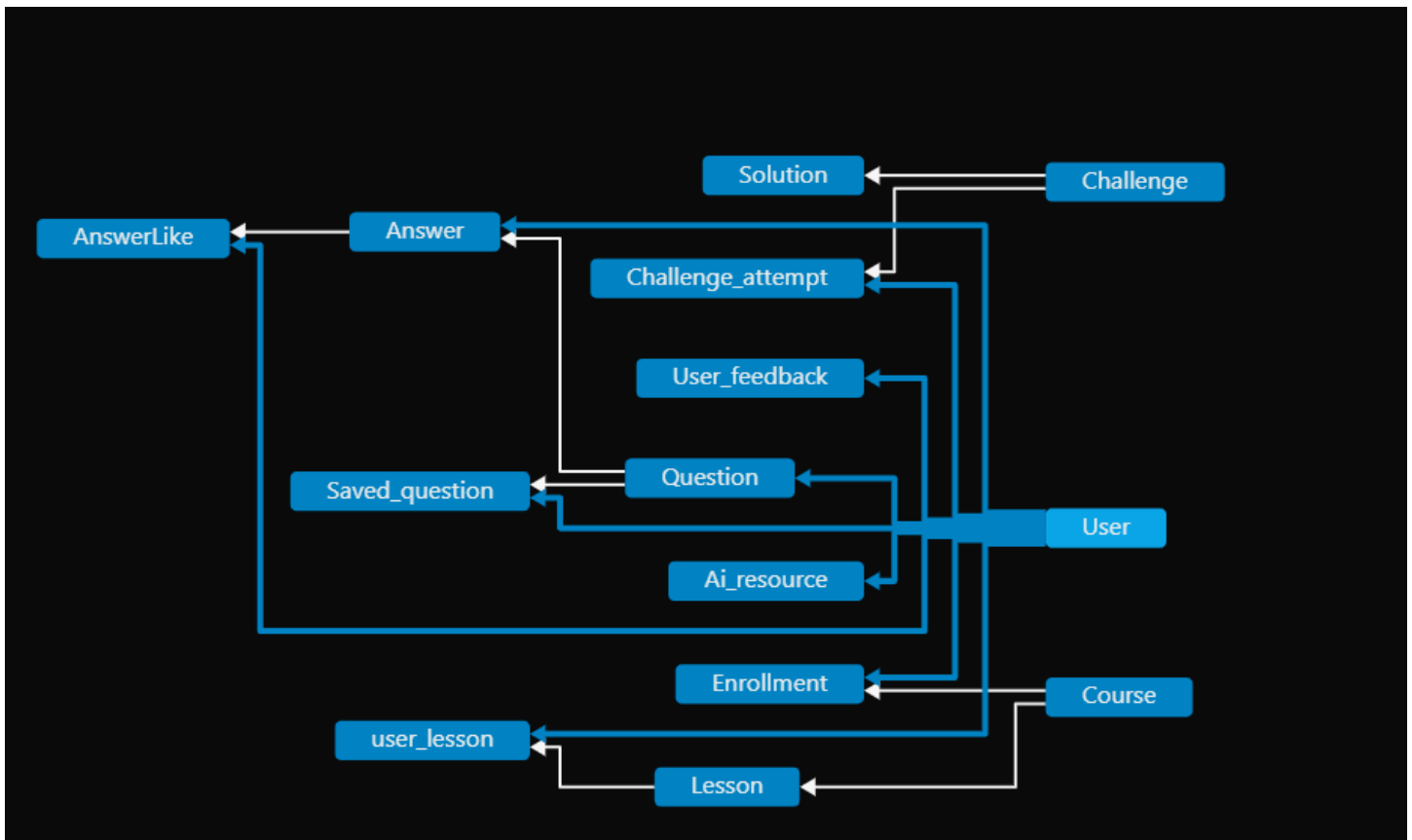
After:-





|   |            |                                          |                                                                          |                                                                                                                                                                                 |
|---|------------|------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | 07/07/2025 | Database -<br>Name of the<br>Reply table | Change the<br>name of the<br>Reply table to<br>Answer table              | Because the Community section is more like a Q&A website, not a chat application, therefore Answer table makes more sense, as this table stores answers for relevant questions. |
| 4 |            | Database -<br>Name of the<br>Chat table  | Change the<br>name of the<br>Chat table to<br>Question table             | Because the Community section is more like Q & A, and I think the Question table is more sensible rather than saying it as a Chat Table                                         |
| 5 |            | Database-<br>Name of the<br>Reply_like   | Change the<br>name of the<br>Reply_like table<br>to AnswerLike<br>table. | Same reason above                                                                                                                                                               |

## Final ER diagram



## Testing

**NOTE:** Tested in different modern browsers, which include Chrome and Microsoft Edge

| Feature/ Module: Login Section |                                                |                                 |                                                 |                                                 |            |       |
|--------------------------------|------------------------------------------------|---------------------------------|-------------------------------------------------|-------------------------------------------------|------------|-------|
| Test ID                        | Test Description                               | Input / How are you testing it  | Expected Output                                 | Actual Output                                   | Pass/ Fail | Notes |
| 001                            | Show the Image of the laptop on the login page | Load the login screen           | Shows the image of the laptop                   | Showed the image of the laptop                  | Pass ▾     | -     |
| 002                            | Show the logo of the app on the Login page     | Load the login screen           | Shows the Application logo on the login page    | Showed the application logo on the login page   | Pass ▾     | -     |
| 003                            | Email input field visibility                   | Load the login screen           | The email field appears with placeholder text   | The email field is visible with the placeholder | Pass ▾     | -     |
| 004                            | Password input field visibility                | Load the login screen           | The password field appears with the placeholder | The password field appears with the placeholder | Pass ▾     | -     |
| 005                            | Placeholder animation (email field)            | Click inside the email field    | Placeholder moves up and shrinks                | Placeholder animation works                     | Pass ▾     | -     |
| 006                            | Placeholder animation (password field)         | Click inside the password field | Placeholder moves up and shrinks                | Placeholder animation works                     | Pass ▾     | -     |
| 007                            | Google Login button styling                    | Load the login screen           | The Google login button appears correctly       | Google login button styled properly             | Pass ▾     | -     |
| 008                            | Login button visibility and style              | Load the login screen           | The login button is visible and styled          | Button displayed properly                       | Pass ▾     | -     |
| 009                            | Input focus effect                             | Click in the                    | Bottom                                          | Visual effect                                   | Pass ▾     | -     |

| Feature/ Module: Login Section |                                                        |                                                                                        |                                  |                                                    |            |       |
|--------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------|------------|-------|
| Test ID                        | Test Description                                       | Input / How are you testing it                                                         | Expected Output                  | Actual Output                                      | Pass/ Fail | Notes |
|                                |                                                        | input field                                                                            | border glows                     | appears as expected                                |            |       |
| 010                            | Hover effect on “Create an account” text and “or” text | Hover the mouse over both texts                                                        | Changes the text colour smoothly | Colour changes as expected                         | Pass ▾     | -     |
| 011                            | Hover effect on the Login button                       | Hover the mouse over the login button                                                  | Button scales up slightly        | Button enlarges with animation                     | Pass ▾     | -     |
| 012                            | Hover effect on the Google login button                | Hover the mouse over the Google login page                                             | Button scales up slightly        | Button enlarges with animation                     | Pass ▾     | -     |
| 013                            | Toggle show password icon                              | Click on the eye icon                                                                  | The password becomes visible     | Password shown                                     | Pass ▾     | -     |
| 014                            | Toggle hide password icon                              | Click on the non-eye icon                                                              | The password becomes hidden      | Password Hidden                                    | Pass ▾     | -     |
| 015                            | Responsive layout (mobile)                             | Resize screen to mobile view                                                           | Layout adapts correctly          | The layout should adapt correctly with no overflow | Pass ▾     | -     |
| 016                            | Log in with valid inputs                               | Email: <a href="mailto:admin@gmail.com">admin@gmail.com</a><br><br>Password: 123456789 | The dashboard will appear        | Dashboard appeared                                 | Pass ▾     | -     |
| 017                            | Log in with invalid inputs                             | Email: <a href="mailto:wrong@gmail.com">wrong@gmail.com</a>                            | An error message will appear     | An error message appeared                          | Pass ▾     | -     |

| Feature/ Module: Login Section |                                                              |                                                                                    |                                                                   |                                                                |            |       |
|--------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------|------------|-------|
| Test ID                        | Test Description                                             | Input / How are you testing it                                                     | Expected Output                                                   | Actual Output                                                  | Pass/ Fail | Notes |
|                                |                                                              | Password: 1234                                                                     | saying "Authentication failed"                                    | saying "Authentication failed"                                 |            |       |
| 018                            | Log in with empty email and password fields                  | Spaces and No value                                                                | An error message will appear saying "Email and password required" | An error message appeared saying "Email and password required" | Pass ▾     | -     |
| 019                            | Log in with a valid email and a wrong password               | Email: <a href="mailto:admin@gmail.com">admin@gmail.com</a><br>Password: 1234      | An error message will appear saying "Authentication failed"       | An error message will appear saying "Authentication failed"    | Pass ▾     | -     |
| 020                            | Log in with an invalid email and an existing password        | Email: <a href="mailto:wrong@gmail.com">wrong@gmail.com</a><br>Password: 123456789 | An error message will appear saying "Authentication failed"       | An error message will appear saying "Authentication failed"    | Pass ▾     | -     |
| 021                            | function of the Google Login button                          | Click on that button                                                               | The page will redirect to the Google login                        | As expected                                                    | Pass ▾     | -     |
| 022                            | Function of the Create an account button                     | Click on it                                                                        | Redirect to the sign-up page                                      | As expected                                                    | Pass ▾     | -     |
| 023                            | Type 2 characters to check the limitation in the email field | type "a@"                                                                          | An error message will appear saying "Invalid Email".              | As expected                                                    | Pass ▾     | -     |

| Feature/ Module: Login Section |                                                                    |                                |                                                                                        |               |            |       |
|--------------------------------|--------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------|---------------|------------|-------|
| Test ID                        | Test Description                                                   | Input / How are you testing it | Expected Output                                                                        | Actual Output | Pass/ Fail | Notes |
| 024                            | Type 51 characters to check the limitation in the email field      | Type "d x 39"@example.com      | An error message will appear saying "Email must be between 3 and 150 characters.       | As expected   | Pass ▾     | -     |
| 025                            | Type 1025 characters to check the limitation in the password field | Type "d x 1025"                | An error message will appear saying "Password must be between 6 and 1024 characters".. | As expected   | Pass ▾     | -     |
| 026                            | Type 50 characters to check the limitation in the email field      | Type "d x 38"@example.com      | Form accepts input and moves to the next step                                          | As expected   | Pass ▾     | -     |
| 027                            | Type 1024 characters to check the limitation in the password field | Type "d x 1024"                | Form accepts input and moves to the next step                                          | As expected   | Pass ▾     | -     |
| 028                            | Type 5 characters to check the limitation in the email field       | Type "a@a.a"                   | Form accepts input and moves to the next step                                          | As expected   | Pass ▾     | -     |

\* All the validations were done with and without the required tag in HTML

NOTE: For the passwords, users can type spaces

| Feature/ Module: Landing page section |                                                      |                                |                                                                           |                                                                                |            |       |
|---------------------------------------|------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------|-------|
| Test ID                               | Test Description                                     | Input / How are you testing it | Expected Output                                                           | Actual Output                                                                  | Pass/ Fail | Notes |
| 001                                   | The logo displays correctly                          | Load page                      | The logo is visible and positioned in the top left of the navigation bar  | The logo was visible and positioned in the top left of the navigation bar      | Pass ▾     | -     |
| 002                                   | The Navigation Bar appears                           | Load page                      | The navigation bar appears                                                | The navigation bar appeared                                                    | Pass ▾     | -     |
| 003                                   | The navigation bar is stuck at the top of the screen | Scroll the page                | The navigation bar was placed at the top when scrolling                   | The navigation bar is placed at the top when scrolling                         | Pass ▾     | -     |
| 004                                   | Navigation bar links scroll to sections              | Click "Why Us" / "Contact"     | Smoothly scrolls to the section                                           | Smoothly scrolled to the section                                               | Pass ▾     | -     |
| 005                                   | Navbar hover animations work                         | Hover over nav links           | Colour changes and underlines                                             | Colour changed and underlined texts                                            | Pass ▾     | -     |
| 006                                   | Theme toggle appears                                 | Load page                      | Dark/light mode toggle is visible depending on the user's previous choice | The dark/light mode toggle was visible depending on the user's previous choice | Pass ▾     | -     |
| 007                                   | Theme switch functionality                           | Click toggle                   | The theme changes between light and dark smoothly                         | The theme changed between light and dark smoothly                              | Pass ▾     | -     |

Feature/ Module: Landing page section

| Test ID | Test Description                                                                    | Input / How are you testing it | Expected Output                                                          | Actual Output                                                              | Pass/ Fail | Notes |
|---------|-------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|------------|-------|
| 008     | Image loading                                                                       | Load page                      | All images display properly                                              | All images displayed properly                                              | Pass ▾     | -     |
| 009     | Animation of the SVG image in the hero section                                      | Load page                      | SVG image's animation works                                              | SVG image's animation worked                                               | Pass ▾     | -     |
| 010     | Buttons hover on the landing page                                                   | Hover over the buttons         | Buttons scale up slightly                                                | Buttons enlarged with animation                                            | Pass ▾     | -     |
| 011     | The Cursor changes when hovering over the buttons                                   | Hover over the buttons         | Cursor changes to pointer                                                | Cursor changed to pointer                                                  | Pass ▾     | -     |
| 012     | Animation appeared when hovering over the cards in the Why Choose use section       | Hover over the cards           | A glow effect follows the mouse, and the box shadow in the cards appears | A glow effect followed the mouse, and the box shadow in the cards appeared | Pass ▾     | -     |
| 013     | Animation when hovering over the language blocks in the Supported Languages section | Hover over the languages block | Language blocks scale up slightly                                        | Language blocks enlarged with animation                                    | Pass ▾     | -     |
| 014     | Animation of moving comment boxes                                                   | Page Load                      | Comment boxes move across the screen continuously                        | Comment boxes moved across the screen continuously                         | Pass ▾     | -     |
| 015     | Stop the animation when hovering over the comment boxes                             | Hover over the comment box     | Stops the animation of the comment boxes                                 | Stopped the animation of the comment boxes                                 | Pass ▾     | -     |
| 016     | Hover over the step                                                                 | Hover over                     | Step blocks                                                              | Step blocks                                                                | Pass ▾     | -     |



Feature/ Module: Landing page section

| Test ID | Test Description                                                      | Input / How are you testing it              | Expected Output                                                | Actual Output                                  | Pass/Fail | Notes |
|---------|-----------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------|------------------------------------------------|-----------|-------|
|         | blocks in the How it works section to test if those blocks enlarge    | the step blocks                             | scale up slightly                                              | enlarged with animation                        |           |       |
| 017     | Responsive layout S(mobile)                                           | Resize screen to mobile view                | Layout adapts correctly                                        | Layout should adapt correctly with no overflow | Pass ▾    | -     |
| 018     | Cards in the Why Us section carousel display correctly on mobile view | Open the app on the mobile screen           | cards appear centred, image + text visible                     | The card displayed as expected in mobile view  | Pass ▾    | -     |
| 019     | Mobile menu opens on hamburger click                                  | Click the hamburger icon on mobile          | Side menu slides open with nav links                           | Menu displayed correctly                       | Pass ▾    | -     |
| 020     | Navigation links are visible in the mobile menu                       | Open menu                                   | All links (Home, Why us, Sign in...) are visible and clickable | Links appear correctly                         | Pass ▾    | -     |
| 021     | The menu closes on clicking the X button                              | Click the close icon (X)                    | Side menu disappears                                           | Menu closed successfully                       | Pass ▾    | -     |
| 022     | Mobile menu supports theme toggle                                     | Click the theme toggle icon inside the menu | Theme switches properly                                        | Toggle works inside the menu                   | Pass ▾    | -     |
| 023     | Clicking the nav links scrolls to correct sections                    | Click "Why Us" in the mobile menu           | Scrolls to "Why Us" section                                    | Smooth scroll occurred                         | Pass ▾    | -     |
| 024     | The menu adapts correctly in light and dark modes                     | Switch themes                               | Menu styles update to match the active theme                   | Styles consistent with the theme               | Pass ▾    | -     |

| Feature/ Module: Sign up section |                                                  |                                |                                                                |                                                |            |       |
|----------------------------------|--------------------------------------------------|--------------------------------|----------------------------------------------------------------|------------------------------------------------|------------|-------|
| Test ID                          | Test Description                                 | Input / How are you testing it | Expected Output                                                | Actual Output                                  | Pass/ Fail | Notes |
| 001                              | The signup page displays correctly on desktop    | Load the page in desktop view  | Page displays with image and input fields properly aligned     | As expected                                    | Pass ▾     | -     |
| 002                              | The signup page displays correctly on mobile     | Load the page in mobile view   | Layout stacks vertically, image resizes properly, no overflow1 | As expected                                    | Pass ▾     | -     |
| 003                              | The signup form input fields appear              | Load page                      | All required fields are visible                                | As expected                                    | Pass ▾     | -     |
| 004                              | Placeholder text appears inside input fields     | Load page                      | Fields show placeholders like "Email", "Name", etc             | As expected                                    | Pass ▾     | -     |
| 005                              | Input focus effect                               | Click in all input fields      | Bottom border glows                                            | Visual effect appears as expected              | Pass ▾     | -     |
| 006                              | Show the Image of the laptop on the login page   | Load page                      | Shows the image of the laptop                                  | Showed the image of the laptop                 | Pass ▾     | -     |
| 007                              | Show the logo of the app on the Login page       | Load page                      | Shows the Application logo on the login page                   | Showed the application logo on the Signup page | Pass ▾     | -     |
| 008                              | Placeholder animation (all fields)               | Click inside all fields        | Placeholder moves up and shrinks                               | Placeholder animation works                    | Pass ▾     | -     |
| 009                              | Toggle the show password icon (both the password | Click on the eye icon          | The password becomes                                           | Password shown                                 | Pass ▾     | -     |

| Feature/ Module: Sign up section |                                                                              |                                                     |                                                          |                                |            |       |
|----------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|--------------------------------|------------|-------|
| Test ID                          | Test Description                                                             | Input / How are you testing it                      | Expected Output                                          | Actual Output                  | Pass/ Fail | Notes |
|                                  | field and the confirm password)                                              |                                                     | visible                                                  |                                |            |       |
| 010                              | Toggle hide password icon (both the password field and the confirm password) | Click on the non-eye icon                           | The password becomes hidden                              | Password Hidden                | Pass ▾     | -     |
| 011                              | Create Account button visibility and style                                   | Load the signup screen                              | The signup button is visible and styled                  | Button displayed properly      | Pass ▾     | -     |
| 012                              | Hover effect on the Create Account button                                    | Hover the mouse over the login button               | Button scales up slightly                                | Button enlarges with animation | Pass ▾     | -     |
| 013                              | Try to make an account with empty fields                                     | No values in the input fields                       | Shows an error saying "All fields are required!"         | As expected                    | Pass ▾     | -     |
| 014                              | If the client changes or removes the input attributions like name, id        | Change and remove input attribution through DevTool | Shows an error saying "All fields are required!"         | As expected                    | Pass ▾     | -     |
| 015                              | Email validation                                                             | Type a non-Email value for the email                | Shows an error saying "Invalid email!"                   | As expected                    | Pass ▾     | -     |
| 016                              | Check if the system takes the existing email                                 | type "admin@gmail.com"                              | Shows an error message saying "Email is already in use." | As expected                    | Pass ▾     | -     |

## Feature/ Module: Sign up section

| Test ID | Test Description                                                    | Input / How are you testing it                       | Expected Output                                                                                     | Actual Output | Pass/ Fail | Notes |
|---------|---------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|------------|-------|
| 017     | Type 51 characters to check the limitation in the email field       | Type "d x 39"@example.com                            | A warning message will appear saying "Display email is too long. Maximum allowed is 50 characters." | As expected   | Pass ▾     | -     |
| 018     | Password matching                                                   | Type two different passwords<br><br>123456<br>654321 | An error message saying "Passwords do not match"                                                    | As expected   | Pass ▾     | -     |
| 019     | Type 5 characters for the passwords                                 | type 12345                                           | A warning message saying "Password is too short"                                                    | As expected   | Pass ▾     | -     |
| 020     | Type 1025 characters to check the limitation in the password fields | Type "d x 1025"                                      | A warning message will appear saying "Password is too long. Maximum allowed is 1,024 characters."   | As expected   | Pass ▾     | -     |
| 021     | Check the character minimum limitation for the name                 | Type less than three characters<br>Type "LL"         | A warning message saying "Name is too short."                                                       | As expected   | Pass ▾     | -     |
| 021     | Check the character                                                 | Type more                                            | A warning                                                                                           | As expected   | Pass ▾     | -     |

## Feature/ Module: Sign up section

| Test ID | Test Description                                                                              | Input / How are you testing it                                                                              | Expected Output                                         | Actual Output | Pass/ Fail | Notes |
|---------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------|------------|-------|
|         | maximum limitation for the name                                                               | than 50 characters<br>"L" x 51                                                                              | message saying<br>"Name is too long"                    |               |            |       |
| 022     | Type correct sign-up details, and try to log in to the web app, to check whether the DB works | Email:<br>"test@gmail.com"<br><br>Name:<br>test<br><br>Password:<br>123456789<br><br>Password:<br>123456789 | Redirect to login page                                  | As expected   | Pass ▾     | -     |
| 023     | Check the image in position                                                                   | Reload the page                                                                                             | Image is appeared                                       | As expected   | Pass ▾     | -     |
| 024     | Type 2 characters to check the limitation in the email field                                  | type "a@"                                                                                                   | An error message will appear saying<br>"Invalid Email". | As expected   | Pass ▾     | -     |
| 025     | Type 50 characters to check the limitation in the email field                                 | Type "d x 38"@example.com                                                                                   | Form accepts input and moves to the next step           | As expected   | Pass ▾     | -     |
| 026     | Type 1024 characters to check the limitation in the password fields                           | Type "d x 1024"                                                                                             | Form accepts input and moves to the next step           | As expected   | Pass ▾     | -     |
| 027     | Type 5 characters to check the limitation in the email field                                  | Type "a@a.a"                                                                                                | Form accepts input and moves to the next step           | As expected   | Pass ▾     | -     |

| Feature/ Module: Sign up section |                                                     |                                |                                                                         |               |            |       |
|----------------------------------|-----------------------------------------------------|--------------------------------|-------------------------------------------------------------------------|---------------|------------|-------|
| Test ID                          | Test Description                                    | Input / How are you testing it | Expected Output                                                         | Actual Output | Pass/ Fail | Notes |
| 028                              | Check the character maximum limitation for the name | Type "d" x 51                  | A warning message "Name is too long. Maximum allowed is 50 characters." | As expected   | Pass ▾     | -     |
| 029                              | Check the character limitation for the name         | Type "d" x 50                  | No error messages, go to the next step                                  | As expected   | Pass ▾     | -     |

| Feature/ Module: Dashboard |                                                                                           |                                                           |                                                                         |                    |            |       |
|----------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------|--------------------|------------|-------|
| Test ID                    | Test Description                                                                          | Input / How are you testing it                            | Expected Output                                                         | Actual Output      | Pass/ Fail | Notes |
| 001                        | Verify background color updates correctly when switching between dark mode and light mode | Toggle theme switch                                       | Background color changes to match selected mode                         | Worked as expected | Pass ▾     | -     |
| 002                        | Redirect unauthenticated users                                                            | Copy the dashboard URL and open it in a different browser | Redirects to /login screen                                              | Worked as expected | Pass ▾     | -     |
| 003                        | Verify hamburger menu button functionality                                                | Tap the hamburger menu icon                               | Verify the element resizes correctly according to the expected behavior | Worked as expected | Pass ▾     | -     |

| Feature/ Module: Dashboard |                                                                                                        |                                                                                    |                                                                       |                    |            |       |
|----------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------|------------|-------|
| Test ID                    | Test Description                                                                                       | Input / How are you testing it                                                     | Expected Output                                                       | Actual Output      | Pass/ Fail | Notes |
| 004                        | Responsive grid                                                                                        | Switch to the developer tools and confirm that clicking the Toggle Fullscreen icon | Layout reflows smoothly on all screen sizes with no overlap.          | Worked as expected | Pass ▾     | -     |
| 005                        | User menu actions                                                                                      | Profile/Settings/Logout                                                            | functions as expected                                                 | Worked as expected | Pass ▾     | -     |
| 006                        | Shows the title bar as Dashboard                                                                       | Load the page                                                                      | Shows as expected                                                     | Shows as expected  | Pass ▾     | -     |
| 007                        | Show the latest course if the user has selected                                                        | Select a course from the My course section                                         | Show the percentage of completion                                     | As expected        | Pass ▾     | -     |
| 008                        | Check the progress bar in the Continue Learning widget                                                 | Complete some lessons in My course Section                                         | Percentage increases, and the progress bar also                       | As expected        | Pass ▾     | -     |
| 009                        | Check the Learning widget at the first login                                                           | First login                                                                        | Shows “No Course selected”                                            | As expected        | Pass ▾     | -     |
| 010                        | Check the number of problems solved in the Problem Solved widget                                       | Reload the page, and first login                                                   | The progress bar is empty, and the down test is 0 questions completed | As expected        | Pass ▾     | -     |
| 011                        | Check the progress bar in the Problem Solved widget to see if it increases when the user completes one | Complete some questions                                                            | The progress bar increases, and the number                            | As expected        | Pass ▾     | -     |

## Feature/ Module: Dashboard

| Test ID | Test Description                                                                         | Input / How are you testing it                                                                                                                                        | Expected Output                                                          | Actual Output | Pass/ Fail | Notes |
|---------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------|------------|-------|
|         |                                                                                          |                                                                                                                                                                       | under the progress bar increases                                         |               |            |       |
| 012     | To check when the user clicks on the Continue Learning, redirects to the relevant course | Click on that                                                                                                                                                         | Redirect to the relevant Course                                          | As expected   | Pass ▾     | -     |
| 013     | Check whether the course details are changing when the user changes the course           | Select a different course<br>"Introduction to JavaScript"                                                                                                             | Course details in the Continuing Learning widget change                  | As expected   | Pass ▾     | -     |
| 014     | Check the strike                                                                         | Stay two days consistently using (adding question, doing challenges ...)                                                                                              | Change the "No current streak...." in the Streak widget, get the value 2 | As expected   | Pass ▾     | -     |
| 015     | Check whether the system shows the question that has the highest number of answers       | Select one question, and give answers. Select the question called "Test 2" for this                                                                                   | It appears in the Community Buzz widget as "Top Question "test 2"        | As expected   | Pass ▾     | -     |
| 016     | Check whether the leaderboard works or not                                               | Creates two different accounts and does some challenges<br><a href="mailto:vidunothap@gmail.com">vidunothap@gmail.com</a> - 2 challenges<br><br>Pasindu Vidunitha - 1 | Shows both names with relevant rank numbers, and font colours            | As expected   | Pass ▾     | -     |



## Feature/ Module: Dashboard

| Test ID | Test Description                                                                      | Input / How are you testing it              | Expected Output                                                                                                    | Actual Output | Pass/ Fail | Notes |
|---------|---------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------|------------|-------|
|         |                                                                                       | challenges                                  |                                                                                                                    |               |            |       |
| 017     | Check whether users can get Bronze Coins, Silver Coins and Gold Coins                 | Do 5 from each                              | Increases by 1 for each 5 challenges<br>5 easy = 1 Bronze coin<br>5 medium = 1 silver coin<br>5 hard = 1 Gold coin | As expected   | Pass ▾     | -     |
| 018     | Check the cursor changes to a pointer from the default when hovering over each widget | Hover over widgets                          | Changes to the pointer from the cursor                                                                             | As expected   | Pass ▾     | -     |
| 019     | Theme toggling                                                                        | Click on the theme icon                     | change from dark to light and light to dark                                                                        | As expected   | Pass ▾     | -     |
| 020     | Mobile responsiveness                                                                 | Through inspect, changes to the mobile view | Layout should be adjusted                                                                                          | As expected   | Pass ▾     | -     |

## Feature/ Module: My Course section - All Courses page

| Test ID | Test Description             | Input / How are you testing it | Expected Output                         | Actual Output | Pass/ Fail | Notes |
|---------|------------------------------|--------------------------------|-----------------------------------------|---------------|------------|-------|
| 001     | Shows the sub navigation bar | Load the page                  | Shows the sub navigation bar at the top | As expected   | Pass ▾     | -     |

Feature/ Module: My Course section - All Courses page

| Test ID | Test Description                                                                                                      | Input / How are you testing it                                    | Expected Output                                                                                                  | Actual Output | Pass/ Fail | Notes |
|---------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------|------------|-------|
| 002     | Sub Navigation bar functionality                                                                                      | Click on the Courses, Completed Courses, and AI-generated Courses | Change the page according to the menu, and change the colour of the text that was clicked from lighter to darker | As expected   | Pass ▾     | -     |
| 003     | Checks the animation of the Course card                                                                               | Hover over the card                                               | A play button appears, and the background becomes darker                                                         | As expected   | Pass ▾     | -     |
| 004     | Check the functionality of the Course cards                                                                           | Click on the cards.                                               | Redirect to relevant courses                                                                                     | As expected   | Pass ▾     | -     |
| 005     | Search bar functionality (search bar for both All Courses and Completed Courses; therefore, it gives the same result) | Type and search                                                   | Shows the relevant courses                                                                                       | As expected   | Pass ▾     | -     |
| 006     | Theme switching                                                                                                       | Click on the theme icon                                           | Change the theme                                                                                                 | As expected   | Pass ▾     | -     |
| 007     | Mobile responsive                                                                                                     | Through the inspect, change the device                            | Adjusts the layout                                                                                               | As expected   | Pass ▾     | -     |

### Feature/ Module: My Course section - Completed Courses

| Test ID | Test Description                                                      | Input / How are you testing it                          | Expected Output                        | Actual Output | Pass/ Fail | Notes |
|---------|-----------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------|---------------|------------|-------|
| 001     | Checks whether there are any courses, without completing them         | Click on the completed Course in the sub navigation bar | Empty Page                             | As expected   | Pass ▾     | -     |
| 002     | Checks whether completed courses are showing in the Completed Courses | Complete the introduction to Python course              | The relevant completed course is there | As expected   | Pass ▾     | -     |
| 003     | Theme switching                                                       | Click on the theme icon                                 | Change the theme                       | As expected   | Pass ▾     | -     |
| 004     | Mobile responsive                                                     | Through the inspect, change the device                  | Adjusts the layout                     | As expected   | Pass ▾     | -     |

### Feature/ Module: My Course section - Lesson Page

| Test ID | Test Description                                   | Input / How are you testing it | Expected Output                                                                                     | Actual Output | Pass/ Fail | Notes |
|---------|----------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|---------------|------------|-------|
| 001     | Checks the code blocks' style in the lesson page   | Click on a course              | The code blocks should have a different font family, font colour, and a different background colour | As expected   | Pass ▾     | -     |
| 002     | Functionality of the arrow keys in the lesson page | Click on the buttons           | Able to go to the next lesson and the previous lesson                                               | As expected   | Pass ▾     | -     |

Feature/ Module: My Course section - Lesson Page

| Test ID | Test Description                                      | Input / How are you testing it                                                                | Expected Output                                                                                    | Actual Output | Pass/Fail | Notes |
|---------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------|-----------|-------|
| 003     | Search functionality in the lesson page               | Type something in the search bar                                                              | Keywords are highlighted                                                                           | As expected   | Pass ▾    | -     |
| 004     | Functionality of the Done Button in the lesson page   | Click on the button                                                                           | Move to the next lesson; if it is the last lesson, it stays the same, and the percentage increases | As expected   | Pass ▾    | -     |
| 005     | Read aloud the capability of the lesson page          | Click the button that has a wave icon                                                         | Start to read the text on the screen, and when you click it again, stop reading                    | As expected   | Pass ▾    | -     |
| 006     | Functionality of the Open editor button               | Opens the code editor with the relevant programming language.                                 | can see the code editor                                                                            | As expected   | Pass ▾    | -     |
| 007     | Theme switching                                       | Click on the theme icon                                                                       | Change the theme                                                                                   | As expected   | Pass ▾    | -     |
| 008     | Mobile responsive                                     | Through the inspect, change the device                                                        | Adjusts the layout                                                                                 | As expected   | Pass ▾    | -     |
| 009     | Type invalid lesson ID, when the course ID is correct | Type http://127.0.0.1:5000/my-courses/7da7fd5f-8ff3-4ef7-9431-20c42961f16e/123 in the url bar | 404 page                                                                                           | As expected   | Pass ▾    | -     |

## Feature/ Module: My Course section - Lesson Page

| Test ID | Test Description                                      | Input / How are you testing it                                                                                   | Expected Output          | Actual Output | Pass/ Fail | Notes |
|---------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|------------|-------|
| 010     | Type invalid course ID, when the lesson ID is correct | Type http://127.0.0.1:5000/my-courses/123/20446059-ce1d-45f2-ac1f-278e1fe518ed in the url bar                    | 404 page                 | As expected   | Pass ▾     | -     |
| 011     | Type validated Course ID, and Lesson ID               | Type http://127.0.0.1:5000/my-courses/7da7fd5f-8ff3-4ef7-9431-20c42961f16e//20446059-ce1d-45f2-ac1f-278e1fe518ed | Relevant content appears | As expected   | Pass ▾     | -     |

## Feature/ Module: My Course section - Code Editor

| Test ID | Test Description          | Input / How are you testing it         | Expected Output     | Actual Output | Pass/ Fail | Notes |
|---------|---------------------------|----------------------------------------|---------------------|---------------|------------|-------|
| 001     | Theme switching           | Click on the theme icon                | Change the theme    | As expected   | Pass ▾     | -     |
| 002     | Mobile responsive         | Through the inspect, change the device | Adjusts the layout  | As expected   | Pass ▾     | -     |
| 003     | Programming language name | Relaod                                 | Correct programming | As expected   | Pass ▾     | -     |

Feature/ Module: My Course section - Code Editor

| Test ID | Test Description                          | Input / How are you testing it                                                           | Expected Output                                                | Actual Output | Pass/ Fail | Notes |
|---------|-------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------|------------|-------|
|         |                                           |                                                                                          | language name                                                  |               |            |       |
| 004     | Check the compiler                        | Run some codes in different languages                                                    | Get output                                                     | As expected   | Pass ▾     | -     |
| 005     | Character limitations in the code         | Type more than 15000 characters                                                          | A warning message<br>"Input too large (max 15,000 chars).<br>" | As expected   | Pass ▾     | -     |
| 006     | Character limitation in the AI chat       | Type more than 1500 characters                                                           | A warning message<br>"message too long..."                     | As expected   | Pass ▾     | -     |
| 007     | Use an input value and test the compliler | type "x = input(">>")<br>print(x)" in the code edit, and type the value "d" in the input | print d in the output                                          | As expected   | Pass ▾     | -     |

### Feature/ Module: My Course - AI generated Course

| Test ID | Test Description                                                         | Input / How are you testing it                                     | Expected Output                                                                                                      | Actual Output | Pass/ Fail | Notes |
|---------|--------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------|------------|-------|
| 001     | Check that the instructions are showing when there is no created courses | Reload                                                             | Showing instructions                                                                                                 | as expected   | Pass ▾     | -     |
| 002     | Plus mark functionality                                                  | Click on it                                                        | The background becomes darker, the warning message shows that code editor is not providing, and the input bar appear | as expected   | Pass ▾     | -     |
| 003     | Boundary testing (all fields)                                            | Tested minimum/m aximum lengths, empty fields, and invalid formats | Valid inputs accepted, invalid rejected                                                                              | same          | Pass ▾     | -     |

### Feature/ Module: Practice Hub Section - Uncomplete Questions

| Test ID | Test Description      | Input / How are you testing it           | Expected Output  | Actual Output      | Pass/ Fail | Notes |
|---------|-----------------------|------------------------------------------|------------------|--------------------|------------|-------|
| 001     | Started task tracking | Start “Sum of List”                      | Status = Started | As expcted         | Pass ▾     | -     |
| 002     | Not started tasks     | Leave “Prime Number Generator” untouched | Status = blank   | Status shows blank | Pass ▾     | -     |

### Feature/ Module: Practice Hub Section - Uncomplete Questions

| Test ID | Test Description    | Input / How are you testing it                          | Expected Output       | Actual Output           | Pass/ Fail | Notes |
|---------|---------------------|---------------------------------------------------------|-----------------------|-------------------------|------------|-------|
| 003     | Mixed task states   | Have both started & not started                         | Correct mix displayed | Matches expected states | Pass ▾     | -     |
| 004     | Check the solutions | change the programming language and check the solutions | give relevant answers | as expected             | Pass ▾     | -     |

### Feature/ Module: Practice Hub Section - Question

| Test ID | Test Description                        | Input / How are you testing it                                                                          | Expected Output                         | Actual Output | Pass/ Fail | Notes |
|---------|-----------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------|------------|-------|
| 001     | Boundary testing (all fields)           | Tested minimum/m aximum lengths, empty fields, and invalid formats across username, email, and password | Valid inputs accepted, invalid rejected | same          | Pass ▾     | -     |
| 002     | Check the correct challenge detail      | reload the page                                                                                         | detail is shown                         | as expected   | Pass ▾     | -     |
| 003     | Ask question from AI chat               | Say hi                                                                                                  | get a response                          | as expected   | Pass ▾     | -     |
| 004     | Check code running in various languages | type a code in various languages                                                                        | get the relevant output                 | as expected   | Pass ▾     | -     |
| 005     | Check the solutions                     | change the programming language and check the                                                           | give relevant answers                   | as expected   | Pass ▾     | -     |



### Feature/ Module: Practice Hub Section - Question

| Test ID | Test Description               | Input / How are you testing it                                               | Expected Output  | Actual Output | Pass/Fail | Notes |
|---------|--------------------------------|------------------------------------------------------------------------------|------------------|---------------|-----------|-------|
|         |                                | solutions                                                                    |                  |               |           |       |
| 006     | Type invalid ID in the URL bar | type http://127.0.0.1:5000/practice-hub/123                                  | 404 page         | as expected   | Pass ▾    | -     |
| 007     | Type valid ID in the URL bar   | type http://127.0.0.1:5000/practice-hub/48461c57-b266-468c-b1be-bc83f86024d3 | Relevant content | ""            | Pass ▾    | -     |

### Feature/ Module: Practice Hub Completed Tasks

| Test ID | Test Description              | Input / How are you testing it                          | Expected Output         | Actual Output               | Pass/Fail | Notes |
|---------|-------------------------------|---------------------------------------------------------|-------------------------|-----------------------------|-----------|-------|
| 001     | Verify completed task appears | Complete "Print Hello World"                            | Task marked completed   | Task displayed in list      | Pass ▾    | -     |
| 002     | Multiple completions          | Complete 2 tasks                                        | Both show in list       | Both listed with timestamps | Pass ▾    | -     |
| 003     | Timestamp validation          | Check completion time                                   | Shows correct date/time | Date/time correct           | Pass ▾    | -     |
| 004     | Check the solutions           | change the programming language and check the solutions | give relevant answers   | as expected                 | Pass ▾    | -     |

## Feature/ Module: Community Section

| Test ID | Test Description                                                                                 | Input / How are you testing it                                                                                   | Expected Output                                                | Actual Output | Pass/ Fail | Notes |
|---------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------|------------|-------|
| 001     | All the question are there                                                                       | Reload                                                                                                           | With number of answers, posted user, profile imae are ther     | As expected   | Pass ▾     |       |
| 002     | Functional test (normal input)                                                                   | Enter a valid question (100 chars)<br>Question posted successfully<br>Question posted successfully               | Question posted successfully                                   | ""            | Pass ▾     | -     |
| 003     | Boundary testing (minimum, maximum, below, above, empty spaces) both question and answers fields | 29 chars (too short), 30 chars (min valid), 30,000 chars (max valid), 30,001 chars (too long), " " (only spaces) | Accept valid (30–30,000). Reject invalid (<30, >30,000, empty) | ""            | Pass ▾     | -     |
| 004     | Check relevant questions to relevant categories                                                  | navigate through sub nav                                                                                         | shows relevant questions                                       | ""            | Pass ▾     | -     |
| 005     | Type invalid ID in the URL bar                                                                   | type http://127.0.0.1:5000/community/123/discussions                                                             | 404 page                                                       | as expected   | Pass ▾     | -     |
| 006     | Type validID in the URL bar                                                                      | type http://127.0.0.1:5000/community/ebf85f0a-115a-4f8c-87df-b3f72d0717a9/di                                     | Relvant content                                                | ""            | Pass ▾     | -     |

## Feature/ Module: Community Section

| Test ID | Test Description | Input / How are you testing it | Expected Output | Actual Output | Pass/ Fail | Notes |
|---------|------------------|--------------------------------|-----------------|---------------|------------|-------|
|         |                  | scussions                      |                 |               |            |       |

## Feature/ Module: Settings (Profile &amp; Password)

| Test ID | Test Description                | Input / How are you testing it                                                                          | Expected Output                         | Actual Output             | Pass/ Fail | Notes |
|---------|---------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------|------------|-------|
| 001     | Profile details display         | View settings                                                                                           | Show username & email                   | Username/e mail displayed | Pass ▾     |       |
| 002     | Password change valid           | Enter old + new password                                                                                | Password updated                        | Update success            | Pass ▾     | -     |
| 003     | Password change invalid         | Wrong current password                                                                                  | Error message shown                     | Error shown               | Pass ▾     |       |
| 004     | Delete account                  | Click delete                                                                                            | Account removed                         | Account removed           | Pass ▾     | -     |
| 005     | Feedback adding                 | put a feedback and send                                                                                 | get the success message                 | same                      | Pass ▾     |       |
| 006     | Boundary testing (all fields)   | Tested minimum/m aximum lengths, empty fields, and invalid formats across username, email, and password | Valid inputs accepted, invalid rejected | same                      | Pass ▾     | -     |
| 007     | Try to add invalid image format | upload a txt file                                                                                       | as error message                        | as expected               | Pass ▾     | -     |

## Feature/ Module: Settings (Profile & Password)

| Test ID | Test Description    | Input / How are you testing it | Expected Output  | Actual Output | Pass/Fail | Notes |
|---------|---------------------|--------------------------------|------------------|---------------|-----------|-------|
| 008     | Profile image chage | use a valid image              | change the image | ""            | Pass      | -     |

NOTE: Theme toggles has tested, it works as I am using the same base.html

## Data Intergirty Testing

Test ID: 001

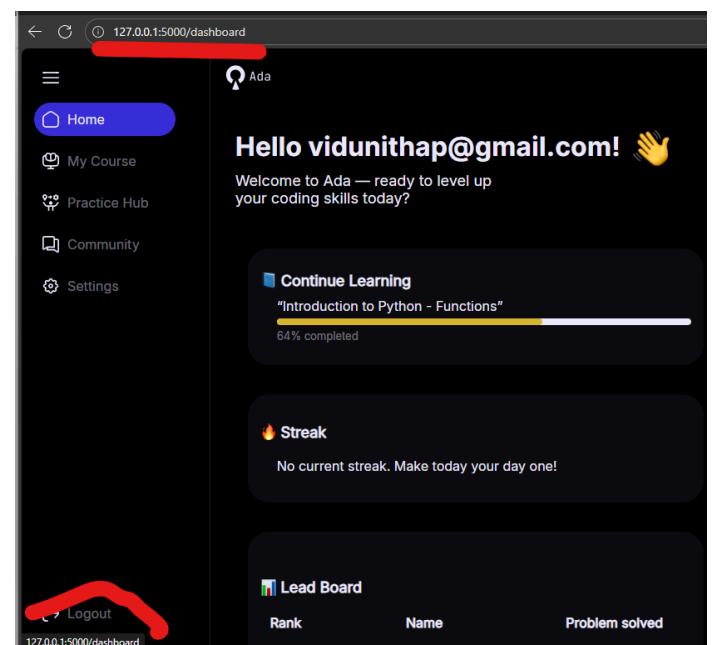
Goal: Ensure the Home link points to the dashboard.

Method: Hovered on *Home* in the side navigation.

Evidence: Browser status bar shows

[127.0.0.1:5000/dashboard](http://127.0.0.1:5000/dashboard) (see screenshot).

Result: Passed – Home correctly directs to the dashboard page.



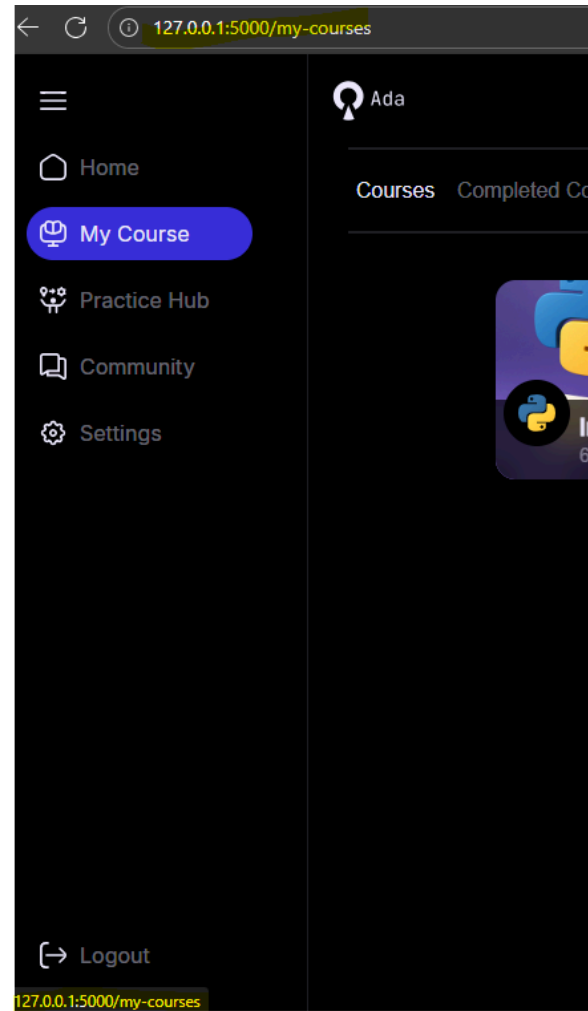
Test ID: 002

Goal: Ensure the My Course link goes to the correct page.

Method: Hovered on *My Course* in the side navigation.

Evidence: Browser status bar shows [127.0.0.1:5000/my-courses](http://127.0.0.1:5000/my-courses) (see screenshot).

Result: Passed – Link correctly loads the *My Course* page.



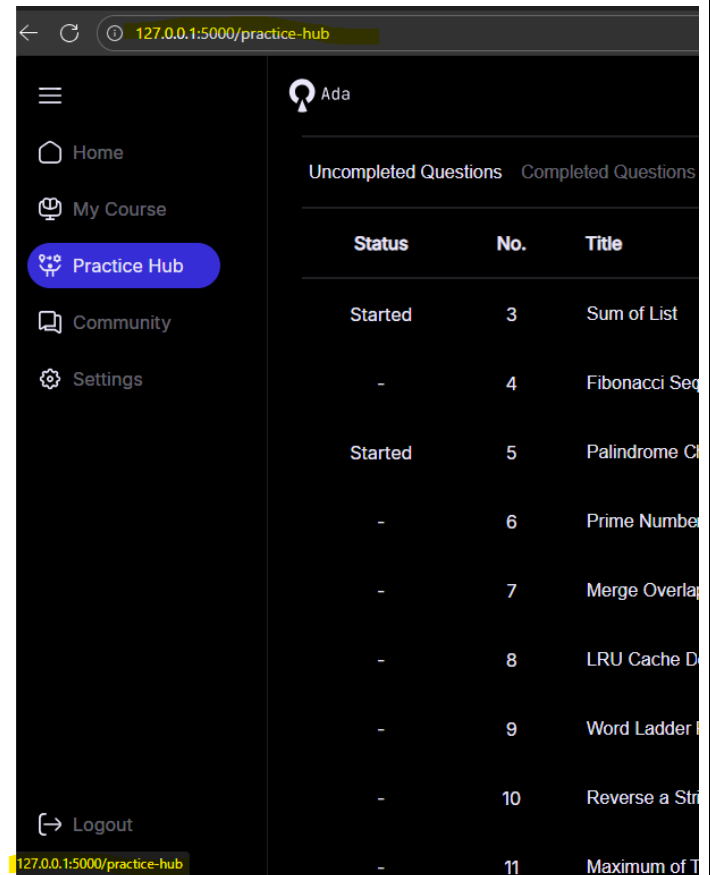
Test ID: 003

Goal: Ensure the Practice Hub link loads correctly.

Method: Hovered on *Practice Hub* in the side navigation.

Evidence: Browser status bar shows [127.0.0.1:5000/practice-hub](http://127.0.0.1:5000/practice-hub) (see screenshot).

Result: Passed – Link correctly opens the Practice Hub page.



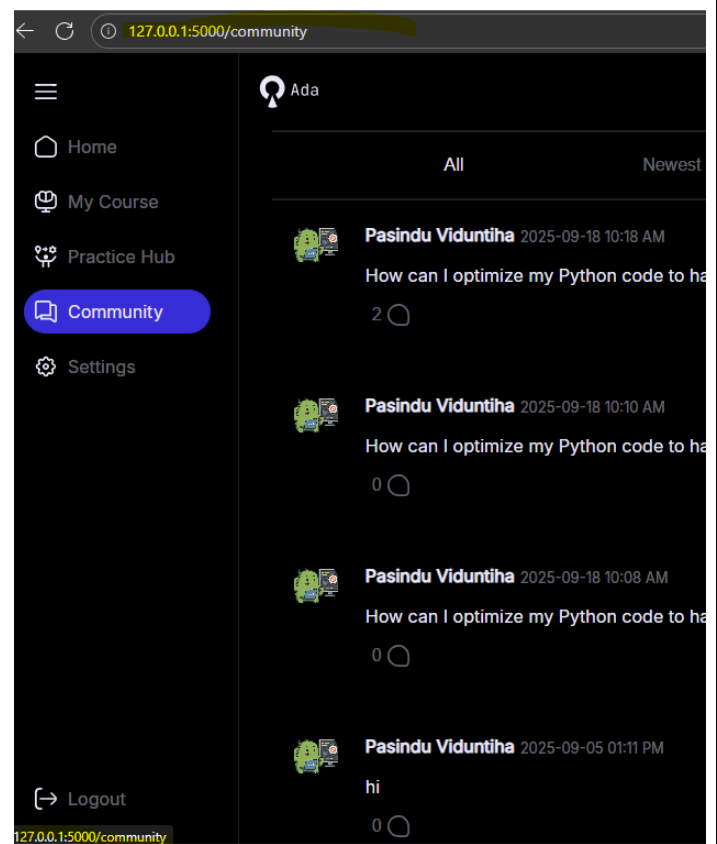
Test ID: 004

Goal: Ensure the Community link loads correctly.

Method: Hovered on *Community* in the side navigation.

Evidence: Browser status bar shows [127.0.0.1:5000/community](http://127.0.0.1:5000/community) (see screenshot).

Result: Passed – Link correctly opens the Community page.



Test ID: 005

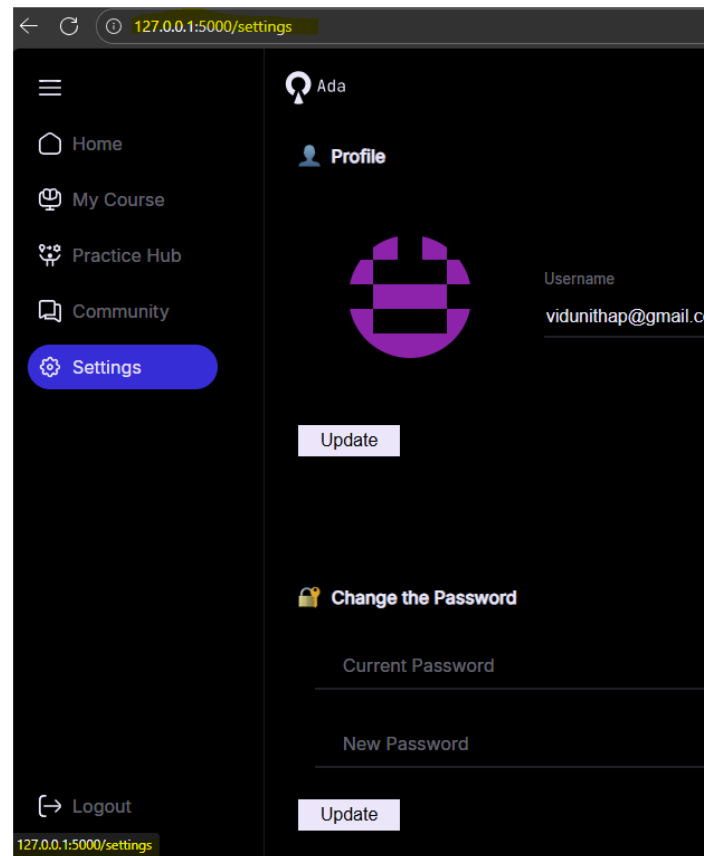
Goal: Ensure the Settings link loads correctly.

Method: Hovered on *Settings* in the side navigation.

Evidence: Browser status bar shows

127.0.0.1:5000/settings (see screenshot).

Result: Passed – Link correctly opens the Settings page.



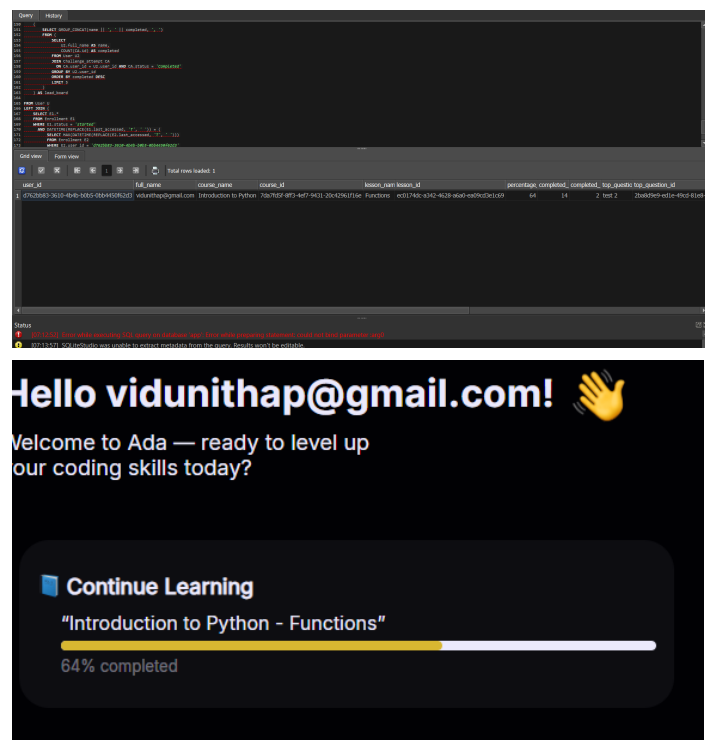
Test ID: 006

Goal: Ensure course progress in the UI matches the database.

Method: Ran SQL query for user **d762bb83-3610-4b4b-b0b5-0bb4450f62d3** to fetch current course and progress.

Evidence:

- SQL shows *Introduction to Python – Functions* with 64% completed.
- Dashboard UI shows the same course with 64% completed (see screenshot).  
Result: Passed – Database values are correctly displayed on the dashboard.







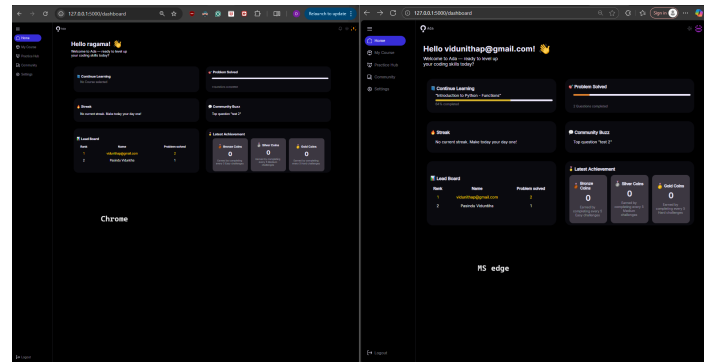
#### Test ID: 008

Goal: Ensure dashboard data displays consistently across browsers.

Method: Opened the dashboard in Chrome and Microsoft Edge with the same user.

Evidence: Screenshot comparison shows:

- Same user email **vidunithap@gmail.com** displayed.
- Same leaderboard data (Rank 1 with 2 problems solved).
- Same achievements (0 bronze, 0 silver, 0 gold).
- Same streak status = “No current streak”.  
Result: Passed – Data is consistent across Chrome and Edge, confirming correct web rendering and no browser-specific data issues.

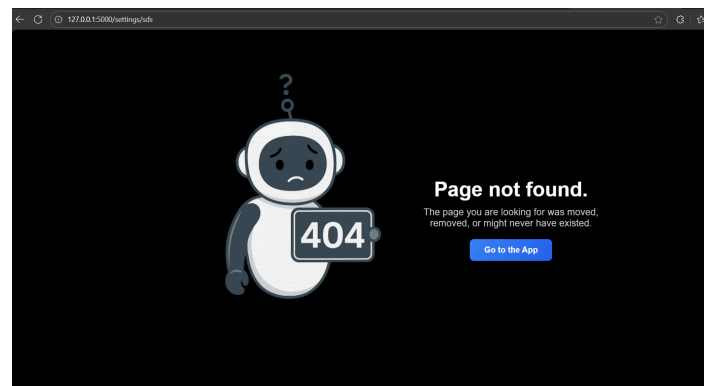


1. Chrome
2. MS edge

#### Test ID: 009

- Goal: Ensure the system handles invalid URLs correctly.
- Method: Entered an incorrect URL (**/settings/sds**) in the browser.
- Expected Result: User is shown a custom error page (404) with a friendly message and a link to return to the app.
- Actual Result: A 404 error page displayed with the message “*Page not found*” and a button to return.
- Result: Passed – The application handles invalid URLs properly without crashing.

NOTE: The same custom error handling is also implemented and tested for 403 (Forbidden) and 405 (Method Not Allowed). These return the correct error pages with friendly messages, ensuring users are not misdirected or exposed to unhandled errors



Test ID: 010

Goal: Ensure Community feed data (questions, answers, save status, user info, timestamps, and profile images) is consistent between the database and the frontend UI for viewer **d762bb83-3610-4b4b-b0b5-0bb4450f62d3**.

Method:

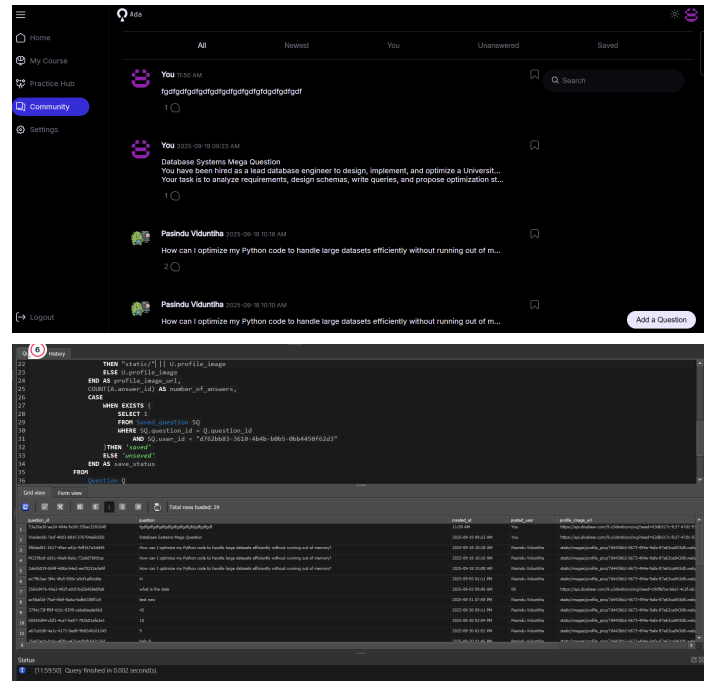
Ran SQL query joining **Question**, **Answer**, **User**, and **Saved\_question** to fetch:

- **question\_id**, **question text**, **created\_at**
- **posted\_user** (CASE: “You” if viewer’s own post, else display\_name)
- **profile\_image\_url** (CASE: Dicebear URL or static path)
- **number\_of\_answers** (COUNT(\*) from Answer)
- **save\_status** (saved/unsaved via EXISTS on Saved\_question)

Compared results against the UI tabs: *All*, *Newest*, *You*, *Unanswered*, *Saved*.

Evidence:

- Content: SQL shows questions “Database Systems Mega Question”, “How can I optimize my Python code...”, and recent test posts → all appear in *All* feed with identical text and timestamps.
- Save status: Items marked **saved** in SQL show filled bookmark in UI; **unsaved** show unfilled.
- Answer counts: SQL counts match UI indicators (e.g., 2 answers vs 0 answers).
- User labels: Viewer’s own posts display as “You”; others show correct usernames.
- Profile images: DB URLs (Dicebear absolute or static path) match avatars in UI.
- Sorting: “Newest” tab ordering matches **created\_at DESC**.
- Filters:
  - *You* tab shows only viewer’s posts.
  - *Unanswered* tab excludes questions with answers.



- *Saved* tab matches rows in *Saved\_question* for this user.

Result: Passed – All Community feed data from the database is displayed accurately and consistently on the frontend.

Test ID: 011

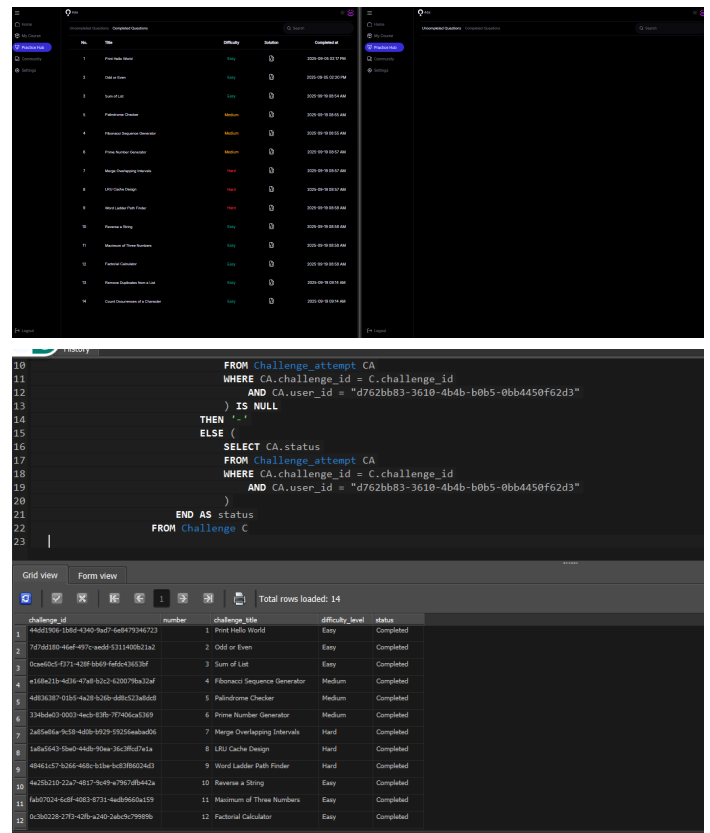
Goal: Ensure all available coding challenges and their statuses (Completed/Uncompleted) are displayed consistently between the database and the Practice Hub UI for user *d762bb83-3610-4b4b-b0b5-0bb4450f62d3*.

Method: Verified challenge list and statuses from the database against the frontend UI (*Uncompleted Questions* / *Completed Questions* tabs).

Evidence:

- Database output (screenshot): Shows 14 challenges, each with *status = Completed* (e.g., *Print Hello World*, *Odd or Even*, *Sum of List*, *Palindrome Checker*, *Prime Number Generator*, *LRU Cache Design*, etc.).
- UI (screenshot): *Completed Questions* tab displays the exact same 14 challenges with matching titles, difficulty levels, and “Completed at” timestamps.
- UI (screenshot): *Uncompleted Questions* tab is empty, correctly reflecting that no challenges remain uncompleted for this user.

Result: Passed – All challenge data and statuses are accurately displayed on the frontend, consistent with the database records.



The screenshot displays a database query result for a specific user. The query is a SQL statement that selects the status of 14 challenges from the *Challenge\_attempt* table, where the challenge ID and user ID match the criteria. The result is shown in a table with columns: *challenge\_id*, *number*, *challenge\_title*, *difficulty\_level*, and *status*. All 14 challenges listed have a status of 'Completed'.

| challenge_id                         | number | challenge_title              | difficulty_level | status    |
|--------------------------------------|--------|------------------------------|------------------|-----------|
| 44d21906-186d-4340-9a27-6a9479346723 | 1      | Print Hello World            | Easy             | Completed |
| 7d7d5180-46af-497c-ae8d-5111400b21a2 | 2      | Odd or Even                  | Easy             | Completed |
| 0c9a00c1-f371-428f-bb69-fd6a436c38f  | 3      | Sum of List                  | Easy             | Completed |
| e1d9a219-4d16-47a8-b2c2-620079ba32af | 4      | Fibonacci Sequence Generator | Medium           | Completed |
| 4d836387-01d6-4a28-b2b8-a88c521a6d8  | 5      | Palindrome Checker           | Medium           | Completed |
| 334bda63-0003-4aeb-83b8-7f7406ca5169 | 6      | Prime Number Generator       | Medium           | Completed |
| 2a85a8fa-9c18-4d38-b929-59256a8a0d6  | 7      | Merge Overlapping Intervals  | Hard             | Completed |
| 1a8a5643-5b40-44b8-b0ea-36c38fd7a1a  | 8      | LRU Cache Design             | Hard             | Completed |
| 4b415c71-b3a6-468a-b3ba-9d386a224d3  | 9      | Word Ladder Path Finder      | Hard             | Completed |
| 4a2b5210-22a7-4817-9a49-a79c78b442a  | 10     | Reverse a String             | Easy             | Completed |
| 1a070241-6d8f-4883-8721-4a4b9660a159 | 11     | Maximum of Three Numbers     | Easy             | Completed |
| 6c300228-27f3-42b5-a4d0-2abdc579889b | 12     | Factorial Calculator         | Easy             | Completed |

Test ID: 012

Goal: Ensure AI generated courses for the user **d762bb83-3610-4b4b-b0b5-0bb4450f62d3** are accurately displayed on the frontend with correct title, status, and created date.

Method: Queried the **Ai\_resource** table for this user and compared results against the UI under *AI generated Courses*.

Evidence:

- Database output (screenshot):
  - *Introduction to Python Programming* → Status: *Haven't Done*, Date: 2025-09-19
  - *Python Essentials* → Status: *Haven't Done*, Date: 2025-09-24
  - *Java Programming Basics* → Status: *Haven't Done*, Date: 2025-09-24
- UI (screenshot): Displays the exact same three courses with matching titles, status (*Haven't Done*), and created dates (2025-09-19 / 2025-09-24).

Result: Passed – Database course data is accurately displayed in the frontend under *AI generated Courses*.

The screenshot displays the 'AI generated Courses' section of a web application. It shows three course cards: 'Introduction to Python Programming' (created 2025-09-19), 'Python Essentials' (created 2025-09-24), and 'Java Programming Basics' (created 2025-09-24). All courses have a status of 'Haven't Done' and a 'Continue' button.

Below the UI, a SQL query is shown, selecting course details for a specific user ID from the 'Ai\_resource' table. The query is as follows:

```
1 SELECT
2     resource_id,
3     title,
4     status,
5     SUBSTR(generated_at, 1, instr(generated_at, 'T') - 1)
6     AS generated_date
7 FROM
8     Ai_resource
9 WHERE user_id = "d762bb83-3610-4b4b-b0b5-0bb4450f62d3"
```

The database output is shown in a table with 4 columns: resource\_id, title, status, and generated\_date. It contains 3 rows of data, matching the UI display.

| resource_id                          | title                              | status       | generated_date |
|--------------------------------------|------------------------------------|--------------|----------------|
| f15243a9-6987-4d7b-a5d1-9168d4bde4   | Introduction to Python Programming | Haven't Done | 2025-09-19     |
| 5402a35e-6a62-45c4-8959-922a02b2665a | Python Essentials                  | Haven't Done | 2025-09-24     |
| 93a2398b-9c59-4847-970e-3d72a67e0d07 | Java Programming Basics            | Haven't Done | 2025-09-24     |

Test ID: 013

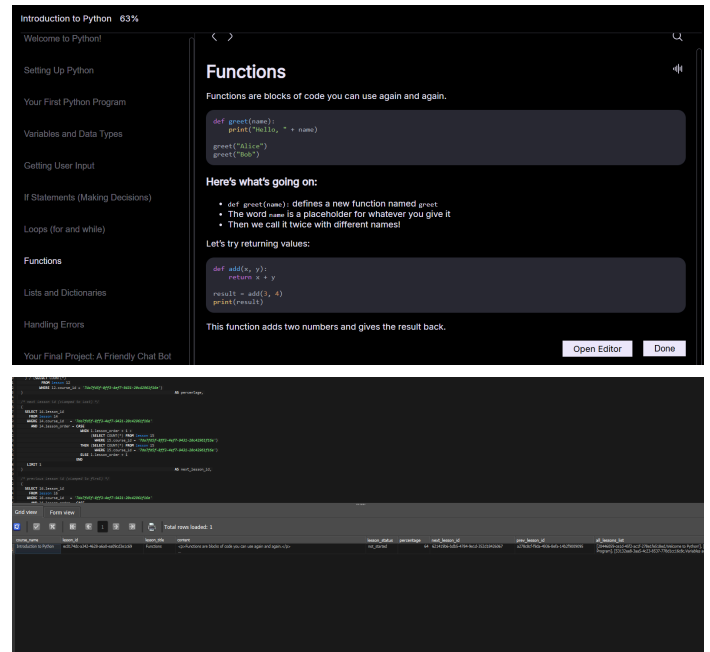
Goal: Verify the lesson view for user `d762bb83-3610-4b4b-b0b5-0bb4450f62d3` in course `7da7fd5f-8ff3-4ef7-9431-20c42961f16e` shows the correct lesson content, navigation (prev/next), lesson list, and overall completion %.

Method: Ran the course/lesson detail query for lesson `ec0174dc-a342-4628-a6a0-ea09cd3e1c69` and compared to the UI lesson page.

Evidence (from screenshots):

- Database output:
  - `course_name`: *Introduction to Python*
  - `lesson_id`: `ec0174dc-a342-4628-a6a0-ea09cd3e1c69` → `lesson_title`: *Functions*
  - `content`: HTML begins with “Functions are blocks of code you can use again and again.”
  - `lesson_status`: *not\_started*
  - `percentage`: 64
  - `next_lesson_id`: `621419b5-bdb5-4784-9e1d-352d18426067`
  - `prev_lesson_id`: `a278ec8f-9dda-4936-8ef8-14b2f8090995`
  - `all_lessons_list`: IDs/titles list (Welcome → ... → *Functions* → ...)
- UI lesson page:
  - Header shows course title *Introduction to Python* with progress 63% (1% lower due to UI rounding).
  - Sidebar shows the same lesson sequence; Functions is open/active.
  - Main panel text matches the DB content (“Functions are blocks of code you can use again and again.”) with the same examples.
  - Navigation controls present (*Open Editor / Done*), consistent with a `not_started` lesson.

Result: Passed – The lesson view faithfully reflects DB data (titles, content, lesson position, prev/next, and



overall progress).

Note on %: The 64% (DB) vs 63% (UI) difference is a rounding/formatting difference only; underlying counts match.

Test ID: 014

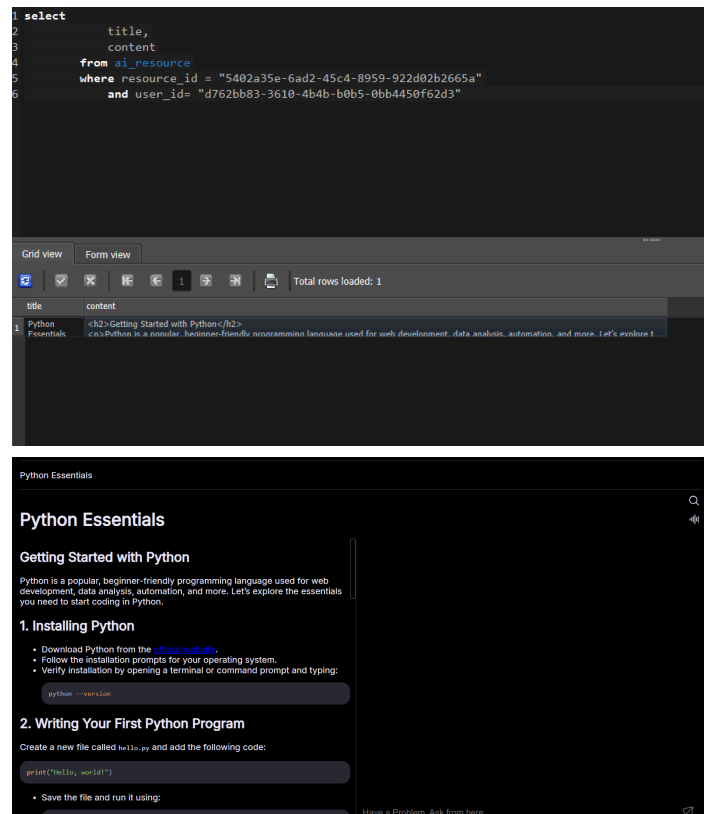
Goal: Verify that AI-generated course content is displayed correctly on the frontend for user `d762bb83-3610-4b4b-b0b5-0bb4450f62d3`.

Method: Queried `ai_resource` for `resource_id = 5402a35e-6ad2-45c4-8959-922d02b2665a` and compared results against the UI course view.

Evidence:

- Database output:
  - title: *Python Essentials*
  - content: `<h2>Getting Started with Python</h2><p>Python is a popular, beginner-friendly programming language ... (HTML formatted content including installation steps and code examples).`
- UI (screenshot):
  - Course page heading: *Python Essentials*
  - Section: *Getting Started with Python*
  - Body text matches the database content ("Python is a popular, beginner-friendly programming language used for web development, data analysis, automation...").
  - Includes identical steps for installing Python and writing the first program (`print("Hello, world!")`).

Result: Passed – AI-generated content is displayed in the UI exactly as stored in the database for the given resource.



Test ID: 015

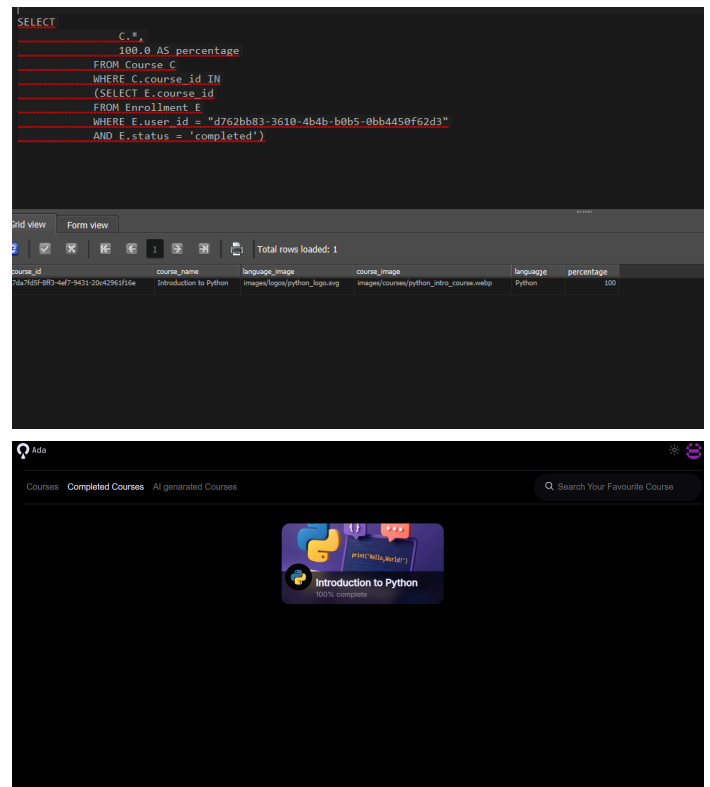
Goal: Verify that completed courses are displayed correctly on the frontend with 100% completion for user `d762bb83-3610-4b4b-b0b5-0bb4450f62d3`.

Method: Queried `Course` and `Enrollment` tables to return courses where the user's enrollment status is `completed` and compared results against the *Completed Courses* tab in the UI.

Evidence:

- Database output (screenshot):
  - `course_name`: *Introduction to Python*
  - `language`: *Python*
  - `course_image`: `images/courses/python_intro_course.webp`
  - `percentage`: 100
- UI (screenshot):
  - *Completed Courses* tab shows *Introduction to Python* with the same course image.
  - Progress displayed as 100% complete.

Result: Passed – The course marked as completed in the database is accurately displayed as 100% complete in the UI under *Completed Courses*.

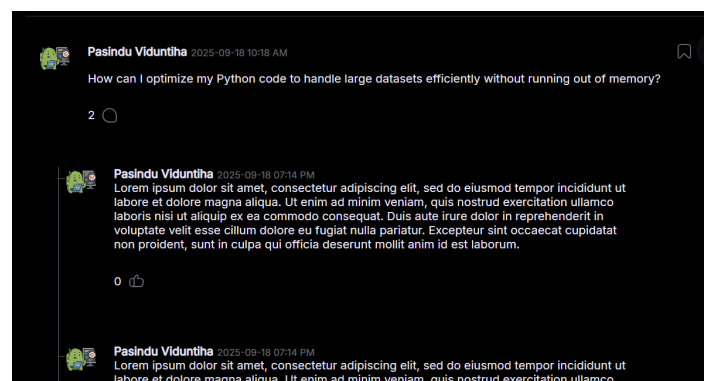


Test ID: 016

Goal: Verify that the *answers for a specific community question* are correctly counted and displayed on the UI for user `d762bb83-3610-4b4b-b0b5-0bb4450f62d3`.

Method: Used the question detail query (the one that joins `Question` with `Answer` and returns `number_of_ans` plus metadata) for question\_id `5fddad81-1617-49ac-a62c-9df167a3dd46`. Compared the result with the Community thread view.

Evidence (screenshots):



- Database output:
  - Question: *“How can I optimize my Python code to handle large datasets efficiently without running out of memory?”*
  - Posted user: *Pasindu Vidunitha*
  - **number\_of\_ans**: 2
  - **save\_status**: *unsaved*
- UI thread:
  - The same question text and poster appear.
  - Reply bubble shows 2.
  - Two answers are rendered under the question (the “Lorem ipsum...” bodies in your screenshot), matching the DB count.

Result: Passed – The same query you ran (which already returns **number\_of\_ans**) is sufficient to verify answers: the DB shows 2 answers and the UI displays exactly two answers for the thread.

NOTE: Answers are coming via AJAX, therefore we can’t get them from this query

```

CASE
WHEN EXISTS (
  SELECT 1
  FROM saved_question SQ
  WHERE SQ.question_id = Q.question_id
        AND SQ.user_id = 'd762bb83-3610-4b4b-b0b5-0bb4450f62d3'
) THEN 'saved'
ELSE 'unsaved'
END AS save_status
FROM question Q
JOIN user U ON U.user_id = Q.user_id
LEFT JOIN answer A ON A.question_id = Q.question_id
WHERE Q.question_id = '5f66a081-1617-49ac-a63c-9df767a3d446'
GROUP BY Q.question_id;

```

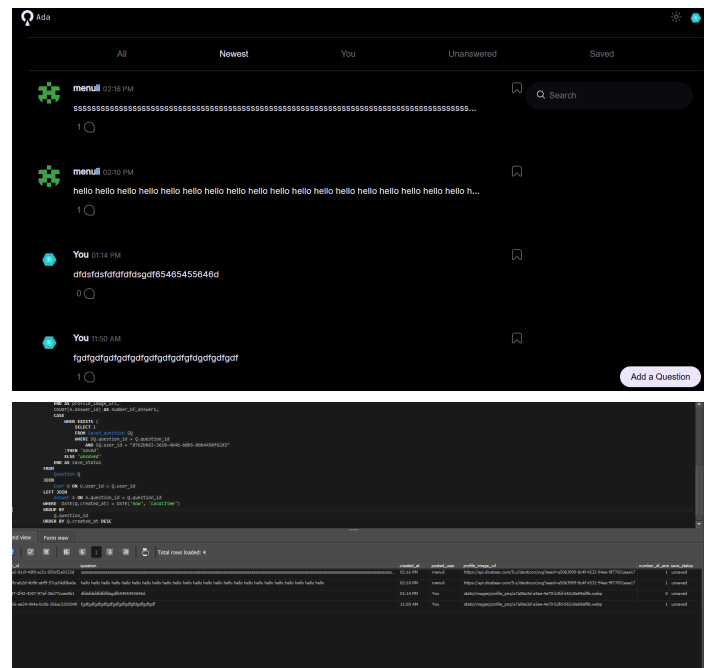
Test ID: 017

Goal: Verify the Newest feed shows today’s questions with correct poster, timestamp order, avatars, answer counts, and save state for user **d762bb83-3610-4b4b-b0b5-0bb4450f62d3**.

Method: Checked today’s questions in the DB (query result grid with 4 rows) and compared to the Newest tab UI.

Evidence (screenshots):

- DB: 4 questions for today sorted by **created\_at DESC**. Top two by *menuli* (02:16 PM, 02:10 PM) with Dicebear avatar URLs; next two by *You* (01:14 PM, 11:50 AM) with local **static/...webp** avatar paths. **number\_of\_ans** shows 1,1,0,1 respectively. All **save\_status** = *unsaved*.





- Result: Passed – The Newest feed exactly mirrors the database for ordering, poster, avatar source, answer counts, and save status.

Goal: Ensure the “You” feed correctly displays only the logged-in user’s questions (d762bb83-3610-4b4b-b0b5-0bb4450f62d3) with accurate timestamps, avatars, answers count, and save status.

Evidence (screenshots):

-

- 11:50 AM → shows same text, 1 answer
- 09:23 AM → “Database Systems Mega Question” matches DB text snippet, 1 answer  
Bookmark icons unfilled (unsaved), avatar consistent.

Result: Passed – The “You” tab correctly mirrors database content for the logged-in user: matching questions, timestamps, avatars, answer counts, and save state.

### Test ID: 019

Goal: Confirm the Unanswered tab only shows questions with 0 answers, with correct ordering, poster, avatars, and save state for user **d762bb83-3610-4b4b-b0b5-0bb4450f62d3**.

Method: Compared today’s unanswered questions from the DB result (19 rows, filtered where **answer\_id IS NULL**, ordered by **created\_at DESC**) with the Unanswered tab in the UI.

Evidence (screenshots):

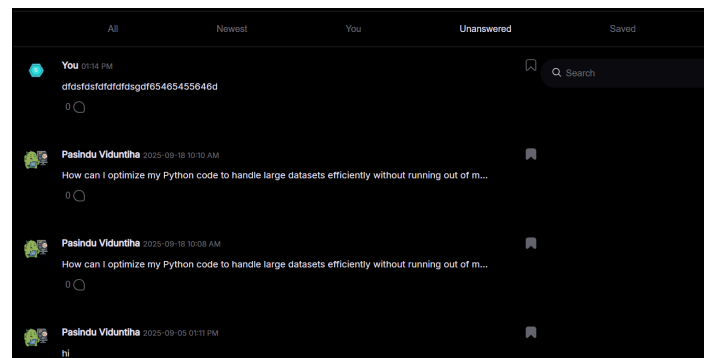
- Database: Top rows include
  - You · 01:14 PM → **dfdsfsdf...** → **number\_of\_ans = 0** → **unsaved**
  - Pasindu Vidunitha · 2025-09-18 10:10 AM → long “How can I optimize my Python...” → **0** → **saved**
  - Pasindu Vidunitha · 2025-09-18 10:08 AM → same title → **0** → **saved**
  - Pasindu Vidunitha · 2025-09-05 01:11 PM → **hi** → **0** → **saved**  
Avatars: Dicebear URLs for Pasindu; static path for You.
- UI (Unanswered tab): Shows the same four items in the same order/times, each with 0 in the reply bubble. Bookmark icons reflect the saved/unsaved states, and avatars match (Dicebear vs static).

```

SELECT *
FROM unanswered
WHERE (question_id = 0 OR answer_id = 0 OR answer_id IS NULL)
ORDER BY created_at DESC
LIMIT 10

```

| question_id | question                                                                                              | created_at          | posted_user       | profile_image_url              | number_of_answers | status  |
|-------------|-------------------------------------------------------------------------------------------------------|---------------------|-------------------|--------------------------------|-------------------|---------|
| 1000001     | dfdsfsdf...                                                                                           | 2025-09-05 01:14 PM | You               | static/path/you-avatar.png     | 0                 | unsaved |
| 1000002     | How can I optimize my Python code to handle large datasets efficiently without running out of memory? | 2025-09-18 10:10 AM | Pasindu Vidunitha | static/path/pasindu-avatar.png | 0                 | saved   |
| 1000003     | How can I optimize my Python code to handle large datasets efficiently without running out of memory? | 2025-09-18 10:08 AM | Pasindu Vidunitha | static/path/pasindu-avatar.png | 0                 | saved   |
| 1000004     | hi                                                                                                    | 2025-09-05 01:11 PM | Pasindu Vidunitha | static/path/pasindu-avatar.png | 0                 | saved   |



Result: Passed – The Unanswered feed accurately mirrors the database: only zero-answer questions are listed, with matching order, posters, avatars, and save status.

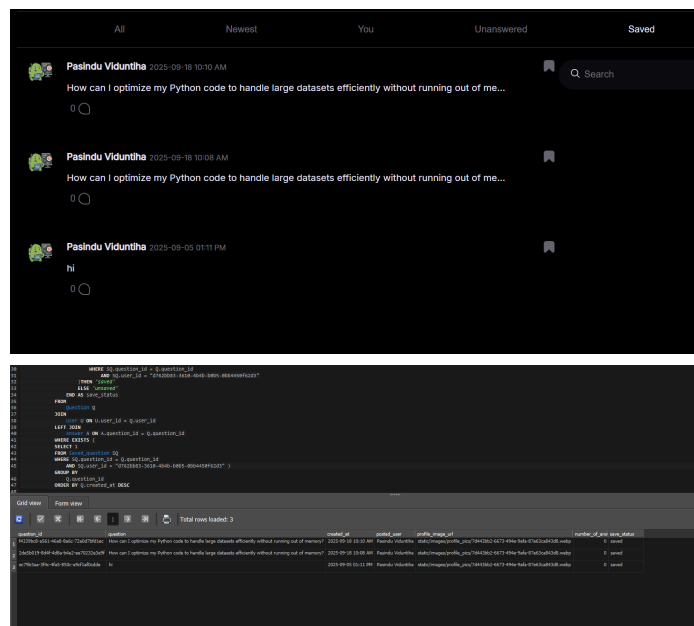
### Test ID: 020

Goal: Confirm the *Saved* tab only shows questions that the user has explicitly saved, with correct ordering, poster, avatars, and save state for user `d762bb83-3610-4b4b-b0b5-0bb4450f62d3`.

Method: Compared today's saved questions from the DB result (3 rows, filtered by `EXISTS (SELECT 1 FROM Saved_question SQ WHERE SQ.question_id = Q.question_id AND SQ.user_id = 'd762bb83-3610-4b4b-b0b5-0bb4450f62d3')`, ordered by `created_at DESC`) with the Saved tab in the UI.

Evidence (screenshots):

- Database: Top rows include
  - Pasindu Viduntiha · 2025-09-18 10:10 AM → “How can I optimize my Python code to handle large datasets...” → `number_of_ans = 0` → saved
  - Pasindu Viduntiha · 2025-09-18 10:08 AM → same title → `0` → saved
  - Pasindu Viduntiha · 2025-09-05 01:11 PM → “hi” → `0` → saved
  - Avatars: Dicebear URLs consistent for Pasindu
- UI (Saved tab):
  - Shows the same 3 questions with same content and avatars.
  - All entries show bookmark icon (saved state).
  - Ordering matches `created_at DESC` from DB.



|                                                                                                                     |  |
|---------------------------------------------------------------------------------------------------------------------|--|
| Result: Pass – Saved tab correctly reflects saved questions only, with accurate ordering, avatars, and save status. |  |
|---------------------------------------------------------------------------------------------------------------------|--|

## Feedback Integration

User testing included two technical and two non-technical testers. I have recorded their exact feedback in quotes and described how I responded.

Technical person (My father) - 1

Feedback: *"The app should not allow simple passwords.", "Add an email verification for the system"*

Rating: ★★★★★★★★☆☆ 8/10

My response: This is a great idea. I will try to do this in the next sprint if I have enough time to do so; otherwise, I will keep this idea for future improvements. Secondly, he said that adding an email verification to the authentication is also another great idea, and I am considering putting this idea into the future improvements section, because of the limited time.

Technical person (Kyle - A classmate) - 2

Feedback: *"Make the navigation bar draggable and reduce its width.", "Tells the user what was wrong " (in the login page)*

Rating: ★★★★★★★★☆☆ 9/10

My response: This is a good UI idea, but with my limited time, I don't think that I will be able to do this; however, I will keep this idea for future improvements. In the sign-up page, he said that it tells what the user has gone wrong, for example, the Username is not correct, or the Password is not correct. I thought about this when I was developing the signing page, but I thought that this would explore user information on the client side. If someone is trying to hack a user account, it will be easy for them to get the user data. Therefore, I will keep my current error message. Furthermore, he said that a hovering effect should be applied to the widgets in the dashboard to make them clickable, and he said that a tick should be placed in front of each completed lesson to inform the user that they have completed the lesson. I feel like this is a great UI improvement, but first of all, I have to complete all the core features of my application.

Non-technical person (Tathsilu - A Sri Lankan Friend) - 1

Feedback: *"Add a user guide to the AI-generating courses page, because it takes a little time to understand the system.",*

*"Add the title Welcome to the login page",*

*"Keep the code editor in the lesson page, so we do not need to go back and forward"*

Rating: ★★★★★★★★☆☆ 8/10

My response: The first impression of the application is nice. He said to change the title to "Welcome" on the login page instead of keeping the name of the application. I think it is good feedback because I checked a lot of modern login pages, most of the UIs have "Welcome " or some kind of greeting message; therefore, I am hoping to add that to future improvements. Also, he said, keep the code editor on the lesson page. This was my previous UI idea, but as I need a responsive design, it is hard to keep all the widgets on the same page, and it is getting messy, and the widgets would be smaller, which affects user experience badly; therefore, I will neglect this idea. Furthermore, he said that he would add a user guide to the AI-generating courses page, because he said that it takes a little time to understand the system. To make the application more user-friendly, I did this in the 8th sprint. Also, he said that he will add a restriction for AI-generated courses, as current people can ask, how to cook, and AI is generating a course on that, but it is not relevant to my application theme; therefore, I added a restriction to the AI model in the 8th sprint to generate only technical-based courses.

Non-technical person (My mother) - 2

Feedback: *"Add a brief explanation about the app on the landing page."*

Rating: ★★★★★★★★★★ 10/10

Feedback: She said that the app was a superb tool for learning programming languages, and it was very user-friendly. For the improvements, she said that adding a brief explanation about the app in the

Landing page, although I have added a brief explanation in the landing page, maybe it is not noticeable to users, so I will consider increasing the font size.

Overall, there is much positive feedback from both sides (technical and non-technical). I will try to add as many of these improvements to the application in the next sprint, but firstly, I should complete the core REST features of the application in the next sprint.

## PROGRAMMING CONVENTIONS AND STANDARDS

To ensure my program is maintainable, consistent and easy to collaborate with others, I used Python programming conventions and standards throughout my Python code. I followed the PEP8 style guide. I used snake case for naming conventions (eg,- user\_id). I tried my best to reduce the repetition of the code, using custom helper functions (eg, db\_execute(), validate\_email\_address()). I used clear, meaningful variable names in my code. For the error handling, I used try/except or if statements. I used Flake8 and Pylint tools to check my mistakes in my Python code. Below, I have shown an example of my Python code to show the indentation, descriptive naming, and error handling.

Python

```
def validate_email_address(email):
    try:
        email_info = validate_email(email, check_deliverability=False)
        if email_info:
            return "valid"
        else:
            return "invalid"
    except EmailNotValidError:
        return "invalid"
```

## DESIGN CHOICE JUSTIFICATION

During development, I tried various technologies and methods before deciding on the suitable options for my application. These choices were made based on efficiency, maintenance and usability. At the beginning of this project, my backend code was in one Python file. Later, as the project became more complex and bigger, I chose to divide the code into blueprints. This helped me a lot because I can focus on specific parts when I have an error, and it will also help with future maintenance. Initially, I handled SQL joins with Python, but this increased the code complexity, reduced the readability of the code, and reduced the performance of the application. Then I decided to use joins directly in SQL code. I used a blue colour-based theme for my whole application because it gives trust, relaxation, less stress, intelligence and stability according to the psychological resources. This is best for a developer-focused interface; this aligns with other developer tools like VS Code, GitHub. In the early stages, I wrote a long block of inline code, which increased the repetition and was hard to maintain; therefore, I created a reusable helper function like db\_execute().

## HCI CONVENTIONS JUSTIFICATION

When designing the interface, I applied HCI conventions to make the system clear, accessible, and easy to use across different devices (PCs and Mobile devices). I kept my designs very minimalistic and clean, as evidenced by my feedback. I added a light theme and a dark theme to my application for user preference. I used different kinds of fonts for the landing page to keep the user, but after logging in to the application, I used the “Inter”

font throughout the application. I used a consistent font throughout the application to keep my designs minimalistic and reduce overwhelming feelings for the end users. By following these HCI guidelines, I ensure that my application is both user-friendly and accessible. I used icons and language that most users are familiar with (eg, in the navigation bar, I used Home, My Course, Settings, etc). After the user signed up for the application, I used the same format for the application (base.html). I ensured to stick to my colour palette always. This shows that I applied HCI conventions such as consistency, minimalism, and flexibility, which align with Nielsen's usability heuristics.

## IMPLICATIONS OF THE DIGITAL OUTCOME

### Database Implications

#### Ethical – Protecting User Accounts

**Meaning:** Protecting user accounts means keeping login details private and secure, so users' personal information is not stolen or exposed to the outside world.

**How I expressed it:** If passwords are stored in plain text, they can be hacked easily, which is unfair and affects user privacy also. To reduce these risks, I used Argon2 password hashing, which makes the passwords more secure. I chose this because this is one of the best ways to protect passwords compared to other methods like hashing and encrypting. I have not yet added stronger password rules due to time limits, but that would be a future improvement to make the system more secure.

#### Ethical – Account Deletion

**Meaning:** Account deletion relates to respecting the users' choice to leave a system. Ethically, if a user wants to delete their account, that means all their personal information should be removed.

**How I expressed it:** Some systems are doing only 'soft delete' accounts, meaning the data is hidden, but still keeping users' information in databases. I chose to use permanent deletion, so when a user deletes their accounts, all the data relevant to that user will be deleted. This respects their decision and supports privacy. In the future, I could add a short recovery period (like 7 days) in case the deletion was a mistake, but the main principle is to respect the user's choice.

### Web Implications

#### Legal – Copyright and Attribution

**Meaning:** Copyright means respecting ownership of creative work, such as icons, images, and written content. Using someone else's work without permission can lead to legal issues.

**How I expressed it:** I used icons from Flaticon, which are free to use with attribution. To follow this rule, I added credit to them in my project documentation and my landing page (footer). I also used AI-generated images and AI course content, which do not have copyright restrictions. This shows my application respects copyright law and avoids plagiarism. For profile picture avatars I used Dicebear, which provides free, CC0-licensed illustrations, so no attribution is required and copyright is respected.

#### Social – Usability and Accessibility

**Meaning:** A social implication is how easy the system is to use for different kinds of users. If the user interface is confusing, users may struggle to use the application.

**How I expressed it:** Feedback from non-technical users helps a lot, because they don't have much development experience, but for my application, most users have basic knowledge of programming, but I used

people who do not have any technical background, because I need to make the application as easy to use as possible. Feedback from these non-technical users helped me a lot to create an accessible, user-friendly application for a wider range of users.

## CREDITS

- Icons designed by Flaticon ([www.flaticon.com](https://www.flaticon.com))
- Some images generated using ChatGPT (OpenAI)
- Course content generated with the help of ChatGPT AI